CO-RANK 1 ARITHMETIC SIEGEL–WEIL I: LOCAL NON-ARCHIMEDEAN

RYAN C. CHEN

Abstract. This is the first in a sequence of four papers, where we prove the arithmetic Siegel–Weil formula in co-rank 1 for Kudla–Rapoport special cycles on exotic smooth integral models of unitary Shimura varieties of arbitrarily large even arithmetic dimension. Our arithmetic Siegel–Weil formula implies that degrees of Kudla–Rapoport arithmetic special 1-cycles are encoded in near-central first derivatives of unitary Eisenstein series Fourier coefficients.

The crucial input is a new local limiting method at all places. In this paper, we formulate and prove the key local theorems at all non-Archimedean places. On the analytic side, the limit relates local Whittaker functions on different groups. On the geometric side at nonsplit non-Archimedean places, the limit relates degrees of 0-cycles on Rapoport–Zink spaces and local contributions to heights of 1-cycles in mixed characteristic.

Contents

Preliminary 3

1. Introduction 3
1.1. Non-Archimedean local main theorems . . . 4
1.2. Strategy . . . 6
1.3. Sketch . . . 8
1.4. Eisenstein series . . . 11
1.5. Arithmetic Siegel–Weil . . . 12
1.6. Previous work . . . 13
1.7. Outline . . . 15
1.8. Acknowledgements . . . 15

2. Conventions on Hermitian spaces and lattices 15
2.1. Hermitian, alternating, symmetric . . . 15
2.2. Lattices for local fields . . . 17

Part 1. Local special cycles 19

3. Moduli spaces of p-divisible groups 19

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3.1. Rapoport–Zink spaces . . . 19
3.2. Local special cycles . . . 24
3.3. Actions on Rapoport–Zink spaces . . . 25
3.4. Similitude variants . . . 25
3.5. Discrete reduced subschemes . . . 28
3.6. Horizontal and vertical decomposition . . . 30
3.7. Serre tensor and signature $(1,1)$ . . . 33

4. More on moduli of p-divisible groups: split 35
4.1. Lifting theory for ordinary p-divisible groups . . . 35
4.2. Quasi-canonical lifting cycles: split . . . 37

5. Canonical and quasi-canonical liftings 42
5.1. Canonical liftings . . . 42
5.2. Quasi-canonical liftings . . . 43
5.3. Quasi-canonical lifting cycles . . . 45

Part 2. Local main theorems 47

6. Local Whittaker functions 47
6.1. Additional notation on $U(m,m)$ . . . 47
6.2. Principal series . . . 47
6.3. Normalized local Whittaker functions . . . 48
6.4. Limits . . . 50

7. Non-Archimedean local main theorems 50
7.1. Statement of local main theorem . . . 51
7.2. Horizontal identity . . . 53
7.3. Induction formula . . . 55
7.4. More on induction formula: split . . . 59
7.5. Limits . . . 63

References 68

Preliminary

1. Introduction

The landmark work of Gross and Zagier [GZ86] showed that Néron–Tate heights of Heegner points on elliptic curves over $\mathbb{Q}$ are encoded in the first central derivatives of associated Rankin–Selberg L-functions. After the work of Gross and Keating [GK93] on arithmetic intersection numbers for modular correspondences, Kudla proposed to recast such formulas in the language of special cycles on higher-dimensional Shimura varieties. This was originally formulated for integral models of orthogonal Shimura varieties in low dimensions by Kudla [Kud97a; Kud97b; Kud04] and the subsequent work of Kudla and Rapoport [KR99; KR00], where they pioneered the moduli definition of special cycles on integral models. Later, the attention was shifted to unitary Shimura varieties by Kudla–Rapoport in [KR11; KR14]. Along with other closely related predictions about special cycles (e.g. modularity of generating series), these ideas are now called *Kudla's program*. Kudla's program has played a role in a range of works, such as Gross–Zagier formulas on Shimura curves [YZZ13], the averaged Colmez conjecture [AGHMP18; YZ18], the arithmetic fundamental lemma [Zha21], results on the Beilinson–Bloch conjecture [LL21], and Picard rank jumps for K3 surfaces [SSTT22]. We refer to Li's excellent surveys [Li23; Li24] for more.

Our work is about *arithmetic Siegel–Weil formulas* in Kudla's program [Kud04, Problem 6], which conjecturally relate the first derivatives of Siegel Eisenstein series for unitary or symplectic groups with "arithmetic theta series" formed from special cycles on integral models of unitary or orthogonal Shimura varieties, as in (1.5.3) below. These are closely parallel to the classical (resp. geometric) Siegel–Weil formulas, which state that special values of Eisenstein series encode representation numbers (resp. complex volumes) for lattices (resp. complex special cycles on Shimura varieties). For the interested reader, we refer to the introduction of our paper [Che24c] where formulations of classical/geometric/arithmetic Siegel–Weil are compared in more detail.

Our contribution to Kudla's program is to prove an arithmetic Siegel–Weil formula relating arithmetic volumes of arithmetic 1-cycle classes and *near-center* derivatives of Eisenstein series. This can also be viewed[1] as a result about *central* derivatives for "singular Fourier coefficients of co-rank 1", hence the title of this work.

The analogous result for arithmetic 0-cycle classes and *central* derivatives for *nonsingular* Fourier coefficients is by now known, thanks to the non-Archimedean work of Li–Zhang [LZ22a], Li–Liu [LL22], and the Archimedean work of Liu [Liu11]. In the non-Archimedean case, the approach is to first decompose the global intersection problem into local intersection problems on Rapoport–Zink spaces at each place, as in [KR14]. Li–Zhang and Li–Liu then solve the problem of relating degrees of local special cycles on these spaces with central derivatives of local Whittaker functions. In another direction, for the "most singular" Fourier coefficient (the constant term), Hörmann [Hör14] and Bruinier–Howard [BH21] have made progress via completely different methods, using the theory of Borcherds products. If the Shimura varieties are compact, the Archimedean aspects are also understood by the work of Garcia–Sankaran [GS19], using the theory of superconnections.

[1] There are some subtleties here about linear invariance and non-diagonalizability, etc., which are discussed in [Che24c, Section 9.1].

Our non-Archimedean work on arithmetic 1-cycle classes takes place in a setting closest to Li–Zhang and Li–Liu, in the sense that we also make a local decomposition over all places. For this, we introduce in [Che24b] a suitably canonical decomposition of Arakelov heights, in order to formulate our non-Archimedean local theorems in terms of local special cycles on Rapoport–Zink spaces and "local height" contributions.

The present paper proves these key "local arithmetic Siegel–Weil" theorems at all non-Archimedean places. We first explain the formulation of our local theorems, and then sketch our strategy: we take the theorems on 0-cycle classes from [LZ22a; LL22] as starting inputs and produce our theorems on 1-cycle classes by "degeneration". Afterwards, we give context by overviewing our main global arithmetic Siegel–Weil results for our four-paper sequence, consisting of this paper along with [Che24a; Che24b; Che24c]. Our proofs of these global results are completed in [Che24c] by patching together our local main theorems.

1.1. **Non-Archimedean local main theorems.** Our non-Archimedean local main theorems relate Kudla–Rapoport local special cycles with *near-center* derivatives of local Whittaker functions.

Let F be a finite étale $\mathbb{Q}_p$-algebra of degree 2. In the introduction, we assume $p \neq 2$ unless $F/\mathbb{Q}_p$ is split (i.e. $F \cong \mathbb{Q}_p \times \mathbb{Q}_p$). If F is a field, we write $\breve{F}$ for the completion of the maximal unramified extension of F. If $F = \mathbb{Q}_p \times \mathbb{Q}_p$, we set $\breve{F} \coloneqq \breve{\mathbb{Q}}_p$ and make a choice of $F \to \breve{F}$.

Given any integer $n \geq 1$, there is an associated *Rapoport–Zink space* $\mathcal{N} \coloneqq \mathcal{N}(n-1,1)$, parameterizing certain polarized p-divisible groups X of height $2n$ and dimension n, with $\mathcal{O}_F$-action of signature $(n-1,1)$ and some additional conditions (Section 3.1). Here, we assume n is even if $F/\mathbb{Q}_p$ is ramified. This Rapoport–Zink space $\mathcal{N}$ is a formal scheme over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$, equidimensional of dimension n. Given framing p-divisible groups $\mathbf{X}_0$ and $\mathbf{X}$ over $\overline{\mathbb{F}}_p$, of dimensions 1 and n respectively, there is a space of *local special quasi-homomorphisms*

$$\mathbf{W} \coloneqq \operatorname{Hom}^0_F(\mathbf{X}_0, \mathbf{X}), \tag{1.1.1}$$

as in Section 3.2. This $\mathbf{W}$ is a non-degenerate $F/\mathbb{Q}_p$ Hermitian space of dimension n if $F/\mathbb{Q}_p$ is nonsplit and dimension $n-1$ if $F/\mathbb{Q}_p$ is split. Given any integer $m \geq 1$ and any m-tuple $\underline{\mathbf{x}} \in \mathbf{W}^m$, there is an associated *Kudla–Rapoport local special cycle* $\mathcal{Z}(\underline{\mathbf{x}}) \hookrightarrow \mathcal{N}$, which is the (closed) locus where every $\mathbf{x} \in \underline{\mathbf{x}}$ deforms to a homomorphism of p-divisible groups (Section 3.2).

The "expected codimension" of $\mathcal{Z}(\underline{\mathbf{x}}) \hookrightarrow \mathcal{N}$ is m, so we consider *derived local special cycle (classes)* ${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}) \in \operatorname{gr}^m_{\mathcal{N}} K'_0(\mathcal{Z}(\underline{\mathbf{x}}))_{\mathbb{Q}}$ in codimension m (Section 3.6). If all elements of the tuple $\underline{\mathbf{x}} = [\mathbf{x}_1, \dots, \mathbf{x}_m]$ are nonzero, then we have

$${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}) \coloneqq \mathcal{O}_{\mathcal{Z}(\mathbf{x}_1)} \otimes^{\mathbb{L}} \cdots \otimes^{\mathbb{L}} \mathcal{O}_{\mathcal{Z}(\mathbf{x}_m)}. \tag{1.1.2}$$

For any $\mathbf{x}_i = 0$, it is usual to make a similar definition by replacing $\mathcal{O}_{\mathcal{Z}(\mathbf{x}_i)}$ with $[\mathcal{O}_{\mathcal{N}}] - [\mathcal{E}]$ for a certain *tautological bundle* $\mathcal{E}$ on $\mathcal{N}$ (Definition 3.1.7). For any closed subscheme $Z \subseteq \mathcal{N}$ which is proper over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$, there is a degree map $\deg_{\overline{\mathbb{F}}_p} \colon \operatorname{gr}^n_{\mathcal{N}} K'_0(Z)_{\mathbb{Q}} \to \mathbb{Q}$ given by taking Euler characteristics of coherent sheaves (3.6.15).

We are interested in "height contributions" for some local 1-cycle classes, namely ${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)$ for $\underline{\mathbf{x}}^\flat \in \mathbf{W}^{n-1}$ with nonsingular Gram matrix. For this, write $\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}} \subseteq \mathcal{Z}(\underline{\mathbf{x}}^\flat)$ for the largest closed formal subscheme which is flat over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$ ("horizontal") and write ${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{V}} \in \operatorname{gr}^{n-1}_{\mathcal{N}} K'_0(\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\overline{\mathbb{F}}_p})_{\mathbb{Q}}$

for the projection of ${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)$ to the special fiber ("vertical") in the sense of Section 3.6. One can show

$$ {}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat) = [\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}] + {}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{V}}, \tag{1.1.3} $$

using [Zha21, Lemma B.1, Lemma B.2] and their proofs. This decomposition of ${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)$ was introduced in the inert case by Li–Zhang [LZ22a, Definition 5.2.1], and also used in the ramified case by Li–Liu [LL22, Section 2.6]. For such $\underline{\mathbf{x}}^\flat$, the structure map $\mathcal{Z}(\underline{\mathbf{x}}^\flat) \to \operatorname{Spf} \mathcal{O}_{\breve{F}}$ is adic and proper [Che24b, Lemma 6.7.3], as also used by [LZ22a] (inert) and [LL22] (ramified). This properness is special to the situation of 1-cycle and 0-cycle classes.

The "vertical height contribution" will simply be the algebro-geometric intersection number

$$ \deg_{\overline{\mathbb{F}}_p}({}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{V}} \cdot \mathcal{E}^\vee) \cdot \log p \tag{1.1.4} $$

as in (3.6.14).

The "horizontal height contribution" is more delicate. The horizontal part $\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}$ is adic finite flat over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$, and we define a "horizontal height contribution" for each component $\mathcal{Z} \hookrightarrow \mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}$ of (the scheme associated to) $\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}$.

One can show that the universal p-divisible group over each $\mathcal{Z}$ has CM. When $n \geq 2$, the universal p-divisible group over $\mathcal{Z}$ decomposes as $\mathfrak{X}_s^2 \times \mathfrak{X}_0^{n-2}$ for CM p-divisible groups $\mathfrak{X}_s$ and $\mathfrak{X}_0$ of height 2 and dimension 1, where $\operatorname{End}(\mathfrak{X}_0) \cong \mathcal{O}_F$ and where $\operatorname{End}(\mathfrak{X}_s) \subseteq \mathcal{O}_F$ is the unique order of index p^s (Section 5.3).

The $\mathcal{O}_F$-module $\operatorname{Hom}(\mathfrak{X}_0, \mathfrak{X}_s)$ (by pre-composition) is free of rank one, with $\psi_s \in \operatorname{Hom}(\mathfrak{X}_0, \mathfrak{X}_s)$ being a generator if and only if ψ_s is an isogeny of degree p^s ("minimal degree"). Choosing any generator ψ_s, we then define the "*local change of tautological height*"

$$ \breve{\delta}_{\mathrm{tau}}(\psi_s) \coloneqq \delta_{\mathrm{tau}}(s) \cdot \log p \coloneqq \left(-\frac{1}{2} v_p(\deg \psi_s) + \frac{1}{[\breve{E}_s : \breve{\mathbb{Q}}_p]} \operatorname{length}_{\mathcal{O}_{\breve{E}_s}} (e^* \Omega^1_{\ker \psi_s / \operatorname{Spec} \mathcal{O}_{\breve{E}_s}}) \right) \log p \tag{1.1.5} $$

where $\breve{E}_s$ is the "field of definition" for $\mathfrak{X}_s$ (Section 5.2) and e^* denotes pullback along the identity section. We have $\delta_{\mathrm{tau}}(s) \in \mathbb{Q}$; this quantity only depends on $s \in \mathbb{Z}_{\geq 0}$, the prime p, and whether $F/\mathbb{Q}_p$ is inert/split/ramified. We then take $\delta_{\mathrm{tau}}(\mathcal{Z}) \coloneqq \delta_{\mathrm{tau}}(s)$, where s could certainly depend on $\mathcal{Z}$.

Definition 1.1.1. For each horizontal component $\mathcal{Z} \hookrightarrow \mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}$ of the special cycle, we take the "horizontal height contribution" for $\mathcal{Z}$ to be

$$ \deg(\mathcal{Z}) \cdot \delta_{\mathrm{tau}}(\mathcal{Z}) \cdot \log p \tag{1.1.6} $$

where $\deg(\mathcal{Z})$ is the degree of $\mathcal{Z} \to \operatorname{Spf} \mathcal{O}_{\breve{F}}$.

In the body of this text, the quantity in (7.1.3) is twice that of Definition 1.1.1. The formula in (1.1.5) is reminiscent of the usual global formula for change of stable Faltings heights along an isogeny of abelian varieties. Unlike the global formula, which does not depend on the choice of isogeny (Faltings height is intrinsic), the quantity $\breve{\delta}_{\mathrm{tau}}(\psi_s)$ may change in general if ψ_s is not chosen to be an isogeny of minimal degree. For this reason, it is somewhat involved for us to verify that the "local change" $\breve{\delta}_{\mathrm{tau}}(\psi_s)$ for minimal degree ψ_s will indeed compute the "correct" local

contribution to a global height, for our eventual global applications. This subtlety is treated in [Che24b, Part 2]. A similar formulation for "local change of height" also appears for different special cycles in our forthcoming joint work [CLZ] on subleading terms of adjoint L-functions. There, such "local change of heights" appear in our formulation of new "near-center arithmetic fundamental lemmas" for near-center derivatives of orbital integrals.

On the analytic side, for integers $m \geq 0$, we consider a quasi-split unitary group $H \coloneqq U(m,m)$ for an $F/\mathbb{Q}_p$ Hermitian space of dimension $2m$ (Section 6.1). For standard sections $\Phi \in I(s,\chi)$ of a certain (local) degenerate principal series for the Siegel parabolic in H, we have *local Whittaker functions* $W_T(h,s,\Phi)$ associated to Hermitian matrices $T \in \mathrm{Herm}_m(\mathbb{Q}_p)$, with $h \in H(\mathbb{Q}_p)$ and $s \in \mathbb{C}$ (Section 6.3).[2] These local Whittaker functions appear in Fourier coefficients of Eisenstein series. For the normalized spherical standard section Φ°, we then consider a *normalized spherical local Whittaker function*

$$W_T^*(s)_n^\circ \coloneqq \Lambda_T(s)_n^\circ W_T(1,s,\Phi^\circ) \tag{1.1.7}$$

for a *local normalizing factor* $\Lambda_T(s)_n^\circ$ (consisting of local L-factors, etc.) which produces a symmetric functional equation $s \leftrightarrow -s$ (Section 6.3) for nonsingular T.

With setup as above, our non-Archimedean "local arithmetic Siegel–Weil" theorem is the following.

Theorem A. *For any nonsingular $T^\flat \in \mathrm{Herm}_{n-1}(\mathbb{Q}_p)$ and any $\underline{\mathbf{x}}^\flat \in \mathbf{W}^{n-1}$ with Gram matrix $T^\flat$, we have*

$$-[\breve{F}:\breve{\mathbb{Q}}_p]\frac{d}{ds}\bigg|_{s=1/2} W_{T^\flat}^*(s)_n^\circ = \left(\frac{2}{[\breve{F}:\breve{\mathbb{Q}}_p]}\deg_{\overline{\mathbb{F}}_p}(\mathcal{E}^\vee \cdot {}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{V}}) + 2\sum_{\mathcal{Z}\hookrightarrow\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}} \deg(\mathcal{Z})\cdot\delta_{\mathrm{tau}}(\mathcal{Z})\right)\cdot\log p. \tag{1.1.8}$$

This appears below as Theorem 7.1.2, there stated in terms of local densities. When $n=1$, both sides of (1.1.8) are zero.

We highlight the fact that $s=1/2$ is *not* the central point of the functional equation for the local Whittaker function $W_{T^\flat}^*(s)_n^\circ$, so Theorem A concerns a *near-center* derivative. Moreover, the value of $W_{T^\flat}^*(s)_n^\circ$ at $s=1/2$ (generally nonzero) *simultaneously* has geometric meaning: it equals $[\breve{F}:\breve{\mathbb{Q}}_p]\deg \mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}$, where $\deg \mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}$ denotes the degree of $\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}} \to \mathrm{Spf}\,\mathcal{O}_{\breve{F}}$ (Lemma 7.1.3). This "geometric local Siegel–Weil" formula for the value at $s=1/2$ was previously observed by Li–Zhang when $F/\mathbb{Q}_p$ is inert [LZ22a, Corollary 4.6.1]. Our Theorem A thus concerns a *subleading* term in the power series expansion at $s=1/2$. These phenomena reappear globally in [Che24c, Theorem A].

1.2. **Strategy.** We now describe our key local strategy: "take a limit" (Figure 1). In fact, this is our strategy at both non-Archimedean and Archimedean places. The Archimedean case is the subject of [Che24a].

[2]The notation $\mathrm{Herm}_m(\mathbb{Q}_p)$ means Hermitian matrices with F-coefficients, i.e. Herm_m is a scheme over $\mathbb{Q}_p$. We make similar notation elsewhere, e.g. $\mathrm{Herm}_m(\mathbb{R})$ will typically mean complex Hermitian matrices. See conventions at the end of Section 2.1.

In Figure 1 below, for $F/F_0 = \mathbb{C}/\mathbb{R}$ or $F/F_0 = F/\mathbb{Q}_p$, we consider $T^\flat \in \mathrm{Herm}_{n-1}(F_0)$ with $\det T^\flat \neq 0$, and $T = \mathrm{diag}(t, T^\flat)$ for suitable nonzero $t \in F_0$. On the left, the limit refers to $t \to 0$ in the natural topology on F_0 (real in the Archimedean case, and p-adic in the non-Archimedean case). The upper horizontal arrow is Theorem A above (a local version of (1.5.3) below, at the non-central point $s_0 = 1/2$), and the lower horizontal arrow should be understood as a (known) local version of (1.5.3) below, at the central point $s_0 = 0$. The local Whittaker functions depend on F/F_0.

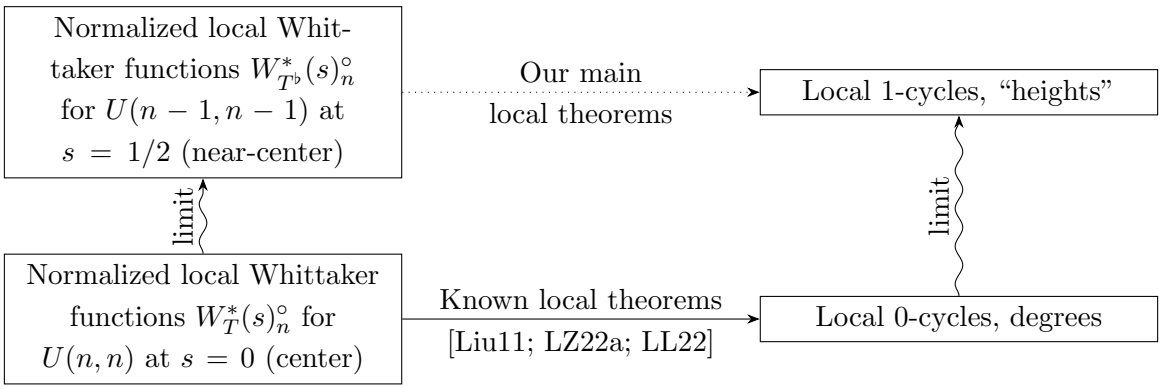


FIGURE 1. A local limiting method

This limiting method is the main conceptual novelty in our work, and is the key idea driving our main results. In Figure 1, the left vertical arrow and upper horizontal arrow are new in this work. In the right vertical arrow, the relation between limits and Faltings heights is also new in this work.

It is striking that the limiting method plays a similar role at all places, Archimedean and non-Archimedean. In the purely Archimedean case, i.e. when $F_0 = \mathbb{R}$ with $T^\flat$ nonsingular and not positive definite, we are able to run our limiting argument for special cycles (currents) in arbitrary dimension. This is why our result in the purely Archimedean case (global version stated as in [Che24c, Theorem B]) applies in arbitrary codimension.

It is interesting to ask whether the limiting method in Figure 1 can be adapted to the case of higher dimensional special cycles (corresponding to $T^\flat$ of smaller rank). Indeed, it is not even clear how to formulate a non-Archimedean "local arithmetic Siegel–Weil formula" in higher co-rank. The analytic side (normalized local Whittaker functions, one might guess) still appears to make sense, but the geometric side will involve flat components of special cycles in higher dimension, non-proper local special cycles, and global special cycles not lying entirely inside a single Rapoport–Zink space. All of these present obstacles for extracting local intersection numbers under current approaches, e.g. it is not clear how to piece together intersection-theoretic information from multiple Rapoport–Zink spaces, or whether this is even a correct approach at all. To generalize our limit strategy, one might need to locally decompose the mixed characteristic global heights for higher-dimensional cycles in a *canonical* way; for local 1-cycles, we accomplished this via "local change of heights", but it is not clear how to do this in general.

We also mention a slight difference if $F_0 = \mathbb{Q}_p$ and $F/\mathbb{Q}_p$ is split. The known local theorems [Liu11; LZ22a; LL22] apply in the Archimedean, inert, and ramified cases respectively. In the split case, the lower left corner of Figure 1 will involve the special value $W_T^*(0)_n^\circ$ (while the derivative at $s = 0$ appears in the Archimedean, inert, and ramified cases). In the split case, the "known local theorem" in Figure 1 refers to a certain vanishing statement for a certain contribution to $W_T^*(0)_n^\circ$ ("vertical part" via an analogue of Cho–Yamauchi's formula; the vanishing is proved in Lemma 7.5.1) and emptiness of local 0-cycles.

1.3. **Sketch.** For illustration purposes, we now sketch the above local limiting strategy at a finer level of detail. We first sketch the non-Archimedean case with $p \neq 2$ and $F/\mathbb{Q}_p$ inert. We then sketch an Archimedean case to highlight the similarity. In the main text, the non-Archimedean inert/ramified/split cases are treated in parallel (Section 7).

Non-Archimedean case, $F/\mathbb{Q}_p$ inert and $p \neq 2$. Suppose $T^\flat \in \mathrm{Herm}_{n-1}(\mathbb{Q}_p)$ with $\det T^\flat \neq 0$, and consider $t \in \mathbb{Q}_p$ such that $T \coloneqq \mathrm{diag}(t, T^\flat)$ has $\varepsilon(T) = -1$ (i.e. T defines a nonsplit Hermitian space; see Section 2.2). For the normalized local Whittaker functions defined in Section 6 below, we will prove the limit formula

$$\left.\frac{d}{ds}\right|_{s=-1/2} W_{T^\flat}^*(s)_n^\circ = \lim_{t\to 0}\left(\left.\frac{d}{ds}\right|_{s=0} W_T^*(s)_n^\circ + (\log|t|_p - \log p) W_{T^\flat}^*(-1/2)_n^\circ\right). \tag{1.3.1}$$

This appears as Proposition 7.5.2 below. Note the similarity with the Archimedean version (1.3.8) (see also Section 6.4 for comparisons with the ramified and split versions).

Recall the Rapoport–Zink space $\mathcal{N}$ and its special cycles as mentioned in Section 1.1 (see Section 3 for more details). If $\underline{\mathbf{x}}$ is a basis for $\mathbf{W}$, then the special cycle $\mathcal{Z}(\underline{\mathbf{x}})$ is a scheme with structure morphism $\mathcal{Z}(\underline{\mathbf{x}}) \to \mathrm{Spf}\, \breve{\mathbb{Z}}_p$ which is adic and proper [LZ22a, Lemma 2.10.1]. In this case, $\mathcal{Z}(\underline{\mathbf{x}})$ is thus a finite order thickening of its (proper) special fiber $\mathcal{Z}(\underline{\mathbf{x}})_{\overline{\mathbb{F}}_p}$, so we have a degree map $\deg_{\overline{\mathbb{F}}_p} : \mathrm{gr}_0 K_0'(\mathcal{Z}(\underline{\mathbf{x}}))_{\mathbb{Q}} \to \mathbb{Q}$ as in (3.6.15).

Let $\underline{\mathbf{x}}^\flat = [\mathbf{x}_1^\flat, \dots, \mathbf{x}_{n-1}^\flat] \in \mathbf{W}^{n-1}$ be a tuple with Gram matrix $T^\flat$ and consider nonzero $\mathbf{x} \in \mathrm{span}_F(\underline{\mathbf{x}}^\flat)^\perp \subseteq \mathbf{W}$ in the orthogonal complement. Set

$$\underline{\mathbf{x}} = [\mathbf{x}, \mathbf{x}_1^\flat, \dots, \mathbf{x}_{n-1}^\flat] \qquad t \coloneqq (\mathbf{x}, \mathbf{x}) \qquad T \coloneqq \mathrm{diag}(t, T^\flat). \tag{1.3.2}$$

Li–Zhang's inert Kudla–Rapoport theorem [LZ22a, Theorem 1.2.1] is

$$(\deg_{\overline{\mathbb{F}}_p} {}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}})) \cdot \log p = \frac{1}{2}\left.\frac{d}{ds}\right|_{s=0} W_T^*(s)_n^\circ. \tag{1.3.3}$$

As an element of $\mathrm{gr}_{\mathcal{N}}^n K_0'(\mathcal{Z}(\underline{\mathbf{x}}))_{\mathbb{Q}}$, the derived tensor product unfolds as

$${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}) = [{}^{\mathbb{L}}\mathcal{Z}(\mathbf{x}) \otimes^{\mathbb{L}}_{\mathcal{O}_{\mathcal{N}}} {}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)] = [{}^{\mathbb{L}}\mathcal{Z}(\mathbf{x}) \otimes^{\mathbb{L}}_{\mathcal{O}_{\mathcal{N}}} {}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{V}}] + [{}^{\mathbb{L}}\mathcal{Z}(\mathbf{x}) \otimes^{\mathbb{L}}_{\mathcal{O}_{\mathcal{N}}} {}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}]. \tag{1.3.4}$$

Here ${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}} = [\mathcal{O}_{\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}}] \in \mathrm{gr}_{\mathcal{N}}^{n-1} K_0'(\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}})_{\mathbb{Q}}$ is the "horizontal part" of ${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)$ and ${}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{V}} \in \mathrm{gr}_{\mathcal{N}}^{n-1} K_0'(\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\overline{\mathbb{F}}_p})_{\mathbb{Q}}$ is the "vertical part" (notation as sketched in Section 1.1).

We then prove the limit formulas

$$\lim_{\mathbf{x}\to 0}(\deg_{\overline{\mathbb{F}}_p}[{}^{\mathbb{L}}\mathcal{Z}(\mathbf{x})\otimes^{\mathbb{L}}_{\mathcal{O}_{\mathcal{N}}}{}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{V}}]) = \deg_{\overline{\mathbb{F}}_p}(\mathcal{E}^\vee\cdot{}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{V}}) \tag{1.3.5}$$

$$\lim_{\mathbf{x}\to 0}\left((\deg_{\overline{\mathbb{F}}_p}[{}^{\mathbb{L}}\mathcal{Z}(\mathbf{x})\otimes^{\mathbb{L}}_{\mathcal{O}_{\mathcal{N}}}{}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}])\cdot\log p+\frac{1}{2}(\log|t|_p-\log p)\cdot\deg(\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}})\right) \tag{1.3.6}$$

$$=\sum_{\mathcal{Z}\hookrightarrow\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}}\deg(\mathcal{Z})\cdot\delta_{\mathrm{tau}}(\mathcal{Z})\cdot\log p$$

with notation as in Section 1.1.

The "vertical" limit formula in (1.3.5) follows from a Grothendieck–Messing theory argument, see Lemma 7.5.4. Such vertical limiting behavior was observed in the inert case by [LZ22a] via computation, and later in the ramified case by [LL22] via a linear-invariance argument. We prove the "horizontal" limit formula in (1.3.6) componentwise, i.e. we prove a refined limiting formula for each component $\mathcal{Z}\hookrightarrow\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}$ (Remark 7.5.5). Each $\mathcal{Z}$ embeds into a smaller Rapoport–Zink space of dimension 2, where we make a computation in terms of quasi-canonical liftings.

We have the formula $W^*_{T^\flat}(-1/2)^\circ_n=\deg(\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}\to\operatorname{Spf}\breve{\mathbb{Z}}_p)$ ("local geometric Siegel–Weil") as in Lemma 7.1.3 (observed in the inert case by Li–Zhang [LZ22a, Corollary 4.6.1]). Using the functional equation $W^*_{T^\flat}(s)^\circ_n=W^*_{T^\flat}(-s)^\circ_n$, the limit formula in (1.3.1) now implies

$$-\frac{d}{ds}\Big|_{s=1/2}W^*_{T^\flat}(s)^\circ_n=\left(2\deg_{\overline{\mathbb{F}}_p}(\mathcal{E}^\vee\cdot{}^{\mathbb{L}}\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{V}})+2\sum_{\mathcal{Z}\hookrightarrow\mathcal{Z}(\underline{\mathbf{x}}^\flat)_{\mathscr{H}}}\deg(\mathcal{Z})\cdot\delta_{\mathrm{tau}}(\mathcal{Z})\right)\cdot\log p. \tag{1.3.7}$$

This is our main non-Archimedean local theorem for odd inert p (i.e. the dotted arrow in Figure 1), and appeared above as a special case of Theorem A. Limiting on the geometric side of Figure 1 was provided by the formulas in (1.3.5) and (1.3.6). Section 7 treats the inert/split/ramified cases simultaneously.

The geometry at split primes is less well-studied in the literature than the inert/ramified cases (we need uniformization in a non-supersingular situation at split primes). Section 4 contains some new results on decomposing local special cycles into quasi-canonical lifting cycles at split primes in the case of ordinary reduction, which we need for our main results. These are analogous to known results at inert and ramified primes (Section 5.3), though our method of proof is different. That the split places and non-supersingular reduction play a nontrivial role on the geometric side is a new feature of our "co-rank 1 situation", not visible in the prior work on (1.5.3) in the case of nonsingular T and the central derivative at $s_0=0$.

Archimedean case. Take any $n\geq 1$. For our Archimedean main local theorems, we run an argument similar in spirit to the non-Archimedean case, where the derived tensor products of complexes of coherent sheaves on Rapoport–Zink spaces is replaced by star products of Green currents on the Hermitian symmetric space. For purposes of exposition, here we consider $T^\flat\in\mathrm{Herm}_{n-1}(\mathbb{R})_{>0}$ and $t\in\mathbb{R}_{<0}$. Set $T\coloneqq\mathrm{diag}(t,T^\flat)$. Let V be any signature $(n-1,1)$ non-degenerate $F/\mathbb{Q}$ Hermitian space with pairing denoted $(-,-)$, and consider the complex Hermitian space $V_{\mathbb{R}}$.

We prove the limiting identity (left vertical arrow in Figure 1)

$$\left.\frac{d}{ds}\right|_{s=-1/2} W^*_{T^\flat}(s)^\circ_n = \lim_{t\to 0^-}\left(\left.\frac{d}{ds}\right|_{s=0} W^*_T(s)^\circ_n + (\log|t|_\infty + \log(4\pi e^\gamma))W^*_{T^\flat}(-1/2)^\circ_n\right) \tag{1.3.8}$$

where γ is the Euler–Mascheroni constant, and $|-|_\infty$ denotes the usual real norm. This formula appears (more generally) as [Che24a, Proposition 4.1.2]. The proof of this limiting formula is the bulk of the work at the Archimedean place. Note the similarity with the non-Archimedean version (1.3.1) (see also Section 6.4 for more comparisons).

On the geometric side of Figure 1, the local 0-cycles should be interpreted as Green currents of top degree $(n-1,n-1)$ on the associated Hermitian symmetric domain $\mathcal{D}$ parameterizing maximal negative definite $\mathbb{C}$-linear subspaces of $V_\mathbb{R}$. The local 1-cycles should be interpreted as Green currents of degree $(n-2,n-2)$. Any tuple $\underline{x}\in V_\mathbb{R}^m$ with nonsingular Gram matrix has an associated Kudla Green current $[\xi(\underline{x})]$, studied by Liu [Liu11] in the unitary case. There is a certain local special cycle $\mathcal{D}(\underline{x})\subseteq\mathcal{D}$ (complex submanifold which is the locus of $\mathbb{C}$-lines $z\in\mathcal{D}$ which are perpendicular to all elements of the tuple $\underline{x}$), arising in the complex uniformization of global special cycles.

Let $\underline{x}^\flat = [x_1^\flat,\dots,x_{n-1}^\flat]\in V_\mathbb{R}^{n-1}$ be a tuple with Gram matrix $T^\flat$ and consider nonzero $x\in \mathrm{span}_\mathbb{C}(\underline{x}^\flat)^\perp\subseteq V_\mathbb{R}$ in the orthogonal complement. Set

$$\underline{x} = [x, x_1^\flat,\dots,x_{n-1}^\flat] \qquad t\coloneqq (x,x) \qquad T\coloneqq \mathrm{diag}(t,T^\flat). \tag{1.3.9}$$

Liu's Archimedean local theorem [Liu11, Theorem 4.1.7] is

$$\int_\mathcal{D}[\xi(\underline{x})] = \left.\frac{d}{ds}\right|_{s=0} W^*_T(s)^\circ_n. \tag{1.3.10}$$

We are using the star product construction of $[\xi(\underline{x})]$, which unfolds as

$$[\xi(\underline{x})] = [\xi(x)]*[\xi(\underline{x}^\flat)] = \omega(x)\wedge[\xi(\underline{x}^\flat)] + [\xi(x)]\wedge\delta_{\mathcal{D}(\underline{x}^\flat)} \tag{1.3.11}$$

where $\omega(x)$ is a $(1,1)$-form associated with x (Kudla–Millson form up to a normalization), $\delta_{\mathcal{D}(\underline{x}^\flat)}$ is a Dirac delta current, and $\xi(x)$ is a certain function on $\mathcal{D}$ with logarithmic singularity along $\mathcal{D}(x)$. The function $\xi(x)$ is expressed in terms of the exponential integral Ei. We have $\int_\mathcal{D}[\xi(x)]\wedge\delta_{\mathcal{D}(\underline{x}^\flat)} = -\mathrm{Ei}(4\pi t)$ and the limit formulas

$$\lim_{x\to 0}\omega(x) = c_1(\widehat{\mathcal{E}}^\vee) \qquad \lim_{u\to 0^-}(\mathrm{Ei}(u)-\log|u|) = \gamma \tag{1.3.12}$$

where $c_1(\widehat{\mathcal{E}}^\vee)$ denotes the Chern form of dual tautological bundle on $\mathcal{D}$ (as in [Che24a, Section 2]). Under the assumption that $T^\flat$ is positive definite, we have $W^*_{T^\flat}(-1/2)^\circ_n = \deg\mathcal{D}(\underline{x}^\flat) = 1$ ("local geometric Siegel–Weil", i.e. $\mathcal{D}(\underline{x}^\flat)$ is a single point). We thus have

$$\int_\mathcal{D} c_1(\widehat{\mathcal{E}}^\vee)\wedge[\xi(\underline{x}^\flat)] = \lim_{x\to 0}\left(\left(\int_\mathcal{D}[\xi(\underline{x})]\right)+\log|t|+\log(4\pi e^\gamma)\right). \tag{1.3.13}$$

Using the functional equation $W^*_{T^\flat}(s)^\circ_n = W^*_{T^\flat}(-s)^\circ_n$, the limit formula in (1.3.8) now implies the identity

$$-\left.\frac{d}{ds}\right|_{s=1/2} W^*_{T^\flat}(s)^\circ_n = \int_\mathcal{D} c_1(\widehat{\mathcal{E}}^\vee)\wedge[\xi(\underline{x}^\flat)] \tag{1.3.14}$$

which is our Archimedean local version of Theorem A for positive definite $T^\flat$ (i.e. the dotted arrow in Figure 1). Its proof appears in [Che24a, Theorem 4.1.1], which also includes a version for non positive definite $T^\flat$. Limiting on the geometric side of Figure 1 was provided by a limiting property of the (normalized) Kudla–Millson form, i.e. $\omega(x) \to c_1(\widehat{\mathcal{E}}^\vee)$ as $x \to 0$. To compare with the limit of Whittaker function derivatives, we used the special value formula $W^*_{T^\flat}(-1/2)^\circ_n = 1$ ("local geometric Siegel–Weil") and the asymptotics of Ei.

1.4. **Eisenstein series.** For context, we briefly introduce the main global results which are completed in [Che24a; Che24b; Che24c]. We take the unitary/Hermitian case (literature review in Section 1.6 also discusses the symplectic/quadratic case). We now switch to global notation, and fix an imaginary quadratic field $F/\mathbb{Q}$ and ring of integers $\mathcal{O}_F$. For the introduction, we assume that the discriminant Δ is odd. Given $m \in \mathbb{Z}_{\geq 0}$ and an even integer $n \in \mathbb{Z}$, we consider the (normalized) *Siegel Eisenstein series*

$$E^*(z,s)^\circ_n \coloneqq \Lambda_m(s)^\circ_n \sum_{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in P_1(\mathbb{Z}) \backslash SU(m,m)(\mathbb{Z})} \frac{\det(y)^{s-s_0}}{\det(cz+d)^n |\det(cz+d)|^{2(s-s_0)}} \tag{1.4.1}$$

for the group

$$H \coloneqq U(m,m) \coloneqq \left\{ h \in \mathrm{Res}_{\mathcal{O}_F/\mathbb{Z}} \mathrm{GL}_{2m} : {}^t\overline{h} \begin{pmatrix} 0 & 1_m \\ -1_m & 0 \end{pmatrix} h = \begin{pmatrix} 0 & 1_m \\ -1_m & 0 \end{pmatrix} \right\} \tag{1.4.2}$$

where $\Lambda_m(s)^\circ_n$ is the normalizing factor

$$\begin{aligned} \Lambda_m(s)^\circ_n \coloneqq \frac{(2\pi)^{m(m-1)/2}}{(-2\pi i)^{nm}} \pi^{m(-s+s_0)} |\Delta|^{m(m-1)/4 + \lfloor m/2 \rfloor (s+s_0)} \\ \cdot \left(\prod_{j=0}^{m-1} \Gamma(s - s_0 + n - j) \cdot L(2s + m - j, \eta^{j+n}) \right). \end{aligned} \tag{1.4.3}$$

In (1.4.2), the notation 1_m stands for the $m \times m$ identity matrix, we wrote $SU(m,m) \subseteq U(m,m)$ for the determinant 1 subgroup, and we set $P_1 \coloneqq P \cap SU(m,m)$ for the Siegel parabolic $P \subseteq U(m,m)$ (consisting of $m \times m$ block upper triangular matrices). The variable $s \in \mathbb{C}$ is a complex parameter, we set $s_0 = (n-m)/2$, and the element $z = x + iy$ lies in Hermitian upper half-space (i.e. $x \in \mathrm{Herm}_m(\mathbb{R})$ and $y \in \mathrm{Herm}_m(\mathbb{R})_{>0}$; the latter means that y is positive definite). The symbol η denotes the quadratic character associated to $F/\mathbb{Q}$ (via class field theory). The sum in (1.4.1) is convergent for $\mathrm{Re}(s) > m/2$ and admits meromorphic continuation to all $s \in \mathbb{C}$. When $m = 1$, the expression in (1.4.1) is a classical Eisenstein series on the usual upper half-plane.

The normalized Eisenstein series has a symmetric functional equation

$$E^*(z,s)^\circ_n = (-1)^{m(m-1)(n-m-1)/2} E^*(z,-s)^\circ_n \tag{1.4.4}$$

as in [Che24c, Section 6.1]. Our definition of the normalizing factor $\Lambda_m(s)^\circ_n$ is motivated by symmetry of global and local functional equations, along with certain local special value formulas; see [Che24c, Part 1] for further discussion. The function $\Lambda_m(s)^\circ_n$ should be closely related with the L-function of an Artin–Tate motive attached to the group $U(m,m)$, in the sense of Gross [Gro97] (see [BH21, Remark 1.1.1]).

Given $T \in \mathrm{Herm}_m(\mathbb{Q})$, the Eisenstein series $E^*(z,s)^\circ_n$ has *T-th Fourier coefficient*

$$E^*_T(y,s)^\circ_n \coloneqq 2^{m(m-1)/2}|\Delta|^{-m(m-1)/4} \int_{\mathrm{Herm}_m(\mathbb{Z})\backslash \mathrm{Herm}_m(\mathbb{R})} E^*(z,s)^\circ_n e^{-2\pi i \mathrm{tr}(Tz)}\, dx \tag{1.4.5}$$

for $z = x + iy$ in Hermitian upper half-space, where this integral is taken with respect to the Euclidean measure[3] on $\mathrm{Herm}_m(\mathbb{R})$. The integral is convergent for $\mathrm{Re}(s) > m/2$, and admits meromorphic continuation to all $s \in \mathbb{C}$. When $\det T \neq 0$, there is a factorization into normalized *local Whittaker functions*

$$E^*_T(y,s)^\circ_n = W^*_{T,\infty}(y,s)^\circ_n \prod_p W^*_{T,p}(s)^\circ_n \tag{1.4.6}$$

over all places, see [Che24c, Section 6.1]. We suppress the definitions of local Whittaker functions in the introduction; see Section 6 for the non-Archimedean cases (and [Che24c, Section 4] more generally).

1.5. **Arithmetic Siegel–Weil.** *Arithmetic Siegel–Weil formulas* predict that the derivative of $E^*_T(y,s)^\circ_n$ at $s = s_0$ should encode *arithmetic degrees/volumes* of special cycles on integral models of Shimura varieties.

Kudla–Rapoport [KR14] introduced the study of special cycles $\mathcal{Z}(T) \to \mathcal{M}$ over (stacky) integral models $\mathcal{M} \to \mathrm{Spec}\, \mathcal{O}_F$ for Shimura varieties associated to $G' \coloneqq \mathrm{Res}_{F/\mathbb{Q}} \mathbb{G}_m \times U(V)$, for signature $(n-1,1)$ non-degenerate $F/\mathbb{Q}$ Hermitian spaces V with pairing $(-,-)$. For the purpose of this introduction, we assume V contains a full-rank self-dual[4] $\mathcal{O}_F$-lattice, and we take $\mathcal{M} \to \mathrm{Spec}\, \mathcal{O}_F$ to be the "exotic smooth" Rapoport–Smithling–Zhang (RSZ) integral model of relative dimension $n-1$ [RSZ21, §6].[5] When $n = 2$, the stack $\mathcal{M}$ is essentially a disjoint union of (stacky) modular curves (see [Che24b, Example 2.2.2]).

The stack $\mathcal{M}$ admits a moduli description: it parameterizes tuples $(A_0, \iota_0, \lambda_0, A, \iota, \lambda)$ where A_0 and A are abelian schemes (dimensions 1 and n respectively) with $\mathcal{O}_F$-actions ι_0 and ι, and with compatible quasi-polarizations λ_0 and λ. The datum $(A_0, \iota_0, \lambda_0, A, \iota, \lambda)$ satisfies a few additional conditions; we refer to [Che24b, Part 1] for further details.

The moduli stack $\mathcal{M}$ carries a natural family of Hermitian $\mathcal{O}_F$-lattices

$$\mathcal{L}at \to \mathcal{M} \qquad \mathcal{L}at \coloneqq \underline{\mathrm{Hom}}_{\mathcal{O}_F}(A_0, A), \tag{1.5.1}$$

e.g. as explained in [Che24b, Section 2.3 and Appendix B.1]. Given any $T \in \mathrm{Herm}_m(\mathbb{Q})$, the associated *Kudla–Rapoport* special cycle $\mathcal{Z}(T) \to \mathcal{M}$ is defined as the substack

$$\mathcal{Z}(T) \coloneqq \{\underline{x} \in \mathcal{L}at^m : (\underline{x}, \underline{x}) = T\} \subseteq \mathcal{L}at^m \tag{1.5.2}$$

consisting of m-tuples $\underline{x}$ with Gram matrix T; see [Che24b, Section 2.3] for more details. This is in close analogy with classical Siegel–Weil: there one considers $\mathcal{O}_F$-lattices varying in a given *genus*

[3]The factor $2^{m(m-1)/2}|\Delta|^{-m(m-1)/4}$ disappears in the (usual) equivalent adèlic formulation, upon taking a certain self-dual Haar measure. The adèlic formulations of (1.4.1) and (1.4.5) are used in [Che24c, Section 2].

[4]We always mean *self-dual* for the bilinear *trace pairing* $\mathrm{tr}_{F/\mathbb{Q}}(v,w)$ unless otherwise specified; see conventions in Section 2.1.

[5]Our assumption on V forces $n \equiv 2 \pmod 4$. We allow a slightly more general setup in [Che24b; Che24c] for general V, at the cost of throwing out finitely many primes (particularly the ramified primes when n is odd). This does not affect the essential ideas of our method, which is local in nature.

and counts tuples with a fixed Gram matrix; here we are considering $\mathcal{O}_F$-lattices varying over the moduli stack $\mathcal{M}$. The morphism $\mathcal{Z}(T) \to \operatorname{Spec} \mathcal{O}_F$ is smooth of relative dimension $n-1-\operatorname{rank}(T)$ in the generic fiber over $\operatorname{Spec} F$. If T is not positive semi-definite, then $\mathcal{Z}(T)$ is empty.

An *arithmetic Siegel–Weil formula* is an identity roughly of the shape

$$\frac{h_F}{w_F}\frac{d}{ds}\Big|_{s=s_0} \frac{2\Lambda_n(s-s_0)_n^\circ}{\kappa\Lambda_m(s)_n^\circ} E_T^*(y,s)_n^\circ \stackrel{?}{=} \widehat{\operatorname{vol}}_{\widehat{\mathcal{E}}^\vee}([\widehat{\mathcal{Z}}(T)]). \tag{1.5.3}$$

Here we set $\kappa = 1$ if $m \neq n$, and set $\kappa = 2$ if $m = n$. The right-hand side of (1.5.3) denotes an *arithmetic volume*, which is a real number "defined" by an arithmetic intersection product

$$\text{"}\widehat{\operatorname{vol}}_{\widehat{\mathcal{E}}^\vee}([\widehat{\mathcal{Z}}(T)])\text{"} := \text{"}\widehat{\deg}([\widehat{\mathcal{Z}}(T)] \cdot \widehat{c}_1(\widehat{\mathcal{E}}^\vee)^{n-m}) \tag{1.5.4}$$

in an arithmetic Chow ring $\widehat{\operatorname{Ch}}^*(\mathcal{M})_{\mathbb{Q}}$ (roughly in the sense of Gillet–Soulé [GS87]) for a certain metrized tautological bundle $\widehat{\mathcal{E}}^\vee$ on $\mathcal{M}$ (the bundle $\widehat{\mathcal{E}}^\vee$ is discussed in [Che24b, Sections 2.2 and 3.3]). The notation $[\widehat{\mathcal{Z}}(T)]$ indicates a class in $\widehat{\operatorname{Ch}}^m(\mathcal{M})_{\mathbb{Q}}$, which is expected to involve $\mathcal{Z}(T)$ and some additional Archimedean data (e.g. from a Green current on the complex Shimura variety), as appearing in arithmetic intersection theory.

We sketched the arithmetic Siegel–Weil formula as a rough expectation, because precise formulations remain open in the general case [Li24, Remark 4.4.2] on both the analytic and geometric sides. These issues are studied further in [Che24b; Che24c].

Our main global result is a proof of (1.5.3) when $m = n$ and $T \in \operatorname{Herm}_n(\mathbb{Q})$ is singular of co-rank 1, as well as the closely related special case $m = n-1$ and T is nonsingular. This is completed in [Che24c, Theorem A]. The main result of this paper (Theorem A) is the non-Archimedean part of this theorem.

1.6. **Previous work.** We summarize in more detail what was previously known on arithmetic Siegel–Weil formulas. These were originally formulated for GSpin Shimura varieties, as opposed to the unitary Shimura varieties considered in Section 1.5; we call these the *orthogonal* and *unitary* cases respectively. In both cases, we write n for the arithmetic dimension of the Shimura varieties (i.e. complex dimension $n-1$).

The problem was initially studied in low-dimensional situations. For quaternionic Shimura curves, the full arithmetic Siegel–Weil formula has been proved in the influential work of Kudla–Rapoport–Yang [KRY04; KRY06]. For modular curves, the formula has been proved in the papers [Yan04; BF06; DY19; SSY23; Zhu23a; Zhu23b].

For Shimura varieties of complex dimension > 1, results on arithmetic Siegel–Weil formulas are currently incomplete. Most of the available results concern the case $m = n$ and $\det T \neq 0$; we restrict to this case for the moment. Then $s_0 = 0$ is the central point and the special cycle $\mathcal{Z}(T) \to \mathcal{M}$ is empty in the generic fiber. The arithmetic cycle class $[\widehat{\mathcal{Z}}(T)]$ is thus "purely vertical", i.e. either purely in positive characteristic (non-Archimedean), or with $\mathcal{Z}(T)$ being empty with possibly nontrivial Green current (Archimedean).

The purely Archimedean case (with $\det T \neq 0$ and $s_0 = 0$) was proved by [Liu11; BY21] (unitary and orthogonal, respectively) using different methods. Garcia–Sankaran's Archimedean results apply here as well if the Shimura varieties are compact (more discussion below).

For unitary groups (with $\det T \neq 0$ and $s_0 = 0$), the purely non-Archimedean case for hyperspecial level was first proposed and studied by Kudla–Rapoport [KR11; KR14] at an odd inert prime, where they proved the formula when $\mathcal{Z}(T)$ has dimension 0 (reducing locally to the case $n = 2$). The case $n = 3$ at an odd inert prime was solved by Terstiege [Ter13]. The case of arbitrary n at odd inert primes was solved in the breakthrough work of Li and Zhang [LZ22a] by an inductive "uncertainty principle" strategy. This strategy was later adapted to solve the analogous problem at odd ramified primes [LL22; HLSY23]. We mention that the problem formulation itself needed to be resolved at ramified primes in the presence of bad reduction, and this was done in [HSY23] for the Krämer model. Split primes play a relatively trivial role when $\det T \neq 0$ and $s_0 = 0$. The timeline for non-Archimedean aspects of the GSpin arithmetic Siegel–Weil formula is similar, i.e. results for $\mathcal{Z}(T)$ of dimension 0 were obtained by Kudla–Rapoport and Bruinier–Yang [KR99; KR00; BY21], the case $n = 3$ was resolved by Terstiege [Ter11], and the case of general n at hyperspecial level was resolved by Li and Zhang using (a modified version of) their "uncertainty principle" strategy [LZ22b].

We now drop the restrictions $\det T \neq 0$ and $s_0 = 0$. For the purpose of arithmetic theta lifting, it is desirable to also understand the special cycle classes $[\widehat{\mathcal{Z}}(T)]$ when $\det T = 0$, to fill out the complete arithmetic theta series. Much less is known about this case, which presents new difficulties on both the analytic and geometric sides. It also presents new opportunities: our arithmetic Siegel–Weil result for singular T relates Faltings heights and derivatives of Eisenstein series. Such formulas were observed by Kudla–Rapoport–Yang on Shimura curves [KRY04]; our result applies on unitary Shimura varieties of arbitrarily high dimension. These mixed characteristic phenomena are not visible from arithmetic Siegel–Weil for nonsingular T at the central point $s_0 = 0$ (which was "purely vertical").

We mention known partial results for singular T, besides the previously mentioned work on Shimura curves. There is concrete progress on the case $T = 0$, where the expected geometric side ("arithmetic volumes") has been computed for certain levels in the work of Hörmann and Bruinier–Howard [Hör14; BH21], with some partial results on the comparison with Eisenstein series. In the general case, an important advance was made by Garcia and Sankaran [GS19], who defined Green currents via superconnections and proved a purely Archimedean version of the arithmetic Siegel–Weil formula on compact Shimura varieties (e.g. when T is not positive semi-definite, giving an empty special cycle with possibly nontrivial Green current) via the classical Siegel–Weil formula.

Besides the mentioned results for $T = 0$, we are not aware of any previous arithmetic Siegel–Weil results which treat non-Archimedean (or combined Archimedean and non-Archimedean) aspects for singular T on Shimura varieties of complex dimension > 1. This is closely related to the following open problem: for Shimura varieties of complex dimension > 1, we are also unaware of any fully global arithmetic Siegel–Weil results (incorporating non-Archimedean places) at a non-central point $s_0 \neq 0$, besides the results in [BH21, Theorem C] (there for certain nonzero 1×1 matrices $T \in \mathbb{Z}$). As discussed at the end of Section 1.5, our main theorems make new contributions in both of these directions.

1.7. **Outline.** We briefly summarize the remaining content in this paper, and discuss the relation with [Che24a; Che24b; Che24c]. Further explanations may be found at the beginning of some sections.

We focus on the non-Archimedean local situation in this paper. In Part 1 "Local special cycles" below, we set up Kudla–Rapoport local special cycles on Rapoport–Zink spaces (inert/ramified/split). In Part 2 "Local main theorems", we formulate and prove our main local non-Archimedean theorems. Section 7 in particular contains the heart of our work. Here, we prove our "local arithmetic Siegel–Weil" theorems at inert/ramified/split non-Archimedean places via the local limiting method sketched in Section 1.2. The Archimedean analogue will appear in [Che24a].

In [Che24b], we execute the reduction process from the global intersection numbers (1.5.4) to the geometric quantities appearing in our main local theorems from the present paper (non-Archimedean) and [Che24a] (Archimedean), via Rapoport–Zink (non-Archimedean) uniformization and complex (Archimedean) uniformization. The non-Archimedean part of this process was non-obvious for us, requiring argument on lifting "local change of heights" for minimal degree isogenies to "global change of heights". In [Che24c], we make the corresponding local-to-global reduction for Eisenstein series and show how to patch together our local arithmetic Siegel–Weil theorems at all places to finish the proof of our main global arithmetic Siegel–Weil theorems; see [Che24c, Section 9]. This final proof uses all the main results from the preceding papers.

Our algebro-geometric conventions follow the Stacks project [SProject] unless stated otherwise.

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2. Conventions on Hermitian spaces and lattices

We will use the conventions from Section 2 throughout all of [Che24a; Che24b; Che24c].

2.1. **Hermitian, alternating, symmetric.** Consider a Dedekind domain $\mathcal{O}_{F_0}$ with fraction field F_0. Let F be a finite étale F_0-algebra of degree 2, i.e. either F is a degree 2 separable field extension

of F_0, or $F \cong F_0 \times F_0$. Let $\mathcal{O}_F \subseteq F$ be the integral closure of $\mathcal{O}_{F_0}$ in F. Write $a \mapsto a^\sigma$ for the nontrivial involution of F over F_0, and $\mathrm{tr}\colon F \to F_0$ for the trace map $a \mapsto a + a^\sigma$.

Assume that the different ideal $\mathfrak{d}$ may be generated by one element u such that $u^\sigma = -u$, and fix such a u. This is always possible if $\mathcal{O}_F$ is a free $\mathcal{O}_{F_0}$-module: in this case, if $\{\alpha, \beta\}$ is any $\mathcal{O}_{F_0}$-basis for $\mathcal{O}_F$, we may take $u = \alpha\beta^\sigma - \alpha^\sigma\beta$.

Let L be a finite locally free $\mathcal{O}_F$-module of constant rank. If F_0 has characteristic $\neq 2$, the following data are equivalent.

(1) A *Hermitian pairing* on L, i.e. an $\mathcal{O}_{F_0}$-bilinear map $(-,-)\colon L \times L \to F$ satisfying

$$(x, ay) = a(x, y) \qquad (y, x) = (x, y)^\sigma \tag{2.1.1}$$

for all $a \in \mathcal{O}_F$ and $x, y \in L$.

(2) An *$\mathcal{O}_F$-compatible alternating pairing* on L, i.e. an $\mathcal{O}_{F_0}$-bilinear map $\langle -,-\rangle\colon L \times L \to F_0$ satisfying

$$\langle ax, y\rangle = \langle x, a^\sigma y\rangle \qquad \langle y, x\rangle = -\langle x, y\rangle \tag{2.1.2}$$

for all $a \in \mathcal{O}_F$ and $x, y \in L$.

(3) An *$\mathcal{O}_F$-compatible symmetric pairing* on L, i.e. an $\mathcal{O}_{F_0}$-bilinear map $(\!(-,-)\!)\colon L \times L \to F_0$ satisfying

$$(\!(ax, y)\!) = (\!(x, a^\sigma y)\!) \qquad (\!(y, x)\!) = (\!(x, y)\!) \tag{2.1.3}$$

for all $a \in \mathcal{O}_F$ and $x, y \in L$.

If L is equipped with any of the equivalent data above, we say that L is a *Hermitian $\mathcal{O}_F$-lattice* (or *Hermitian $\mathcal{O}_F$-module*). Note that our Hermitian pairings $(-,-)$ are conjugate linear in the first argument. We pass between these pairings using the formulas (depending on the choice of u)

$$\begin{aligned}
2(x,y) &= (\!(x,y)\!) - u^{-1}(ux,y) & \langle x,y\rangle &= (\!(u^{-1}x,y)\!) & (\!(x,y)\!) &= \mathrm{tr}((x,y)) \\
2(x,y) &= \langle ux,y\rangle - u\langle x,y\rangle & \langle x,y\rangle &= -\mathrm{tr}((x,y)u^{-1}) & (\!(x,y)\!) &= \langle ux,y\rangle
\end{aligned}$$

and this will be freely used in the paper. The choice of u plays a limited role for us, so we generally suppress it.

We say that $(\!(-,-)\!)$ is the associated *trace pairing*, and otherwise avoid the notation $(\!(-,-)\!)$ outside of Section 2.1.

Given any tuple $\underline{x} = [x_1, \ldots, x_m] \in L^m$, its *Gram matrix* is the matrix $(\underline{x}, \underline{x}) = T$ with i,j-th entry $T_{i,j} = (x_i, x_j)$. We write $L_F \coloneqq L \otimes_{\mathcal{O}_F} F$ and say that a Hermitian $\mathcal{O}_F$-module L is *non-degenerate* if the Gram matrix for any F-basis of L_F has nonzero determinant. Given non-degenerate Hermitian F-modules V and V' with Hermitian pairings $(-,-)$ and $(-,-)'$, there is a canonical σ-linear involution of F-modules

$$\mathrm{Hom}_F(V,V') \xrightarrow{f \mapsto f^\dagger} \mathrm{Hom}_F(V',V) \qquad \begin{array}{l}\text{such that } (fx,y')' = (x, f^\dagger y') \\ \text{for all } x \in V \text{ and } y' \in V'.\end{array} \tag{2.1.4}$$

The notation $\mathrm{Hom}_F(V,V')$ and $\mathrm{Hom}_F(V',V)$ does not include any requirement on preserving Hermitian pairings.

Given a non-degenerate Hermitian $\mathcal{O}_F$-lattice L, we always form its *dual lattice* L^* with respect to the trace pairing $(-,-)$, i.e.

$$L^* \coloneqq \{x \in L_F : \mathrm{tr}(x,y) \in \mathcal{O}_{F_0} \text{ for all } y \in L\}. \tag{2.1.5}$$

The dual lattice $L^\vee$ with respect to $(-,-)$ is the same as the dual lattice for $\langle -,-\rangle$. We have $L^\vee = uL^*$ (as sublattices of L_F). If the dual $L^\vee$ with respect to $(-,-)$ or $\langle -,-\rangle$ is intended, we will state this explicitly. We say that L is *self-dual* if $L = L^*$.

As a typical example of passing between $(-,-)$ and $\langle -,-\rangle$, suppose $\mathcal{O}_{F_0} = \mathbb{Z}$ and suppose $\mathcal{O}_F$ is the ring of integers in an imaginary quadratic field $F/\mathbb{Q}$. Let (A,ι,λ) be a *Hermitian abelian variety* (see [Che24b, Definition 2.1.1]) over an algebraically closed field k of characteristic $\neq p$, i.e. A is an abelian variety over k with an action $\iota\colon \mathcal{O}_F \to \mathrm{End}(A)$, and λ is an $\mathcal{O}_F$-compatible quasi-polarization on A. After picking a trivialization $\mathbb{Z}_p(1) \cong \mathbb{Z}_p$ of p-th power roots of unity over k, the polarization λ induces an $(\mathcal{O}_F \otimes_{\mathbb{Z}} \mathbb{Z}_p)$-compatible alternating pairing on the Tate module $T_p(A)$, so we automatically view $T_p(A)$ as a Hermitian $(\mathcal{O}_F \otimes_{\mathbb{Z}} \mathbb{Z}_p)$-lattice without further mention. If (A',ι',λ') is another such Hermitian abelian scheme, note that the induced Hermitian pairing on $\mathrm{Hom}_{\mathcal{O}_F}(T_p(A), T_p(A'))$ does not depend on the choice of trivialization $\mathbb{Z}_p(1) \cong \mathbb{Z}_p$ or the choice of u.

The notation $\mathrm{Herm}_n(\mathcal{O}_{F_0})$ means the set of $n \times n$ Hermitian matrices with coefficients in $\mathcal{O}_F$ (i.e. $T \in M_{n,n}(\mathcal{O}_F)$ satisfying $T = {}^t\overline{T}$ where ${}^t\overline{T}$ means conjugate transpose). Here we are considering the subfunctor $\mathrm{Herm}_n \subseteq \mathrm{Res}_{\mathcal{O}_F/\mathcal{O}_{F_0}} M_{n,n}$ of the Weil restriction (of $n \times n$ matrices $M_{n,n}$). We adhere strictly to this notation (when $\mathcal{O}_F$ is understood), e.g. $\mathrm{Herm}_n(\mathbb{R})$ will typically mean $n \times n$ complex Hermitian matrices when $\mathcal{O}_F/\mathcal{O}_{F_0} = \mathbb{C}/\mathbb{R}$ is understood.

2.2. **Lattices for local fields.** Continuing in the setup of Section 2.1, suppose F_0 is a local field. Let $\eta\colon F_0^\times \to \{\pm 1\}$ be the character associated to F/F_0 by local class field theory. Given a non-degenerate Hermitian F-module V of rank n, define its *local invariant*

$$\varepsilon(V) \coloneqq \eta((-1)^{n(n-1)/2} \det T) \in \{\pm 1\} \tag{2.2.1}$$

where T is the Gram matrix of any basis for V. This is normalized so that $\varepsilon(V) = 1$ for the Hermitian F-module V given by the antidiagonal unit Gram matrix. Rank n non-degenerate Hermitian F-modules V and V' are isomorphic if and only if $\varepsilon(V) = \varepsilon(V')$, except in the case of $F/F_0 = \mathbb{C}/\mathbb{R}$, in which case V is classified up to isomorphism by its signature. Given any (global) CM field, a collection of n-dimensional local Hermitian spaces V_v (over all places v of the index 2 totally real subfield) comes from a global Hermitian space if and only if $\varepsilon(V_v) = +1$ for almost all v and the global product formula $\prod_v \varepsilon(V_v) = +1$ is also satisfied, by Landherr's theorem [Lan35] (also [HK89, §4.2]). If $T \in \mathrm{Herm}_n(F_0)$ is a Hermitian matrix (with entries in F) satisfying $\det T \neq 0$, we set $\varepsilon(T) \coloneqq \eta((-1)^{n(n-1)/2} \det T)$.

Next, assume F_0 is non-Archimedean and that $\mathcal{O}_{F_0} \subseteq F_0$ is its ring of integers. Write q for the residue cardinality of $\mathcal{O}_{F_0}$. If q is even, we require F/F_0 to be unramified. Let $\varpi_0 \in \mathcal{O}_{F_0}$ and $\varpi \in \mathcal{O}_F$ be uniformizers (meaning $\varpi \in \varpi_0 \mathcal{O}_F^\times$ in the unramified cases) satisfying $\varpi^\sigma = -\varpi$. If a non-degenerate Hermitian F-module V contains a full rank self-dual $\mathcal{O}_F$-lattice, then $\varepsilon(V) = 1$.

The "norm" $\|-\|$ on a Hermitian F-module V with pairing $(-,-)$ is given by

$$\|x\| \coloneqq q^{-v_{\varpi_0}((x,x))/2} \tag{2.2.2}$$

where v_{ϖ_0} is the ϖ_0-adic valuation, normalized so that $v_{\varpi_0}(\varpi_0) = 1$.

Given a non-degenerate Hermitian $\mathcal{O}_F$-lattice L of rank n, we set $\varepsilon(L) \coloneqq \varepsilon(L_F)$. By a *lattice* or *sublattice* $L' \subseteq L$, we mean any $\mathcal{O}_F$-submodule which is finite free of constant rank (similarly for lattices or sublattices in L_F). If L' has rank n, we say that L' is *full rank* in L_F. A sublattice $L' \subseteq L$ is *saturated* if $ax \in L'$ with $a \in F^\times$ and $x \in L$ implies $x \in L'$ (equivalently, L' is a direct summand of L).

We say that L is *integral* if $L \subseteq L^*$. If F/F_0 is nonsplit, we say that L is *almost self-dual* if $L \subseteq L^*$ and $\mathrm{length}_{\mathcal{O}_F}(L^*/L) = 1$. We say that a non-degenerate integral lattice L is *maximal integral* if any integral lattice $L' \subseteq L_F$ with $L \subseteq L'$ satisfies $L = L'$.

If L is a non-degenerate Hermitian $\mathcal{O}_F$-lattice, we define the *valuation* $\mathrm{val}(L) \in \frac{1}{2}\mathbb{Z}$ such that

$$q^{-\mathrm{val}(L)} = \mathrm{vol}(L) \tag{2.2.3}$$

where $\mathrm{vol}(L)$ is the volume of L for the self-dual Haar measure on L_F with respect to the pairing $x, y \mapsto \psi(\mathrm{tr}(x,y))$ for any unramified (unitary) additive character $\psi\colon F_0 \to \mathbb{C}^\times$. If L is integral, we have $q^{2\mathrm{val}(L)} = |L^*/L|$. If F/F_0 is unramified, we have $\mathrm{val}(L) \in \mathbb{Z}$. Given $x \in L$, we write $\langle x \rangle \subseteq L$ for the rank one $\mathcal{O}_F$-submodule generated by x. If $(x,x) \neq 0$, we set $\mathrm{val}(x) \coloneqq \mathrm{val}(\langle x \rangle)$ (and otherwise set $\mathrm{val}(x) = \infty$).

Continuing to assume L is non-degenerate and integral, we define its sequence of *fundamental invariants* to be the unique sequence of integers $(a_1, \dots, a_n)$ with $0 \leq a_1 \leq \cdots \leq a_n$ such that $L^*/L \cong \bigoplus_{i=1}^n \mathcal{O}_F/\varpi^{a_i}$ (where n is the rank of L). If F/F_0 is unramified, two non-degenerate integral Hermitian $\mathcal{O}_F$-lattices of the same rank are isomorphic if and only if they have the same sequence of fundamental invariants (follows upon diagonalizing the Hermitian form). In all cases, we set

$$t(L) \coloneqq |\{i : a_i \neq 0\}| \in \mathbb{Z} \qquad a_{\max}(L) \coloneqq a_n \tag{2.2.4}$$

and refer to $t(L)$ as the *type* of L. If F/F_0 is ramified, recall that $t(L)$, $2\mathrm{val}(L)$, and n all have the same parity (follows from [Jac62, Proposition 4.3, Proposition 8.1]).

Given a finite length $\mathcal{O}_F$-module M, we define $\ell(M) \in \mathbb{Z}$ such that

$$q^{\ell(M)} = |M| \tag{2.2.5}$$

where $|M|$ denotes the cardinality of M.

The above terminology is adapted from e.g. [LZ22a] (inert), [FYZ21] (inert and split), [LL22] (ramified). We made slight modifications to give a uniform description (e.g. our $\mathrm{val}(L)$ is half of the $\mathrm{val}(L)$ appearing in [LL22], and our $\ell(M)$ differs by a factor of 2 from some of the references).

Part 1. Local special cycles

3. Moduli spaces of p-divisible groups

We review some unitary Rapoport–Zink spaces [RZ96] and their special cycles. These appear in p-adic uniformization of the moduli stacks appearing in our eventual global results (see [Che24b, Part 3]). Some (freely used) notation on p-divisible groups appears in [Che24b, Appendix C.1].

Fix a prime p and let $F/\mathbb{Q}_p$ be a degree 2 étale algebra, i.e. $F/\mathbb{Q}_p$ is an inert quadratic extension, a ramified quadratic extension, or $F = \mathbb{Q}_p \times \mathbb{Q}_p$. If $F/\mathbb{Q}_p$ is ramified, we assume $p \neq 2$. Write $a \mapsto a^\sigma$ for the nontrivial automorphism σ of F over $\mathbb{Q}_p$. We write $\mathcal{O}_F$ for the integral closure of $\mathbb{Z}_p$ in F (hence $\mathcal{O}_F = \mathbb{Z}_p \times \mathbb{Z}_p$ in the split case). If $F/\mathbb{Q}_p$ is ramified, we write ϖ for a uniformizer of $\mathcal{O}_F$ satisfying $\varpi^\sigma = -\varpi$.

We use the usual notation $\breve{\mathbb{Q}}_p$ for the completion of the maximal unramified extension of $\mathbb{Q}_p$, with ring of integers $\breve{\mathbb{Z}}_p$.

For $F/\mathbb{Q}_p$ nonsplit, let $\breve{F}$ be the completion of the maximal unramified extension of F. If $F/\mathbb{Q}_p$ is split, set $\breve{F} = \breve{\mathbb{Q}}_p$, and view $\breve{F}$ as an F-algebra by choosing one of the two morphisms of $\mathbb{Q}_p$-algebras $F \to \breve{F}$. We also equip $\breve{F}$ with the structure of a $\breve{\mathbb{Q}}_p$-algebra (taking the identity map if $F/\mathbb{Q}_p$ is split).

In all cases, let $\mathcal{O}_{\breve{F}} \subseteq \breve{F}$ be the ring of integers and let $\overline{k}$ be the residue field of $\breve{F}$. There is a canonical map $\mathcal{O}_F \to \mathcal{O}_{\breve{F}}$ (using the above choice of $F \to \breve{F}$ when $F/\mathbb{Q}_p$ is split).

We write $\Delta \subseteq \mathbb{Z}_p$ for the discriminant ideal and $\mathfrak{d} \subseteq \mathcal{O}_F$ for the different ideal. These are $\Delta = \mathbb{Z}_p$ and $\mathfrak{d} = \mathcal{O}_F$ in the split case. We also abuse notation and write $\mathfrak{d}$ for a chosen generator of the different ideal satisfying $\mathfrak{d}^\sigma = -\mathfrak{d}$, taking $\mathfrak{d} = \varpi$ in the ramified case.

In the split case, let e^+ be the nontrivial idempotent in $\mathcal{O}_F$ mapping to $1 \in \mathcal{O}_{\breve{F}}$, and let e^- be the nontrivial idempotent mapping to $0 \in \mathcal{O}_{\breve{F}}$. Given an $\mathcal{O}_F$-module M, we write $M = M^+ \oplus M^-$ where e^+ projects to M^+ and e^- projects to M^-. We use similar notation $f = f^+ \oplus f^-$ for morphisms $f\colon M \to M'$ of $\mathcal{O}_F$-modules. We often use this for p-divisible groups X with $\mathcal{O}_F$ action, e.g. $X = X^+ \times X^-$, and similarly for $\mathcal{O}_F$-linear quasi-homomorphisms.

3.1. Rapoport–Zink spaces.

Definition 3.1.1. Let S be a formal scheme over $\operatorname{Spf} \mathbb{Z}_p$ and let $n \geq 1$ be an integer. By a *Hermitian p-divisible group* over S, we mean a tuple (X, ι, λ) where

X	is a p-divisible group over S of height $2n$ and dimension n
$\iota\colon \mathcal{O}_F \to \operatorname{End}(X)$	is a ring homomorphism
$\lambda\colon X \to X^\vee$	is a quasi-polarization satisfying:

(1) (Action compatibility) The Rosati involution $\dagger$ on $\operatorname{End}^0(X)$ satisfies $\iota(a)^\dagger = \iota(a^\sigma)$ for all $a \in \mathcal{O}_F$.

An *isomorphism* of Hermitian p-divisible groups is an isomorphism of underlying p-divisible groups which respects the $\mathcal{O}_F$-actions and polarizations.

In Sections 3–5, we only consider Hermitian p-divisible groups X over formal schemes S equipped with a morphism $S \to \operatorname{Spf} \mathcal{O}_{\breve{F}}$. We assume that X is supersingular if $F/\mathbb{Q}_p$ is nonsplit, and that X is ordinary if $F/\mathbb{Q}_p$ is split.

We primarily discuss Hermitian p-divisible groups satisfying either of the following two conditions.

(2) (Principal polarization) The quasi-polarization λ is a principal polarization.

(2°) (Polarization condition $\circ$) Assume n is even if $F/\mathbb{Q}_p$ is ramified. The quasi-polarization $\Delta\lambda$ is a polarization, and $\ker(\Delta\lambda) = X[\iota(\mathfrak{d})]$.

In case (2), we say that (X, ι, λ) is *principally polarized*. In case (2°), or if $F/\mathbb{Q}_p$ is unramified in case (2), we say that (X, ι, λ) is $\circ$-*polarized*. In case (2°), we also use the alternative terminology ϖ^{-1}-*modular*.[6]

Given an integer r with $0 \le r \le n$, we next consider

(1) (Kottwitz $(n-r, r)$ signature condition) For all $a \in \mathcal{O}_F$, the characteristic polynomial of $\iota(a)$ acting on $\operatorname{Lie} X$ is $(T-a)^{n-r}(T-a^\sigma)^r \in \mathcal{O}_S[T]$

for pairs (X, ι), i.e. n-dimensional p-divisible groups X over a formal scheme S with action $\iota\colon \mathcal{O}_F \to \operatorname{End}(X)$. Here we view $\mathcal{O}_S$ as an $\mathcal{O}_F$-algebra via $S \to \operatorname{Spf} \mathcal{O}_{\breve{F}} \to \operatorname{Spec} \mathcal{O}_F$.

If (X, ι, λ) is a Hermitian p-divisible group of signature $(n-r, r)$, then $(X^\sigma, \iota^\sigma, \lambda^\sigma)$ with

$$X^\sigma = X \qquad \iota^\sigma = \iota \circ \sigma \qquad \lambda^\sigma = \lambda \tag{3.1.1}$$

is a Hermitian p-divisible group of signature $(r, n-r)$. We use similar notation $(X, \iota) \leftrightarrow (X^\sigma, \iota^\sigma)$ without the presence of a polarization. This allows us to switch between signatures $(n-r, r)$ and signature $(r, n-r)$ (e.g. for comparison with the literature).

From here on, we always implicitly restrict to signature $(n-1, 1)$ (and even n) when discussing $\circ$-polarized Hermitian p-divisible groups for ramified $F/\mathbb{Q}_p$. In this case, we also impose

(2) (Pappas wedge condition) For all $a \in \mathcal{O}_F$, the action of $\iota(a)$ on $\operatorname{Lie} X$ satisfies

$$\bigwedge^2 (\iota(a) - a) = 0 \quad \text{and} \quad \bigwedge^n (\iota(a) - a^\sigma) = 0.$$

(3) (PRRSZ spin condition) For every geometric point $\overline{s}$ of S, the action of $(\iota(a) - a)$ on $\operatorname{Lie} X_{\overline{s}}$ is nonzero for some $a \in \mathcal{O}_F$.

The signature $(n-1, 1)$ condition implies that the equation involving $\bigwedge^n$ in the wedge condition is automatic, and that the wedge condition is empty if $n = 2$.

We temporarily allow $p = 2$ even if $F/\mathbb{Q}_p$ is ramified. Recall that there exists a supersingular p-divisible group $\mathbf{X}_0$ of height 2 and dimension 1 over $\overline{k}$, and that $\mathbf{X}_0$ is unique up to isomorphism. The same holds for "ordinary" instead of "supersingular". These claims also hold for any algebraically closed field κ over $\overline{k}$. In the supersingular case $\mathbf{X}_0$ is given by a Lubin–Tate formal group law, and in the ordinary case we have $\mathbf{X}_0 \cong \boldsymbol{\mu}_{p^\infty} \times \underline{\mathbb{Q}_p/\mathbb{Z}_p}$. We have $\operatorname{End}(\mathbf{X}_0) \cong \mathbb{Z}_p \times \mathbb{Z}_p$ in the ordinary

[6]Strictly speaking, it is more common in the literature to normalize the polarization differently, i.e. our λ corresponds to $\Delta^{-1}\lambda$ from [RSZ17, §3] and [LL22, §2] (this certainly gives an equivalent moduli problem). We prefer our normalization, which seems more natural for our eventual main results on comparison with local Whittaker functions and Eisenstein series. A related remark is [LL22, Footnote 9].

case and $\mathrm{End}(\mathbf{X}_0) \cong \mathcal{O}_D$ in the supersingular case, where $\mathcal{O}_D$ is the unique maximal order in the quaternion division algebra D over $\mathbb{Q}_p$ (e.g. [Gro86] or [Wew07, §1]).

Quasi-polarizations on $\mathbf{X}_0$ exist and are unique up to $\mathbb{Q}_p^\times$ scalar, and there exists a principal polarization $\lambda_{\mathbf{X}_0}$ on $\mathbf{X}_0$ (unique up to $\mathbb{Z}_p^\times$ scalar). The induced Rosati involution on $\mathrm{End}(\mathbf{X}_0)$ is the standard involution (this can be verified on the Dieudonné module, see e.g. [RSZ17, Page 2205]), hence induces the nontrivial Galois involution on F for any embedding $F \hookrightarrow \mathrm{End}^0(\mathbf{X}_0)$ (if such an embedding exists).

From now on, we assume $\mathbf{X}_0$ is supersingular if $F/\mathbb{Q}_p$ is nonsplit, and that $\mathbf{X}_0$ is ordinary if $F/\mathbb{Q}_p$ is split. Then there exists an embedding $j\colon \mathcal{O}_F \hookrightarrow \mathrm{End}(\mathbf{X}_0)$. Given such a j, form $(\mathbf{X}_0, j)$ and $(\mathbf{X}_0^\sigma, j^\sigma)$ as above. There is an $\mathcal{O}_F$-linear isogeny of degree $|\Delta|_p^{-1}$

$$\begin{aligned} \mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F &\longrightarrow \mathbf{X}_0 \times \mathbf{X}_0^\sigma \\ x \otimes a &\longmapsto (j(a)x, j(a^\sigma)x) \end{aligned} \tag{3.1.2}$$

where $\mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F$ is the Serre tensor p-divisible group (as in [Che24b, (C.1.1)]), with its Serre tensor $\mathcal{O}_F$-action. See also [KR11, Lemma 6.2] (there in the inert case for $p \neq 2$, but the version in (3.1.2) allows for $p = 2$).

Suppose $\lambda_{\mathbf{X}_0}$ is a principal polarization of $\mathbf{X}_0$. Under the map in (3.1.2), the ($\mathcal{O}_F$-action compatible) product polarization $\lambda_{\mathbf{X}_0} \times \lambda_{\mathbf{X}_0}^\sigma$ on $\mathbf{X}_0 \times \mathbf{X}_0^\sigma$ pulls back to the polarization

$$\lambda_{\mathbf{X}_0} \otimes \lambda_{\mathrm{tr}}\colon \mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F \to (\mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F^*) \cong \mathbf{X}_0^\vee \otimes_{\mathbb{Z}_p} \mathcal{O}_F^* \tag{3.1.3}$$

where $\mathcal{O}_F^* \coloneqq \mathrm{Hom}_{\mathbb{Z}_p}(\mathcal{O}_F, \mathbb{Z}_p)$ and $\lambda_{\mathrm{tr}}\colon \mathcal{O}_F \to \mathcal{O}_F^*$ is induced by the symmetric bilinear pairing $\mathrm{tr}_{F/\mathbb{Q}_p}(a^\sigma b)$ on $\mathcal{O}_F$. Indeed, after picking a $\mathbb{Z}_p$-basis $\{1, \alpha\}$ for $\mathcal{O}_F$ to identify $\mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F \cong \mathbf{X}_0^2$, the map in (3.1.2) is given by the matrix

$$\phi = \begin{pmatrix} 1 & \alpha \\ 1 & \alpha^\sigma \end{pmatrix} \in M_{2,2}(\mathcal{O}_F) \tag{3.1.4}$$

and $\det \phi$ generates the different ideal of $\mathcal{O}_F/\mathbb{Z}_p$ (so Smith normal form shows $\deg \phi = |\Delta|_p^{-1}$). Identifying $\mathbf{X}_0^\vee \otimes_{\mathbb{Z}_p} \mathcal{O}_F^* \cong \mathbf{X}_0^{\vee 2}$ using the basis of $\mathcal{O}_F^*$ dual to $\{1, \alpha\}$, the preceding claim about pullback polarizations follows because $({}^t\phi^\sigma)\phi$ (where ${}^t\phi^\sigma$ means conjugate transpose) is the Gram matrix for the basis $\{1, \alpha\}$ and the trace pairing on $\mathcal{O}_F$.

If $p \neq 2$ and $\mathbf{X}_0$ is supersingular, the polarization $\lambda_{\mathbf{X}_0} \otimes \lambda_{\mathrm{tr}}$ coincides with the polarization on $\mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F$ described in [KR11, (6.2)] (inert case, with modification as in [RSZ17, Footnote 4]) and [RSZ17, (3.4)] (ramified case, though we normalize differently).

Suppose $\iota_{\mathbf{X}_0}\colon \mathcal{O}_F \hookrightarrow \mathrm{End}(\mathbf{X}_0)$ is an action of signature $(1,0)$. The pair $(\mathbf{X}_0, \iota_{\mathbf{X}_0})$ exists and is unique up to isomorphism. In the split case, the element $\iota_{\mathbf{X}_0}(e^+)$ is projection to $\boldsymbol{\mu}_{p^\infty}$ and $\iota_{\mathbf{X}_0}(e^-)$ is projection to $\underline{\mathbb{Q}_p/\mathbb{Z}_p}$. In the ramified case, note that $(\mathbf{X}_0, \iota_{\mathbf{X}_0})$ is simultaneously of signature $(1,0)$ and $(0,1)$.

Fix $(\mathbf{X}_0, \iota_{\mathbf{X}_0})$ as above, and form $(\mathbf{X}_0^\sigma, \iota_{\mathbf{X}_0}^\sigma)$. We have

$$\mathrm{Hom}_{\mathcal{O}_F}(\mathbf{X}_0, \mathbf{X}_0^\sigma) \cong \begin{cases} \mathcal{O}_F & \text{if } F/\mathbb{Q}_p \text{ is nonsplit} \\ 0 & \text{if } F/\mathbb{Q}_p \text{ is split} \end{cases} \tag{3.1.5}$$

as $\mathcal{O}_F$-modules by precomposition. Using $\mathrm{End}(\mathbf{X}_0) \cong \mathcal{O}_D$ in the nonsplit cases, we find that the $\mathcal{O}_F$-module $\mathrm{Hom}_{\mathcal{O}_F}(\mathbf{X}_0, \mathbf{X}_0^\sigma)$ is generated by any isogeny of degree p if $F/\mathbb{Q}_p$ is inert, and is generated by an isomorphism if $F/\mathbb{Q}_p$ is ramified (namely any element $a \in \mathcal{O}_D^\times$ such that conjugation by a induces the nontrivial Galois involution on F). We have $\mathrm{End}_{\mathcal{O}_F}(\mathbf{X}_0) = \mathcal{O}_F$ in all cases.

Suppose $\lambda_{\mathbf{X}_0}$ is a principal polarization of $\mathbf{X}_0$. If $F/\mathbb{Q}_p$ is unramified, the triple $(\mathbf{X}_0, \iota_{\mathbf{X}_0}, \lambda_{\mathbf{X}_0})$ is unique (up to isomorphism): given another polarization $\lambda'_{\mathbf{X}_0}$, we have $\lambda_{\mathbf{X}_0}^{-1} \circ \lambda'_{\mathbf{X}_0} \in \mathbb{Z}_p^\times$ and know that the norm map $N_{F/\mathbb{Q}_p} \colon \mathcal{O}_F^\times \to \mathbb{Z}_p^\times$ is surjective. If $F/\mathbb{Q}_p$ is ramified, the same reasoning shows that there are two choices of $\lambda_{\mathbf{X}_0}$ (differing by a $\mathbb{Z}_p^\times$ scalar) because $N_{F/\mathbb{Q}_p}(\mathcal{O}_F^\times) \subseteq \mathbb{Z}_p^\times$ has index 2. Fix a choice of $(\mathbf{X}_0, \iota_{\mathbf{X}_0}, \lambda_{\mathbf{X}_0})$.

We now re-impose our running assumption that $p \neq 2$ if $F/\mathbb{Q}_p$ is ramified. Fix any $\circ$-polarized Hermitian p-divisible group $(\mathbf{X}, \iota_{\mathbf{X}}, \lambda_{\mathbf{X}})$ of signature $(n-r, r)$ over $\overline{k}$. Such triples exist and are unique up to F-linear quasi-isogenies preserving polarizations exactly. This uniqueness may be proved via Dieudonné theory: see [Vol10, §1] (inert case, but we allow $p = 2$ by the same proof) and [RSZ17, Proposition 3.1] [RSZ18, §6] (ramified case). In the split case, we have a stronger uniqueness statement.

Lemma 3.1.2. *For $F/\mathbb{Q}_p$ split, the triple $(\mathbf{X}, \iota_{\mathbf{X}}, \lambda_{\mathbf{X}})$ is unique up to isomorphism. This also holds over any algebraically closed field extension κ of $\overline{k}$.*

Proof. Decompose $\mathbf{X} = \mathbf{X}^+ \times \mathbf{X}^-$ using the idempotents in the $\mathcal{O}_F = \mathbb{Z}_p \times \mathbb{Z}_p$ action given by $\iota_{\mathbf{X}}$. Then $\mathbf{X}^+$ and $\mathbf{X}^-$ are the unique ordinary p-divisible groups over κ of height n and the correct dimension ($n-r$ and r, respectively). Uniqueness of $\lambda_{\mathbf{X}}$ (up to isomorphism) corresponds to the following fact: there is a unique self-dual Hermitian $\mathcal{O}_F$-lattice (up to isomorphism) of any given rank. □

For existence in all cases, we may construct $(\mathbf{X}, \iota_{\mathbf{X}}, \lambda_{\mathbf{X}})$ as follows. For $F/\mathbb{Q}_p$ unramified, we can take $\mathbf{X} = (\mathbf{X}_0)^{n-r} \times (\mathbf{X}_0^\sigma)^r$ (with the product $\mathcal{O}_F$-action and polarization).

For $F/\mathbb{Q}_p$ ramified, we can take $\mathbf{X} = (\mathbf{X}_0)^{n-2} \times (\mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F)$ using the Serre tensor construction. The $\mathcal{O}_F$-action $\iota_{\mathbf{X}}$ is diagonal on $(\mathbf{X}_0)^{n-2}$ and given by the Serre tensor $\mathcal{O}_F$-action on $\mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F$. We can take the product quasi-polarization $\lambda_{\mathbf{X}}$ of $\mathbf{X}$ given by

$$\begin{aligned} &-\iota_{\mathbf{X}_0}(\varpi)^{-2} \circ (\lambda_{\mathbf{X}_0} \otimes \lambda_{\mathrm{tr}}) \quad \text{on } \mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F, \text{ and} \\ &\underbrace{\begin{pmatrix} 0 & \lambda_{\mathbf{X}_0} \circ -\iota_{\mathbf{X}_0}(\varpi)^{-1} \\ \lambda_{\mathbf{X}_0} \circ \iota_{\mathbf{X}_0}(\varpi)^{-1} & 0 \end{pmatrix} \times \cdots \times \begin{pmatrix} 0 & \lambda_{\mathbf{X}_0} \circ -\iota_{\mathbf{X}_0}(\varpi)^{-1} \\ \lambda_{\mathbf{X}_0} \circ \iota_{\mathbf{X}_0}(\varpi)^{-1} & 0 \end{pmatrix}}_{(n-2)/2 \text{ times}} \quad \text{on } \mathbf{X}_0^{n-2}. \end{aligned} \tag{3.1.6}$$

This is the construction of [RSZ17, §3.3] (but rescaled).

Given a $\circ$-polarized triple (X, ι, λ) of signature $(n-r, r)$ over some base scheme S, we define a *framing* $\rho \colon X_{\overline{S}} \to \mathbf{X}_{\overline{S}}$ to be an F-linear quasi-isogeny which preserves polarizations, where the subscript indicates base-change to $\overline{S} := S \times_{\mathrm{Spec}\, \mathcal{O}_{\breve{F}}} \mathrm{Spec}\, \overline{k}$. An isomorphism of *framed tuples* $(X, \iota, \lambda, \rho)$ is an $\mathcal{O}_F$-linear isomorphism of p-divisible groups compatible with framings.

Definition 3.1.3. We define the Rapoport–Zink space $\mathcal{N}(n-r,r)$ over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$ to be given by the (set-valued) functor

$$\mathcal{N}(n-r,r)(S) \coloneqq \{\text{isomorphism classes of framed tuples } (X,\iota,\lambda,\rho) \text{ over } S\} \tag{3.1.7}$$

for schemes S over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$.

The space $\mathcal{N}(n-r,r)$ does not depend on the choices of framing object (up to functorial isomorphism).

Lemma 3.1.4. *The functor $\mathcal{N}(n-r,r)$ is represented by a locally Noetherian formal scheme which is formally locally of finite type and separated over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$. Each irreducible component of the reduced subscheme is projective over $\overline{k}$.*

Proof. Representability, local Noetherianity, and formally locally of finite type-ness follow via [RZ96, Theorem 2.16]; various closedness statements can be checked via [RZ96, Proposition 2.9], which holds verbatim with "isogeny" replaced by "homomorphism". Projectivity of the reduced irreducible components follows from [RZ96, Proposition 2.32], also using [RSZ17, Proposition 3.8] in the ramified case. Separatedness now follows because this can be checked on underlying reduced subschemes (then apply the valuative criterion). □

Lemma 3.1.5. *The formal scheme $\mathcal{N}(n-r,r)$ is regular and the structure morphism $\mathcal{N}(n-r,r)\to\operatorname{Spf}\mathcal{O}_{\breve{F}}$ is formally smooth of relative dimension $(n-r)r$.*

Proof. We know the structure map $\mathcal{N}(n-r,r)\to\operatorname{Spf}\mathcal{O}_{\breve{F}}$ is formally smooth of relative dimension $(n-r)r$ via [Mih22, Proposition 1.3] in the inert case (also [Che24b, Section 2.5], where we allow $p=2$ and the split case) and [RSZ17, Proposition 3.8] for the ramified case. We conclude that $\mathcal{N}(n-r,r)$ is regular because the map to $\operatorname{Spf}\mathcal{O}_{\breve{F}}$ is formally smooth and formally locally of finite type. □

For $F/\mathbb{Q}_p$ ramified, the Rapoport–Zink space $\mathcal{N}(n-1,1)$ is often called *exotic smooth* in the literature, following the terminology of [RSZ17].

Definition 3.1.6. By the *canonical lifting* of $(\mathbf{X}_0,\iota_{\mathbf{X}_0},\lambda_{\mathbf{X}_0})$, we mean a tuple $(\mathfrak{X}_0,\iota_{\mathfrak{X}_0},\lambda_{\mathfrak{X}_0},\rho_{\mathfrak{X}_0})$ where $(\mathfrak{X}_0,\iota_{\mathfrak{X}_0},\lambda_{\mathfrak{X}_0})$ is a signature $(1,0)$ principally polarized Hermitian p-divisible group over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$, and $\rho_{\mathfrak{X}_0}\colon \mathfrak{X}_{0,\overline{k}}\to\mathbf{X}_0$ is an $\mathcal{O}_F$-linear isomorphism (necessarily preserving polarizations up to $\mathbb{Z}_p^\times$ scalar).

The canonical lifting $(\mathfrak{X}_0,\iota_{\mathfrak{X}_0},\lambda_{\mathfrak{X}_0},\rho_{\mathfrak{X}_0})$ of Definition 3.1.6 exists and is unique up to unique isomorphism (e.g. see Lemma 3.4.3 and following discussion).

Next consider the locally free sheaf Lie on $\mathcal{N}(n-r,r)$ given by the functorial assignment $(X,\iota,\lambda,\rho)\mapsto\operatorname{Lie}X$. In the case of signature $(n-1,1)$, there is a unique maximal local direct summand $\mathcal{F}\subseteq\operatorname{Lie}$ of rank $n-1$ such that the $\mathcal{O}_F$-action via ι is $\mathcal{O}_F$-linear on $\mathcal{F}$ and σ-linear on $\operatorname{Lie}/\mathcal{F}$. The ramified case is proved in [LL22, Lemma 2.36]. In the unramified case, we have a canonical eigenspace decomposition $\operatorname{Lie}=\mathcal{F}\oplus(\operatorname{Lie}/\mathcal{F})$ for the $\mathcal{O}_F$-action.

Consider the canonical lifting $(\mathfrak{X}_0,\iota_{\mathfrak{X}_0},\lambda_{\mathfrak{X}_0},\rho_{\mathfrak{X}_0})$ over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$ (Definition 3.1.6). Given any $\operatorname{Spf}\mathcal{O}_{\breve{F}}$-scheme S, we write $\mathbb{D}(\mathfrak{X}_{0,S})(S)$ for evaluation of the (covariant) Dieudonné crystal $\mathbb{D}(\mathfrak{X}_{0,S})$

at $\mathrm{id}\colon S \to S$, with associated Hodge filtration step $F^0\mathbb{D}(\mathfrak{X}_{0,S})(S)$. The assignment $S \mapsto F^0\mathbb{D}(\mathfrak{X}_{0,S})(S)$ defines a (trivial) line bundle on $\mathcal{N}(n-1,1)$, which we denote $\mathrm{Lie}_0^\vee$. The principal polarization $\lambda_{\mathfrak{X}_0}$ induces an identification $(\mathrm{Lie}\,\mathfrak{X}_{0,S})^\vee \cong (\mathrm{Lie}\,\mathfrak{X}_{0,S}^\vee)^\vee$ and the latter is $F^0\mathbb{D}(\mathfrak{X}_{0,S})(S)$.

Definition 3.1.7. The *tautological bundle* $\mathcal{E}$ on $\mathcal{N}(n-1,1)$ is the line bundle whose dual is $\mathcal{E}^\vee \coloneqq \underline{\mathrm{Hom}}(\mathrm{Lie}_0^\vee, \mathrm{Lie}\,/\mathcal{F})$.

The definition of $\mathcal{E}$ is taken from [How19, Definition 3.4] (at least in the inert case). The line bundle $\mathcal{E}$ on $\mathcal{N}(n-1,1)$ is a local analogue of the global tautological bundle as in [Che24b, Definition 2.2.6], which pulls back to $\mathcal{E}$ under Rapoport–Zink uniformization [Che24b, Section 6.8].

3.2. **Local special cycles.** We define certain local special cycles on Rapoport–Zink spaces, following [KR11, Definition 3.2] (there in the inert case). Retain notation from Section 3.1.

The space of *local special quasi-homomorphisms* means the F-module

$$\mathbf{W} \coloneqq \mathrm{Hom}_F^0(\mathbf{X}_0, \mathbf{X}). \tag{3.2.1}$$

If $F/\mathbb{Q}_p$ is nonsplit, then $\mathbf{W}$ is free of rank n (see also [RSZ17, Lemma 3.5] in the ramified case). If $F/\mathbb{Q}_p$ is split, then $\mathbf{W}$ is a free F-module of rank $n-r$ (because $\mathrm{Hom}_{\mathcal{O}_F}(\mathbf{X}_0, \mathbf{X}_0^\sigma) = 0$ in the split case, in contrast with $\mathrm{Hom}_{\mathcal{O}_F}(\mathbf{X}_0, \mathbf{X}_0^\sigma) \cong \mathcal{O}_F$ in the nonsplit cases). In the split case only, set $\mathbf{W}^\perp \coloneqq \mathrm{Hom}_F^0(\mathbf{X}_0^\sigma, \mathbf{X})$. In the nonsplit cases, set $\mathbf{W}^\perp = 0$.

Set

$$\mathbf{V} = \mathbf{W} \oplus \mathbf{W}^\perp \qquad \mathbf{V}_0 = \mathrm{Hom}_F^0(\mathbf{X}_0, \mathbf{X}_0). \tag{3.2.2}$$

In all cases, these are free F-modules of rank n and 1, respectively.

We equip $\mathbf{W}$, $\mathbf{W}^\perp$, and $\mathbf{V}_0$ with the non-degenerate Hermitian pairings $(x,y) = x^\dagger y \in \mathrm{End}_F^0(\mathbf{X}_0) = F$. We give $\mathbf{V}$ the Hermitian form making $\mathbf{W}$ and $\mathbf{W}^\perp$ orthogonal. We have $\varepsilon(\mathbf{V}) = (-1)^r$ if $F/\mathbb{Q}_p$ is nonsplit, and have $\varepsilon(\mathbf{V}) = 1$ if $F/\mathbb{Q}_p$ is split. This follows upon inspecting the explicit framing tuples constructed in Section 3.1 (see [RSZ17, Lemma 3.5] for the ramified case).

Definition 3.2.1 (Kudla–Rapoport local special cycles)**.** Given any set $L \subseteq \mathbf{W}$, there is an associated *local special cycle*

$$\tilde{\mathcal{Z}}(L) \subseteq \mathcal{N}(n-r, r) \tag{3.2.3}$$

which is the subfunctor consisting of tuples $(X, \iota, \lambda, \rho)$ over schemes S over $\mathrm{Spf}\,\mathcal{O}_{\breve{F}}$ such that, for all $x \in L$, the quasi-homomorphism

$$\rho^{-1} \circ x_{\overline{S}} \circ \rho_{\mathfrak{X}_0,\overline{S}}\colon \mathfrak{X}_{0,\overline{S}} \to X_{\overline{S}} \tag{3.2.4}$$

lifts to a homomorphism $\mathfrak{X}_{0,S} \to X$.

In the preceding definition, such lifts are unique if they exist by Drinfeld rigidity for quasi-homomorphisms of p-divisible groups. We know that $\tilde{\mathcal{Z}}(L) \subseteq \mathcal{N}(n-r,r)$ is a closed subfunctor (hence a locally Noetherian formal scheme) by [RZ96, Proposition 2.9] for quasi-homomorphisms. From the definition, it is clear that $\tilde{\mathcal{Z}}(L)$ depends only on the $\mathcal{O}_F$-span of L.

Lemma 3.2.2. *Let $L \subseteq \mathbf{W}$ be any subset. If $\tilde{\mathcal{Z}}(L) \neq \emptyset$, then $(x,y) \in \mathfrak{d}^{-1}$ for all $x, y \in L$.*

Proof. If $\tilde{\mathcal{Z}}(L) \neq \emptyset$, then $\tilde{\mathcal{Z}}(L)(\overline{k}) \neq \emptyset$ because $\tilde{\mathcal{Z}}(L) \to \operatorname{Spf} \mathcal{O}_{\breve{F}}$ is formally locally of finite type. If $\tilde{\mathcal{Z}}(L)(\overline{k}) \neq \emptyset$ and $x, y \in L$, we find $\mathfrak{d} x^\dagger y \in \operatorname{End}_{\mathcal{O}_F}(\mathbf{X}_0) = \mathcal{O}_F$ by the $\circ$-polarization condition defining $\mathcal{N}(n-r,r)$, where $\mathfrak{d}$ is the different ideal. □

If $F/\mathbb{Q}_p$ is nonsplit, set $\mathcal{Z}(L) \coloneqq \tilde{\mathcal{Z}}(L)$ for any subset $L \subseteq \mathbf{W}$. If $F/\mathbb{Q}_p$ is split, we will instead define $\mathcal{Z}(L)$ as a certain open and closed subfunctor (see (3.5.4)) for later notational uniformity.

In all cases, we write $\mathcal{Z}(L)_{\mathscr{H}} \subseteq \mathcal{Z}(L)$ (*horizontal special cycle*) for the flat part of $\mathcal{Z}(L)$, i.e. the largest closed formal subscheme which is flat over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$.

3.3. **Actions on Rapoport–Zink spaces.** Consider the group action

$$I_1 \coloneqq \{\gamma \in \operatorname{End}^0_F(\mathbf{X}) : \gamma^\dagger \gamma = 1\} \qquad I_1 \circlearrowright \mathcal{N}(n-r,r) \tag{3.3.1}$$
$$(X, \iota, \lambda, \rho) \mapsto (X, \iota, \lambda, \gamma \circ \rho)$$

and the isomorphism

$$I_1 \xrightarrow{\sim} U(\mathbf{W}) \times U(\mathbf{W}^\perp) \qquad I_1 \circlearrowright \mathbf{V} \tag{3.3.2}$$
$$x \mapsto \gamma \circ x. \tag{3.3.3}$$

For any subset $L \subseteq \mathbf{W}$ with associated local special cycle $\tilde{\mathcal{Z}}(L) \subseteq \mathcal{N}(n-r,r)$, the above action of I_1 satisfies

$$\gamma(\tilde{\mathcal{Z}}(L)) = \tilde{\mathcal{Z}}(\gamma L). \tag{3.3.4}$$

We will also have $\gamma(\mathcal{Z}(L)) = \mathcal{Z}(\gamma(L))$. This was just checked in the nonsplit cases; in the split case, this will be clear from the definition, see (3.5.4).

3.4. **Similitude variants.** As mentioned in Section 1.5, the global applications in [Che24b; Che24c] will involve Shimura varieties for a product G' of a unitary group with a torus factor. For this reason, it is not $\mathcal{N}(n-1,1)$ which appears most naturally in Rapoport–Zink uniformization for these Shimura varieties, but rather a similitude variant $\mathcal{N}'$ to be defined below. On the other hand, $\mathcal{N}'$ will simply be a disjoint union of several copies of $\mathcal{N}(n-1,1)$, as explained in (3.4.2) and (3.4.3) below. For the local theorems of this paper in signature $(n-1,1)$, it turns out to be harmless to formulate and prove our theorems on $\mathcal{N}(n-1,1)$.

As the spaces $\mathcal{N}'$ are used for our global applications, we explain their definition and their special cycles in this section, i.e. analogues of Sections 3.1 to 3.3. For the purpose of the local theorems in this paper only, Section 3.4 can be skipped. Since it makes no difference, we allow arbitrary signature $(n-r,r)$.

We first discuss the Rapoport–Zink spaces, continuing the notation of Section 3.1. Given a principally polarized triple $(X_0, \iota_0, \lambda_0)$ of signature $(1,0)$ over some base scheme S, a *similitude framing* is an F-linear quasi-isogeny $\rho_0 \colon X_{0,\overline{S}} \to \mathbf{X}_{0,\overline{S}}$ such that

$$\rho_0^*(\lambda_{\mathbf{X}_0,\overline{S}}) = b\lambda_{0,\overline{S}} \quad \text{for some } b \in \mathbb{Q}_p^\times, \tag{3.4.1}$$

where $b \in \mathbb{Q}_p^\times$ really means a section of the constant sheaf. We call $(X_0, \iota_0, \lambda_0, \rho_0)$ a *framed similitude* tuple. An *isomorphism* of framed similitude tuples $f \colon (X_0, \iota_0, \lambda_0, \rho_0) \to (X_0', \iota_0', \lambda_0', \rho_0')$ is an $\mathcal{O}_F$-linear isomorphism of p-divisible groups $f \colon X_0 \to X_0'$ such that $f^*(\lambda_0')$ and λ_0 agree up

to $\mathbb{Z}_p^\times$-scalar and also $\rho_0' \circ f_{\overline{S}} = \rho_0$. Given a $\circ$-polarized triple (X, ι, λ) of signature $(n-r, r)$ over some base scheme S, we define a *similitude framing* $\rho\colon X_{\overline{S}} \to \mathbf{X}_{\overline{S}}$ in the same way.

Definition 3.4.1. We consider Rapoport–Zink spaces over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$, given by the (set-valued) functors

$$\mathcal{N}(1,0)'(S) \coloneqq \{\text{isomorphism classes of framed similitude tuples } (X_0, \iota_0, \lambda_0, \rho_0) \text{ over } S\}$$
$$\mathcal{N}(n-r,r)'(S) \coloneqq \{\text{isomorphism classes of framed similitude tuples } (X, \iota, \lambda, \rho) \text{ over } S\}$$

for schemes S over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$. Here, signature $(1,0)$ and principal polarizations are understood for $\mathcal{N}(1,0)'$. Signature $(n-r,r)$ and $\circ$-polarizations are understood for $\mathcal{N}(n-r,r)'$.

As in Definition 3.1.3, these Rapoport–Zink spaces do not depend on the choices of framing objects (up to functorial isomorphism). The variant of $\mathcal{N}(n-r,r)'$ defined by the stricter requirement $b \in \mathbb{Z}_p^\times$ is canonically isomorphic to $\mathcal{N}(n-r,r)$ from Definition 3.1.3; we use this without further mention. The statement and proof of Lemma 3.1.4 hold verbatim with $\mathcal{N}(n-r,r)$ replaced by $\mathcal{N}(1,0)'$ and $\mathcal{N}(n-r,r)'$.

Definition 3.4.2. We define the open and closed subfunctor $\mathcal{N}' \subseteq \mathcal{N}(1,0)' \times \mathcal{N}(n-r,r)'$ as

$$\mathcal{N}'(S) \coloneqq \left\{ (X_0, \iota_0, \lambda_0, \rho_0, X, \iota, \lambda, \rho) : \begin{array}{l} \rho_0^*(\lambda_{\mathbf{X}_0,\overline{S}}) = b_0 \lambda_{0,\overline{S}} \quad \rho^*(\lambda_{\mathbf{X},\overline{S}}) = b\lambda_{\overline{S}} \\ \text{with } b_0 = b \text{ in } \mathbb{Q}_p^\times/\mathbb{Z}_p^\times \end{array} \right\}$$

for schemes S over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$.

Given an object of $\mathcal{N}'(S)$ as in Definition 3.4.2, let $a \in F^\times/\mathcal{O}_F^\times$ correspond to $(X_0, \iota_0, \lambda_0, \rho_0) \in \mathcal{N}(1,0)'(S)$ (by the identification of Lemma 3.4.3). We then have an isomorphism

$$\begin{aligned} \mathcal{N}' &\xrightarrow{\quad\sim\quad} \mathcal{N}(1,0)' \times \mathcal{N}(n-r,r) \\ (X_0, \iota_0, \lambda_0, \rho_0, X, \iota, \lambda, \rho) &\longmapsto (X_0, \iota_0, \lambda_0, \rho_0, X, \iota, \lambda, a^{-1}\rho). \end{aligned} \tag{3.4.2}$$

Whenever we write $(X_0, \iota_0, \lambda_0, \rho_0, X, \iota, \lambda, \rho)$ for a (functorial) point of $\mathcal{N}'$, we mean an object as on the left of (3.4.2) (i.e. ρ preserves polarizations up to $\mathbb{Q}_p^\times$ scalar).

Lemma 3.4.3. *There is an isomorphism*

$$\begin{aligned} F^\times/\mathcal{O}_F^\times &\xrightarrow{\quad\sim\quad} \mathcal{N}(1,0)' \\ a &\longmapsto (\mathbf{X}_0, \iota_{\mathbf{X}_0}, \lambda_{\mathbf{X}_0}, a \cdot \mathrm{id}_{\mathbf{X}_0}) \end{aligned} \tag{3.4.3}$$

where the left-hand side is viewed as a constant formal scheme over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$.

Proof. In the nonsplit case, [How19, Proposition 2.1] states that $\mathcal{N}(1,0)'$ is a disjoint union of copies of $\operatorname{Spf} \mathcal{O}_{\breve{F}}$, so it is enough to check the claim on $\overline{k}$-points. The claim on $\overline{k}$-points follows from uniqueness of the triple $(\mathbf{X}_0, \iota_{\mathbf{X}_0}, \lambda_{\mathbf{X}_0})$ (up to isomorphism preserving polarizations up to $\mathbb{Z}_p^\times$ scalar) and the equality $\mathrm{End}_F^0(\mathbf{X}_0) = F$.

The split case holds via the following similar argument. The map $\mathcal{N}(1,0)'(\kappa) \to \mathcal{N}(1,0)'(\kappa')$ is bijective for any extension of algebraically closed fields $\kappa \subseteq \kappa'$ (essentially by uniqueness of the triple $(\mathbf{X}_0, \iota_{\mathbf{X}_0}, \lambda_{\mathbf{X}_0})$, which holds over any algebraically closed field of characteristic p). So the

reduced subscheme $\mathcal{N}(1,0)'_{\mathrm{red}}$ is isomorphic to a (discrete) disjoint union of copies of $\operatorname{Spec}\overline{k}$. We also see that the map $F^\times/\mathcal{O}_F^\times \to \mathcal{N}(1,0)'$ is bijective on $\overline{k}$-points (this follows as in the nonsplit case). To finish, note that $\mathcal{N}(1,0)'$ is formally étale over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$ (e.g. by Grothendieck–Messing theory as in [Che24b, Section 2.5]). □

The canonical lifting of Definition 3.1.6 corresponds, via Lemma 3.4.3 to the unique section $\operatorname{Spf}\mathcal{O}_{\breve{F}} \to \mathcal{N}(1,0)'$ associated to the element $1 \in F^\times/\mathcal{O}_F^\times$.

We next discuss special cycles, continuing the notation of Section 3.2.

Definition 3.4.4 (Kudla–Rapoport local special cycles)**.** Given any set $L \subseteq \mathbf{W}$, there is an associated *local special cycle*

$$\mathcal{Z}(L)' \subseteq \mathcal{N}' \tag{3.4.4}$$

which is the subfunctor consisting of tuples $(X_0, \iota_0, \lambda_0, \rho_0, X, \iota, \lambda, \rho)$ over schemes S over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$ such that, for all $x \in L$, the quasi-homomorphism

$$\rho^{-1} \circ x_{\overline{S}} \circ \rho_0 \colon X_{0,\overline{S}} \to X_{\overline{S}} \tag{3.4.5}$$

lifts to a homomorphism $X_0 \to X$.

Again, such lifts are unique (if they exist) by Drinfeld rigidity for quasi-homomorphisms, and $\mathcal{Z}(L)' \subseteq \mathcal{N}'$ and is a closed subfunctor (hence a locally Noetherian formal scheme) by [RZ96, Proposition 2.9] for quasi-homomorphisms. It is again clear that $\mathcal{Z}(L)'$ depends only on the $\mathcal{O}_F$-span of L.

The isomorphism $\mathcal{N}' \xrightarrow{\sim} \mathcal{N}(1,0)' \times \mathcal{N}(n-r,r)$ of (3.4.2) induces an isomorphism

$$\mathcal{Z}(L)' \xrightarrow{\sim} \mathcal{N}(1,0)' \times \tilde{\mathcal{Z}}(L). \tag{3.4.6}$$

The statement and proof of Lemma 3.2.2 hold verbatim, with $\mathcal{Z}(L)'$ instead of $\tilde{\mathcal{Z}}(L)$.

We last discuss a group action on $\mathcal{N}'$, continuing in the notation of Section 3.3. Consider the groups

$$I_0 \coloneqq \{\gamma_0 \in \operatorname{End}_F^0(\mathbf{X}_0) : \gamma_0^\dagger\gamma_0 \in \mathbb{Q}_p^\times\} \qquad I \coloneqq \{\gamma \in \operatorname{End}_F^0(\mathbf{X}) : \gamma^\dagger\gamma \in \mathbb{Q}_p^\times\} \tag{3.4.7}$$

$$I' \coloneqq \{(\gamma_0,\gamma) \in I_0 \times I : \gamma_0^\dagger\gamma_0 = \gamma^\dagger\gamma\}. \tag{3.4.8}$$

We have $I_0 = F^\times$ (canonically). Using this identification, there is an isomorphism $I' \to I_0 \times I_1$ given by $(\gamma_0,\gamma) \mapsto (\gamma_0, \gamma_0^{-1}\gamma)$. We have actions

$$\begin{aligned} &I \circlearrowright \mathcal{N}(n-r,r)' && I' \circlearrowright \mathcal{N}' \\ &(X,\iota,\lambda,\rho) \mapsto (X,\iota,\lambda,\gamma\circ\rho) && (X_0,\iota_0,\lambda_0,\rho_0,X,\iota,\lambda,\rho) \mapsto (X_0,\iota_0,\lambda_0,\gamma_0\circ\rho_0,X,\iota,\lambda,\gamma\circ\rho). \end{aligned} \tag{3.4.9}$$

These actions are compatible with the action from Section 3.3, via the isomorphisms $I' \cong I_0 \times I_1$ and $\mathcal{N}' \cong \mathcal{N}(1,0)' \times \mathcal{N}(n-r,r)$. We have an isomorphism

$$I_0 \xrightarrow{\sim} GU(\mathbf{V}_0) \tag{3.4.10}$$

where $\gamma_0 \in I_0$ acts on $\mathbf{V}_0$ as $x \mapsto \gamma_0 \circ x$.

For any subset $L \subseteq \mathbf{W}$ with associated local special cycles $\mathcal{Z}'(L) \subseteq \mathcal{N}'$, the action of I' satisfies

$$(\gamma_0,\gamma)(\mathcal{Z}(L)') = \mathcal{Z}(\gamma L \gamma_0^{-1})'. \tag{3.4.11}$$

3.5. **Discrete reduced subschemes.** In the nonsplit cases (at least if $p \neq 2$), the reduced subscheme $\mathcal{N}(n-1,1)_{\mathrm{red}}$ of $\mathcal{N}(n-1,1)$ admits a stratification by Deligne–Lusztig varieties, described by a certain Bruhat–Tits building [VW11; Wu16]. Later, we will use these results implicitly via citation to [LZ22a; LL22].

In this section, we further discuss some cases where the reduced subscheme is discrete (continuing to allow $p = 2$ if $F/\mathbb{Q}_p$ is unramified).

In the split case, set

$$\mathbf{L} \coloneqq \mathrm{Hom}_{\mathcal{O}_F}(\mathbf{X}_0, \mathbf{X}) \qquad \mathbf{L}^\perp \coloneqq \mathrm{Hom}_{\mathcal{O}_F}(\mathbf{X}_0^\sigma, \mathbf{X}). \tag{3.5.1}$$

In the nonsplit case, define $\mathbf{L}$ in the same way but take $\mathbf{L}^\perp \coloneqq 0$. Let $K_{1,\mathbf{L}} \subseteq U(\mathbf{W})$ and $K_{1,\mathbf{L}^\perp} \subseteq U(\mathbf{W}^\perp)$ be the respective stabilizers.

Lemma 3.5.1. *Take signature $(n-r,r) = (1,1)$ if $F/\mathbb{Q}_p$ is nonsplit, and take any signature $(n-r,r)$ if $F/\mathbb{Q}_p$ is split.*

(1) The framing object $(\mathbf{X}, \iota_{\mathbf{X}}, \lambda_{\mathbf{X}})$ is unique up to isomorphism. This also holds over any algebraically closed field κ over $\overline{k}$, at least if $F/\mathbb{Q}_p$ is unramified.

(2) The reduced scheme $\mathcal{N}(n-r,r)_{\mathrm{red}}$ is discrete, i.e. a disjoint union of copies of $\operatorname{Spec}\overline{k}$. If $F/\mathbb{Q}_p$ is inert, then $\mathcal{N}(1,1)_{\mathrm{red}}$ is one point. If $F/\mathbb{Q}_p$ is ramified, then $\mathcal{N}(1,1)_{\mathrm{red}}$ is two points.

(3) The lattices $\mathbf{L} \subseteq \mathbf{W}$ and $\mathbf{L}^\perp \subseteq \mathbf{W}^\perp$ are maximal integral lattices. In the nonsplit cases, $\mathbf{L} \subseteq \mathbf{W} = \mathbf{V}$ is the unique maximal integral lattice.

(4) The group $I_1 \cong U(\mathbf{W}) \times U(\mathbf{W}^\perp)$ acts transitively on $\mathcal{N}(n-r,r)(\overline{k})$. Consider the resulting surjection

$$\begin{aligned} \mathcal{N}(n-r,r)(\overline{k}) &\longrightarrow U(\mathbf{W})/K_{1,\mathbf{L}} \times U(\mathbf{W}^\perp)/K_{1,\mathbf{L}^\perp} \\ (\mathbf{X}, \iota_{\mathbf{X}}, \lambda_{\mathbf{X}}, (\gamma, \gamma^\perp)) &\longmapsto (\gamma, \gamma^\perp). \end{aligned} \tag{3.5.2}$$

If $F/\mathbb{Q}_p$ is unramified, this map is a bijection. If $F/\mathbb{Q}_p$ is ramified, this map is 2-to-1. If $F/\mathbb{Q}_p$ is nonsplit, the set $U(\mathbf{W})/K_{1,\mathbf{L}} \times U(\mathbf{W}^\perp)/K_{1,\mathbf{L}^\perp}$ has size 1.

(5) Consider the bijective identification

$$U(\mathbf{W})/K_1(\mathbf{L}) \times U(\mathbf{W}^\perp)/K_{1,\mathbf{L}^\perp} \longleftrightarrow \left\{ \begin{array}{l} \text{maximal full-rank integral } \mathcal{O}_F\text{-lattices } N \subseteq \mathbf{V} \\ \text{where } N = M \oplus M^\perp \text{ with} \\ M \subseteq \mathbf{W} \text{ and } M^\perp \subseteq \mathbf{W}^\perp \end{array} \right\}$$

$$(\gamma, \gamma^\perp) \longmapsto \gamma(\mathbf{L}) \oplus \gamma^\perp(\mathbf{L}^\perp).$$

Given any subset $L \subseteq \mathbf{W}$, the subset $\tilde{\mathcal{Z}}(L)(\overline{k}) \subseteq \mathcal{N}(n-r,r)(\overline{k})$ is identified (via (3.5.2)) with the pre-image of the set of lattices $\{N : L \subseteq N\}$.

Proof.

(1) In the inert case, this follows from Dieudonné theory as in [Vol10, Proposition 1.10] (but we allow $p = 2$ by the same method), diagonalizability of Hermitian $\mathcal{O}_F$-lattices, and the

following fact: consider the rank 2 Hermitian $\mathcal{O}_F$-lattice Λ with pairing $(-,-)$ specified by the Gram matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}, \tag{3.5.3}$$

and also write $(-,-)$ for the induced pairing on $\Lambda \otimes_{\mathcal{O}_F} W(\kappa)[1/p]$ (which is "sesquilinear" for the Frobenius on $W(\kappa)[1/p]$). If $x \in \Lambda \otimes_{\mathcal{O}_F} W(\kappa)[1/p]$ is any element with $(x,x) \in W(\kappa)$, then $x \in \Lambda \otimes_{\mathcal{O}_F} W(\kappa)$ (so $\Lambda \otimes_{\mathcal{O}_F} W(\kappa)$ satisfies a certain "unique maximal integral lattice" property). This computation shows that $\mathcal{N}(1,1)(\overline{k})$ is a single point if $F/\mathbb{Q}_p$ is inert.

The ramified case follows from [RSZ17, Lemma 6.1]. The split case was already verified in Lemma 3.1.2.

(2) In the unramified case, part (1) implies $\mathcal{N}(n-r,r)(\overline{k}) \to \mathcal{N}(n-r,r)(\kappa)$ is bijective for every algebraically closed field extension κ over $\overline{k}$. Discreteness then follows because $\mathcal{N}(n-r,r)_{\mathrm{red}} \to \operatorname{Spec} \overline{k}$ is locally of finite type. If $F/\mathbb{Q}_p$ is inert, $\mathcal{N}(1,1)(\overline{k})$ being a single point was already discussed above. The ramified case is [RSZ17, Lemma 6.1].

(3) By part (1), we may assume $(\mathbf{X}, \iota_{\mathbf{X}}, \lambda_{\mathbf{X}})$ is the explicit tuple constructed in Section 3.1. The claim can then be verified explicitly, using (3.1.5) and the surrounding discussion. In the split case, $\mathbf{L}$ and $\mathbf{L}^\perp$ will be self-dual. In the inert case, $\mathbf{L}$ admits a basis with Gram matrix $\mathrm{diag}(1,p)$. In the ramified case, $\mathbf{L}$ admits a Gram matrix with basis $\mathrm{diag}(1,-a)$ for some $a \in \mathbb{Z}_p^\times$ which is a non-norm, i.e. $a \notin N_{F/\mathbb{Q}_p}(F^\times)$. This claim in the ramified case follows from the observation that $\mathbf{L}$ is integral and that $\mathbf{L} \subseteq \mathrm{Hom}_{\mathcal{O}_F}(\mathbf{X}_0, \mathbf{X}_0 \times \mathbf{X}_0^\sigma) \subseteq \varpi^{-1}\mathbf{L}$ (as the isogeny in (3.1.2) has kernel contained in the ϖ-torsion subgroup).

(4) Transitivity of the I_1 action is immediate from part (1). Note also $\mathrm{End}(\mathbf{X}) \cap (U(\mathbf{W}) \times U(\mathbf{W}^\perp)) \subseteq K_{1,\mathbf{L}} \times K_{1,\mathbf{L}^\perp}$ so the displayed map is well-defined. In the nonsplit cases, the assertions follow from parts (2) and (3). Bijectivity in the split case follows because we then have $\mathrm{End}(\mathbf{X}) \cap (U(\mathbf{W}) \times U(\mathbf{W}^\perp)) = K_{1,\mathbf{L}} \times K_{1,\mathbf{L}^\perp}$.

(5) Follows from the previous parts, i.e. $\tilde{\mathcal{Z}}(L)(\overline{k})$ corresponds to $(\gamma, \gamma^\perp)$ such that $L \subseteq \gamma(\mathbf{L})$. □

Suppose $F/\mathbb{Q}_p$ is split and $L \subseteq \mathbf{W}$ is any subset (with arbitrary signature $(n-r,r)$). We take $\mathcal{Z}(L) \subseteq \tilde{\mathcal{Z}}(L)$ to be the open and closed subfunctor corresponding (via Lemma 3.5.1(4)) to the locus where $\gamma^\perp(\mathbf{L}^\perp) = \mathbf{L}^\perp$. By the previous discussion, there is an isomorphism of formal schemes

$$\begin{aligned} \tilde{\mathcal{Z}}(L) &\longrightarrow \mathcal{Z}(L) \times U(\mathbf{W}^\perp)/K_{1,\mathbf{L}^\perp} \\ (\mathbf{X}, \iota_{\mathbf{X}}, \lambda_{\mathbf{X}}, (\gamma, \gamma^\perp)) &\longmapsto ((\mathbf{X}, \iota_{\mathbf{X}}, \lambda_{\mathbf{X}}, (\gamma, 1)), \gamma^\perp). \end{aligned} \tag{3.5.4}$$

In this case, we have a canonical bijection

$$\mathcal{Z}(L)(\overline{k}) = \{M \subseteq \mathbf{W} : \text{full rank self-dual lattice with } L \subseteq M\} \tag{3.5.5}$$

via Lemma 3.5.1.

Lemma 3.5.2. *Suppose $F/\mathbb{Q}_p$ is split. If $L \subseteq \mathbf{W}$ is an $\mathcal{O}_F$-lattice of full rank (i.e. rank $n-r$), then $\mathcal{Z}(L)(\overline{k})$ is a finite set.*

Proof. Our task is to show that the right-hand side of (3.5.5) is finite. For such M, we must have $L \subseteq M \subseteq M^\vee \subseteq L^\vee$ where $L^\vee$ and $M^\vee$ denote the dual lattices. If $L \not\subseteq L^\vee$ then $\mathcal{Z}(L)$ is empty. Otherwise, $L^\vee/L$ is an $\mathcal{O}_F$-module of finite length, so there are only finitely many possibilities for M. □

3.6. **Horizontal and vertical decomposition.** For a locally Noetherian formal scheme $\mathcal{X}$, viewed as a ringed space with structure sheaf $\mathcal{O}_\mathcal{X}$, we write

$$K_0'(\mathcal{X}) \coloneqq K_0(Coh(\mathcal{O}_\mathcal{X})) \qquad F_d K_0'(\mathcal{X}) \subseteq K_0'(\mathcal{X}) \tag{3.6.1}$$

for the K_0 group of coherent $\mathcal{O}_\mathcal{X}$-modules and the subgroup generated by coherent sheaves supported in (formal scheme-theoretic) dimension $\leq d$, respectively. If $\mathcal{X}$ is moreover formally locally of finite type over $\mathrm{Spf}\, R$ for a complete discrete valuation ring R, we say that $\mathcal{X}$ is *equidimensional of dimension n* if every open formal subscheme of $\mathcal{X}$ has dimension n. In this case, if $\mathcal{Z} \to \mathcal{X}$ is an adic finite morphism of locally Noetherian formal schemes, we write

$$F^m_\mathcal{X} K_0'(\mathcal{Z}) \coloneqq F_{n-m} K_0'(\mathcal{Z}) \qquad \mathrm{gr}^m_\mathcal{X} K_0'(\mathcal{Z}) \coloneqq F^m_\mathcal{X} K_0'(\mathcal{Z}) / F^{m+1}_\mathcal{X} K_0'(\mathcal{Z}). \tag{3.6.2}$$

We often work with these groups tensor $\mathbb{Q}$, written as $K_0'(\mathcal{X})_\mathbb{Q}$, etc.

The discussion in Section 3.1 implies that the Rapoport–Zink space $\mathcal{N}(n-r, r)$ is equidimensional of dimension $(n-r)r+1$. For the rest of Section 3.6 we fix signature $(n-1, 1)$ and use the shorthand $\mathcal{N} \coloneqq \mathcal{N}(n-1, 1)$. The material below is a local analogue of [Che24b, Section 3.6].

Assume $F/\mathbb{Q}_p$ is nonsplit for the moment. For any nonzero $x \in \mathbf{W}$, the local special cycle $\mathcal{Z}(x)$ is a Cartier divisor on $\mathcal{N}$, by [KR11, Proposition 3.5] (inert), [How19, Proposition 4.3] (inert allowing $p = 2$), and [LL22, Lemma 2.40] (ramified exotic smooth). For any $x \in \mathbf{W}$, set

$${}^D\mathcal{Z}(x) \coloneqq \begin{cases} \mathcal{O}_{\mathcal{Z}(x)} & \text{if } x \neq 0 \\ (\cdots 0 \to \mathcal{E} \xrightarrow{0} \mathcal{O}_\mathcal{N} \to 0 \cdots) & \text{if } x = 0 \end{cases} \tag{3.6.3}$$

in $D^b_{Coh(\mathcal{O}_{\mathcal{Z}(x)})}(\mathcal{O}_\mathcal{N})$ (bounded derived category of $\mathcal{O}_\mathcal{N}$-modules with cohomology sheaves coherent and supported along $\mathcal{O}_{\mathcal{Z}(x)}$), where the $\mathcal{O}_\mathcal{N}$ term is in degree 0. For any tuple $\underline{x} \in \mathbf{W}^m$, we then consider the *derived local special cycle*

$${}^D\mathcal{Z}(\underline{x}) \coloneqq {}^D\mathcal{Z}(x_1) \otimes^\mathbb{L} \cdots \otimes^\mathbb{L} {}^D\mathcal{Z}(x_m) \in D^b_{Coh(\mathcal{O}_{\mathcal{Z}(\underline{x})})}(\mathcal{O}_\mathcal{N}). \tag{3.6.4}$$

We write ${}^\mathbb{L}\mathcal{Z}(\underline{x}) \in K_0'(\mathcal{Z}(\underline{x}))_\mathbb{Q}$ for the image of ${}^D\mathcal{Z}(\underline{x})$, and have ${}^\mathbb{L}\mathcal{Z}(\underline{x}) \in F^m_\mathcal{N} K_0'(\mathcal{Z}(\underline{x}))_\mathbb{Q}$ by multiplicativity of the codimension filtration[7] and depends only on $\mathrm{span}_{\mathcal{O}_F}(\underline{x})$ ("linear invariance") by [How19, Theorem B] (inert) and [LL22, Proposition 2.33] (ramified exotic smooth).

Continuing to assume $F/\mathbb{Q}_p$ is nonsplit, assume $\underline{x} \in \mathbf{W}^m$ spans a non-degenerate Hermitian $\mathcal{O}_F$-lattice of rank $m^\flat$. We define certain *derived vertical local special cycles* ${}^\mathbb{L}\mathcal{Z}(\underline{x})_\mathscr{V} \in \mathrm{gr}^m_\mathcal{N} K_0'(\mathcal{Z}(\underline{x})_{\overline{k}})_\mathbb{Q}$ as follows.

[7] There is a technicality here, as $\mathcal{N}$ is a formal scheme rather than a scheme. So we instead prove filtration multiplicativity via uniformization ([Che24b, Corollary 6.7.8]) by reducing to the analogous filtration multiplicativity statement for global special cycles. We will make a few other forward references to [Che24b, Section 6.7] where we verify some properties of local special cycles via uniformization.

For integers $e \gg 0$, we have a scheme-theoretic union decomposition (intersect ideal sheaves)

$$\mathcal{Z}(\underline{x}) = \mathcal{Z}(\underline{x})_{\mathscr{H}} \cup \mathcal{Z}(\underline{x})_{\mathscr{V}} \tag{3.6.5}$$

by [Che24b, Lemma 6.7.5], where $\mathcal{Z}(\underline{x})_{\mathscr{H}}$ is the flat part of $\mathcal{Z}(\underline{x})$, i.e. the largest closed formal subscheme which is flat over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$, and $\mathcal{Z}(\underline{x})_{\mathscr{V}} \coloneqq \mathcal{Z}(\underline{x})_{\operatorname{Spf} \mathcal{O}_{\breve{F}}/p^e}$ for $e \gg 0$. Since $\mathcal{Z}(\underline{x})_{\mathscr{H}}$ is equidimensional of dimension $n - m^\flat$ ([Che24b, Lemma 6.7.4]), and since $\mathcal{Z}(\underline{x})_{\mathscr{H}} \cap \mathcal{Z}(\underline{x})_{\mathscr{V}}$ has dimension $\leq n - m^\flat - 1$, there is an induced decomposition

$$\operatorname{gr}_{\mathcal{N}}^{m^\flat} K_0'(\mathcal{Z}(\underline{x})) = \operatorname{gr}_{\mathcal{N}}^{m^\flat} K_0'(\mathcal{Z}(\underline{x})_{\mathscr{H}}) \oplus \operatorname{gr}_{\mathcal{N}}^{m^\flat} K_0'(\mathcal{Z}(\underline{x})_{\overline{k}}) \tag{3.6.6}$$

independent of e (cf. [Zha21, Lemma B.1][8]). Here we have used the pushforward dévissage isomorphism $K_0'(\mathcal{Z}(\underline{x})_{\overline{k}}) \xrightarrow{\sim} K_0'(\mathcal{Z}(\underline{x})_{\mathscr{V}})$ for K_0' groups.

If $m = m^\flat$, we define ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}}$ to be given by the projection

$$\begin{aligned} \operatorname{gr}_{\mathcal{N}}^{m} K_0'(\mathcal{Z}(\underline{x}))_{\mathbb{Q}} &\longrightarrow \operatorname{gr}_{\mathcal{N}}^{m} K_0'(\mathcal{Z}(\underline{x})_{\overline{k}})_{\mathbb{Q}} \\ {}^{\mathbb{L}}\mathcal{Z}(\underline{x}) &\longmapsto {}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}}. \end{aligned} \tag{3.6.7}$$

By the linear invariance property for ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})$ discussed above, the class ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}}$ depends only on $\operatorname{span}_{\mathcal{O}_F}(\underline{x})$.

For possibly $m \neq m^\flat$, we say that $\underline{x} = [x_1, \dots, x_m]$ is in *minimal form* if $\underline{x}^\flat \coloneqq [x_{m-m^\flat+1}, \dots, x_m]$ satisfies $\operatorname{span}_{\mathcal{O}_F}(\underline{x}^\flat) = \operatorname{span}_{\mathcal{O}_F}(\underline{x})$. In this case, set $\underline{x}^\# \coloneqq [x_1, \dots, x_{m-m^\flat}]$ and define[9]

$${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}} \coloneqq {}^{\mathbb{L}}\mathcal{Z}(\underline{x}^\#) \cdot {}^{\mathbb{L}}\mathcal{Z}(\underline{x}^\flat)_{\mathscr{V}} \in \operatorname{gr}_{\mathcal{N}}^{m} K_0'(\mathcal{Z}(\underline{x})_{\overline{k}})_{\mathbb{Q}}. \tag{3.6.8}$$

For $\underline{x}$ possibly not in minimal form, select any $\gamma \in \mathrm{GL}_m(\mathcal{O}_F)$ such that $\underline{x} \cdot \gamma$ is in minimal form, and set ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}} \coloneqq {}^{\mathbb{L}}\mathcal{Z}(\underline{x} \cdot \gamma)_{\mathscr{V}}$ (note $\mathcal{Z}(\underline{x}) = \mathcal{Z}(\underline{x} \cdot \gamma)$).

We claim that ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}}$ depends only on m and $\operatorname{span}_{\mathcal{O}_F}(\underline{x})$, and not on the choice of $\underline{x}$ or a minimal form ("linear invariance"). For $\underline{x}$ in minimal form and with notation as above, we already explained that ${}^{\mathbb{L}}\mathcal{Z}(\underline{x}^\flat)_{\mathscr{V}}$ depends only on $\operatorname{span}_{\mathcal{O}_F}(\underline{x})$. Recall $\mathcal{Z}(\underline{x}) = \mathcal{Z}(\underline{x}^\flat)$. Consider any element x_i of the tuple $\underline{x}^\#$. Then $x_i \in \operatorname{span}_{\mathcal{O}_F}(\underline{x}^\flat) = \operatorname{span}_{\mathcal{O}_F}(\underline{x})$. In particular, $\mathcal{Z}(\underline{x}^\flat) \subseteq \mathcal{Z}(x_i)$. But Grothendieck–Messing theory provides a canonical isomorphism

$$\mathcal{E}|_{\mathcal{Z}(x_i)} \xrightarrow{\sim} \mathcal{I}(x_i)/\mathcal{I}(x_i)^2 \tag{3.6.9}$$

if $x_i \neq 0$, where $\mathcal{I}(x_i) \subseteq \mathcal{O}_{\mathcal{N}}$ is the ideal sheaf of the Cartier divisor $\mathcal{Z}(x_i) \subseteq \mathcal{N}$ (follows from [How19, Definition 4.2] (inert) and [LL22, Lemma 2.39]). In other words, (3.6.9) provides a canonical isomorphism between the complexes

$$(\cdots 0 \to \mathcal{E} \xrightarrow{0} \mathcal{O}_{\mathcal{N}} \to 0 \cdots) \quad \text{and} \quad (\cdots 0 \to \mathcal{I}(x_i) \to \mathcal{O}_{\mathcal{N}} \to 0 \cdots) \tag{3.6.10}$$

[8]Strictly speaking, our setup for K_0' groups may be different from Zhang's in non quasi-compact settings. The proof of the cited lemma is the same in our setup.

[9]One needs to show that the map $\alpha \mapsto {}^{\mathbb{L}}\mathcal{Z}(\underline{x}^\#) \cdot \alpha$ sends $F_{\mathcal{N}}^{m^\flat+1} K_0'(\mathcal{Z}(\underline{x})_{\overline{k}})_{\mathbb{Q}} \to F_{\mathcal{N}}^{m+1} K_0'(\mathcal{Z}(\underline{x})_{\overline{k}})_{\mathbb{Q}}$. This is clear if $m^\flat \geq n-1$, but we do not know a proof of this in general as $\mathcal{N}$ is a formal scheme and not a scheme. Since we are mostly interested in the case $m^\flat \geq n-1$, we do not pursue this point further. Even when $m^\flat = n-1$ and $m = n$, one still needs to check that ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}}$ lies in $F_{\mathcal{N}}^{n}(\mathcal{Z}(\underline{x})_{\overline{k}})_{\mathbb{Q}}$ (rather than $F_{\mathcal{N}}^{n-1}(\mathcal{Z}(\underline{x})_{\overline{k}})_{\mathbb{Q}}$). This follows e.g. because $\mathcal{Z}(\underline{x})_{\overline{k}}$ is a Noetherian scheme ([Che24b, Lemma 6.7.3]) whose reduced irreducible components are projective over $\overline{k}$. The definition of ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}}$ should thus be treated as conditional unless $m = m^\flat$ or $m^\flat \geq n-1$.

after tensoring with $\mathcal{O}_{\mathcal{Z}(\underline{x}^\flat)}$. We thus have a canonical identification

$$ {}^D\mathcal{Z}(\underline{x}^\#)|_{\mathcal{Z}(\underline{x}^\flat)} \cong {}^D\mathcal{Z}(\underline{0}_{m-m^\flat})|_{\mathcal{Z}(\underline{x}^\flat)} \tag{3.6.11} $$

as elements of $D^b_{Coh}(\mathcal{O}_{\mathcal{Z}(\underline{x}^\flat)})$, where both sides are computed by derived pullback, and where $\underline{0}_{m-m^\flat} \in \mathbf{W}^{m-m^\flat}$ is the tuple with all entries equal to 0. The formula in (3.6.11) implies

$$ {}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}} = {}^{\mathbb{L}}\mathcal{Z}(\underline{0}_{m-m^\flat}) \cdot {}^{\mathbb{L}}\mathcal{Z}(\underline{x}^\flat)_{\mathscr{V}} \in \mathrm{gr}^m_{\mathcal{N}} K_0'(\mathcal{Z}(\underline{x})_{\overline{k}})_{\mathbb{Q}}. \tag{3.6.12} $$

We have explained that the right-hand side does not depend on any auxiliary choices.

Next, suppose $F/\mathbb{Q}_p$ is split, and assume $\underline{x} \in \mathbf{W}^m$ has $\mathcal{O}_F$-span which is a lattice of rank $n-1$ (full rank). We have $\mathrm{gr}^{n-1}_{\mathcal{N}} K_0'(\mathcal{Z}(\underline{x})_{\overline{k}}) = 0$ for dimension reasons (the reduced subscheme of $\mathcal{N}$ is dimension 0, see Section 3.5 and [Che24b, Lemma 6.7.3]). Constructing ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}} \in \mathrm{gr}^m_{\mathcal{N}} K_0'(\mathcal{Z}(\underline{x})_{\overline{k}})$ as above gives the *derived vertical local special cycle* ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}} = 0$.

Next, consider $\underline{x} \in \mathbf{W}^m$ which is a basis for its $\mathcal{O}_F$-span $L^\flat := \mathrm{span}_{\mathcal{O}_F}(\underline{x})$. If $F/\mathbb{Q}_p$ is split, we also assume $m = n-1$. In this situation, we set ${}^{\mathbb{L}}\mathcal{Z}(L^\flat)_{\mathscr{V}} := {}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}}$, since the latter depends only on $L^\flat$. If $n = 2$ and $m = 1$, we have ${}^{\mathbb{L}}\mathcal{Z}(\underline{x})_{\mathscr{V}} = 0$ since the reduced subscheme $\mathcal{N}_{\mathrm{red}}$ has dimension 0 (Section 3.5) and since $\mathcal{N}$ has dimension 2 in this case.

Given any proper scheme $\mathcal{Z} \to \mathrm{Spec}\,\overline{k}$ with a closed immersion $\mathcal{Z} \to \mathcal{N}$, any $x \in \mathbf{W}$ defines a map

$$ \begin{aligned} \mathrm{gr}^{n-1}_{\mathcal{N}} K_0'(\mathcal{Z}) &\longrightarrow \mathrm{gr}^n_{\mathcal{N}} K_0'(\mathcal{Z}) \\ \alpha &\longmapsto \alpha \cdot {}^{\mathbb{L}}\mathcal{Z}(x) \end{aligned} \tag{3.6.13} $$

defined by derived tensor product over $\mathcal{O}_{\mathcal{N}}$, descended to K_0'-groups, where we use the fact that $\mathcal{Z}$ is a scheme for compatibility with codimension filtrations (then apply e.g. [SProject, Lemma 0FD0]). Equivalently, we have

$$ \alpha \cdot {}^{\mathbb{L}}\mathcal{Z}(x) = \begin{cases} \alpha - \alpha \otimes \mathcal{I}(x) & \text{if } x \neq 0 \\ \alpha - \alpha \otimes \mathcal{E} & \text{if } x = 0 \end{cases} \tag{3.6.14} $$

as a difference of classes in $K_0'(\mathcal{Z})$, abusing notation and picking any lift of α to $K_0'(\mathcal{Z})$. In the special case of $x = 0$, we abuse notation and write $\alpha \cdot \mathcal{E}^\vee := \alpha \cdot {}^{\mathbb{L}}\mathcal{Z}(0)$.

There is a degree map given by the composition

$$ \deg_{\overline{k}} \colon \mathrm{gr}^n_{\mathcal{N}} K_0'(\mathcal{Z})_{\mathbb{Q}} \xrightarrow{\sim} \mathrm{gr}^n_{\mathcal{N}} K_0'(\mathcal{Z}_{\overline{k}})_{\mathbb{Q}} \to \mathrm{gr}_0 K_0'(\mathrm{Spec}\,\overline{k})_{\mathbb{Q}} = \mathbb{Q} \tag{3.6.15} $$

where the first arrow is induced by the dévissage pushforward isomorphism $K_0'(\mathcal{Z}_{\overline{k}}) \to K_0'(\mathcal{Z})$ and the second arrow is pushforward along $\mathcal{Z}_{\overline{k}} \to \mathrm{Spec}\,\overline{k}$ (e.g. induced by taking Euler characteristics of coherent sheaves on $\mathcal{Z}_{\overline{k}}$).

At the end of Section 7.5, we apply this after tensoring with $\mathbb{Q}$, in the special case of $\alpha = {}^{\mathbb{L}}\mathcal{Z}(L^\flat)_{\mathscr{V}}$ and $\mathcal{Z} = \mathcal{Z}(L^\flat)_{\overline{k}}$ for $L^\flat \subseteq \mathbf{W}$ being a non-degenerate lattice of rank $n-1$, where the properness of such $\mathcal{Z}(L^\flat)_{\overline{k}}$ follows from [Che24b, Lemma 6.7.3].

3.7. **Serre tensor and signature** $(1,1)$**.** The case of signature $(1,1)$ plays an important role for describing local special cycles via the Serre tensor construction.

As above, let $\mathbf{X}_0$ be the unique supersingular p-divisible group over $\overline{k}$ of height 2 and dimension 1 if $F/\mathbb{Q}_p$ is nonsplit, and similarly with "supersingular" replaced by "ordinary" if $F/\mathbb{Q}_p$ is split. For schemes S over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$, we consider pairs (X,ρ) where X is a p-divisible group over S and $\rho\colon X_{\overline{S}} \to \mathbf{X}_{0,\overline{S}}$ is any quasi-isogeny.

We form the Rapoport–Zink space $\widetilde{\mathcal{N}}_{2,1}$ over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$, given by

$$\widetilde{\mathcal{N}}_{2,1}(S) \coloneqq \{\text{isomorphism classes of framed tuples } (X,\rho) \text{ over } S\}. \tag{3.7.1}$$

This is a locally Noetherian formal scheme which is formally locally of finite type over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$ via the standard representability result [RZ96, Theorem 2.16]. There is a bijection $\operatorname{Isog}^0(\mathbf{X}_0) \to \widetilde{\mathcal{N}}_{2,1}(\overline{k})$ given by $\rho \mapsto (\mathbf{X}_0,\rho)$. Indeed, this follows as in the proof of Lemma 3.5.1 by uniqueness of $\mathbf{X}_0$ over any algebraically closed field κ (any quasi-endomorphism of $\mathbf{X}_0$ descends to $\overline{k}$ as well, as may be checked on isocrystals).

We let $\mathcal{N}_{2,1} \subseteq \widetilde{\mathcal{N}}_{2,1}$ be the open and closed locus where the framing ρ is fiberwise an isomorphism. Then $\mathcal{N}_{2,1}$ is representable by a formal scheme, and there is a (non-canonical) isomorphism $\mathcal{N}_{2,1} \cong \operatorname{Spf}\mathcal{O}_{\breve{F}}[\![t]\!]$ (e.g. by Grothendieck–Messing theory).

For arbitrary signature $(n-r,r)$ in the split case, we can form $\widetilde{\mathcal{N}}_{n,r}$ and $\mathcal{N}_{n,r}$ as above, where we replace $\mathbf{X}_0$ with the unique ordinary p-divisible group of height n and dimension r. The previous assertions for $\widetilde{\mathcal{N}}_{2,1}$ and $\mathcal{N}_{2,1}$ hold in this case as well, except we now have $\mathcal{N}_{n,r} \cong \operatorname{Spf}\mathcal{O}_{\breve{F}}[\![t_1,\dots,t_{(n-r)r}]\!]$ (again by Grothendieck–Messing theory).

Lemma 3.7.1. *Given any $(X,\rho) \in \mathcal{N}_{2,1}(S)$, any principal polarization of $\mathbf{X}_0$ lifts uniquely to a principal polarization on X.*

Proof. Uniqueness follows from Drinfeld rigidity. Any two principal polarizations on X differ by $\mathbb{Z}_p^\times$ scalar (since this holds for $\mathbf{X}_0$), so it is enough to show existence of a principal polarization on X. Since $\mathcal{N}_{2,1} \cong \operatorname{Spf}\mathcal{O}_{\breve{F}}[\![t]\!]$, it is enough to check the case where the scheme S is a finite order thickening of $\operatorname{Spec}\overline{k}$. By Serre–Tate, we can view X as the p-divisible group of an elliptic curve over S (deforming an elliptic curve over $\operatorname{Spec}\overline{k}$ with p-divisible group $\mathbf{X}_0$). Any elliptic curve admits a (unique) principal polarization. □

The preceding (possibly standard) argument also appeared in the proof of [RSZ17, Proposition 6.3] (for the same purpose), there in the supersingular case.

Recall the triple $(\mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F, \iota, \lambda_{\mathbf{X}_0} \otimes \lambda_{\mathrm{tr}})$ described in Section 3.1, arising from the Serre tensor construction. For any $(X,\rho) \in \mathcal{N}_{2,1}(S)$, the same construction gives a tuple $(X \otimes_{\mathbb{Z}_p} \mathcal{O}_F, \iota, \lambda_{\mathbf{X}_0} \otimes \lambda_{\mathrm{tr}}, \rho\otimes 1)$ where $\lambda_{\mathbf{X}_0}$ also denotes the unique lift of $\lambda_{\mathbf{X}_0}$ from $\mathbf{X}_0$ to X as in Lemma 3.7.1 (by abuse of notation), and where $\rho\otimes 1\colon X_{\overline{S}} \otimes_{\mathbb{Z}_p} \mathcal{O}_F \to \mathbf{X}_{0,\overline{S}} \otimes_{\mathbb{Z}_p} \mathcal{O}_F$.

Lemma 3.7.2 (Serre tensor isomorphism)**.** *For any F-linear quasi-isogeny $\phi\colon \mathbf{X}_0 \otimes_{\mathbb{Z}_p} \mathcal{O}_F \to \mathbf{X}$ preserving polarizations exactly, the induced map*

$$\begin{aligned} \mathcal{N}_{2,1} &\longrightarrow \mathcal{N}(1,1) \\ (X,\rho) &\longmapsto (X \otimes_{\mathbb{Z}_p} \mathcal{O}_F, \iota, -\mathfrak{d}^{-2}\cdot(\lambda_{\mathbf{X}_0}\otimes\lambda_{\mathrm{tr}}), \phi_{\overline{S}}\circ(\rho\otimes 1)) \end{aligned} \tag{3.7.2}$$

(defined on S-points for schemes S over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$) is an open and closed immersion whose set-theoretic image is a single point.

If $F/\mathbb{Q}_p$ is split, the inverse is given by restricting $(X,\iota,\lambda,\rho)\mapsto(X^-,(\phi_S^-)^{-1}\circ\rho^-)$ to the appropriate component of $\mathcal{N}(1,1)$.

Proof. For the ramified case, we refer to [RSZ17, Proposition 6.3]. In the unramified case, the lemma follows by identifying the deformation theory of $\mathcal{N}_{2,1}$ and $\mathcal{N}(1,1)$ using Grothendieck–Messing theory, using the eigenspace decomposition for the $\mathcal{O}_F$-action on the Dieudonné crystals of objects in $\mathcal{N}(1,1)(S)$ as in [Che24b, (2.5.4)] and surrounding discussion (so the deformation problem for $\mathcal{N}_{2,1}$ identifies with the deformation problem of the "$-$ eigenspace" of the Hodge filtration for objects in $\mathcal{N}(1,1)(S)$ in the notation of loc. cit.). This is essentially how we verified generic formal smoothness of special cycles in loc. cit.. □

When $F/\mathbb{Q}_p$ is split and the signature $(n-r,r)$ is arbitrary, recall that any $(X,\iota,\lambda,\rho)\in\mathcal{N}(n-r,r)$ admits a decomposition $X=X^+\times X^-$ and $\rho=\rho^+\times\rho^-$ where $\rho^\pm\colon X^\pm\to\mathbf{X}^\pm$ using the nontrivial idempotents $e^\pm\in\mathcal{O}_F$.

Lemma 3.7.3. *Suppose $F/\mathbb{Q}_p$ is split, and consider arbitrary signature $(n-r,r)$. For any formal scheme S over $\operatorname{Spf}\mathcal{O}_{\breve{F}}$, the forgetful functor*

$$\left\{\begin{array}{l}\text{groupoid of principally polarized}\\ \text{Hermitian } p\text{-divisible groups } (X,\iota,\lambda)\\ \text{over } S \text{ of signature } (n-r,r)\end{array}\right\}\longrightarrow\left\{\begin{array}{l}\text{groupoid of ordinary } p\text{-divisible groups}\\ \text{over } S \text{ of height } n \text{ and dimension } r\end{array}\right\}$$

$$(X,\iota,\lambda)\longmapsto X^- \tag{3.7.3}$$

is an equivalence of categories. The same holds if we consider the groupoids with morphisms being quasi-isogenies (rather than isomorphisms).

Proof. An explicit quasi-inverse is given by $(X^-)\mapsto(X,\iota,\lambda)$ (over a scheme S) with

$$\begin{aligned}&X=(X^-)^\vee\times X^- \\ &\iota(e^+)\colon X\to(X^-)^\vee \qquad \iota(e^-)\colon X\to X^- \qquad \text{projections}\\ &\lambda=\begin{pmatrix}0&1\\-1&0\end{pmatrix}:(X^-)^\vee\times X^-\to X^-\times(X^-)^\vee.\end{aligned} \tag{3.7.4}$$

This is analogous to the following phenomenon: if L is a free $\mathcal{O}_F$-module of rank n equipped with a perfect Hermitian pairing, then $U(L)\cong\mathrm{GL}(L^-)$ (and similarly with F instead of $\mathcal{O}_F$). □

Remark 3.7.4. Suppose $F/\mathbb{Q}_p$ is split, and consider $S=\operatorname{Spf}\mathcal{O}$ for $\mathcal{O}$ the ring of integers in a finite degree extension of $\breve{F}=\breve{\mathbb{Q}}_p$. Write $(\mathfrak{X}_0,\iota_{\mathfrak{X}_0},\lambda_{\mathfrak{X}_0})$ for the canonical lift (over S) as in Definition 3.1.6, forgetting the framing. Let X^- be any lift of $\mathbf{X}_0$ over S. Suppose $n\geq 2$, and consider $(\mathfrak{X}_0)^{n-2}\times(X^-\otimes_{\mathbb{Z}_p}\mathcal{O}_F)$, with the product $\mathcal{O}_F$-action and product polarization $(\lambda_{\mathfrak{X}_0})^{n-2}\times(\lambda_{\mathbf{X}_0}\otimes\lambda_{\mathrm{tr}})$.

Then, under the equivalence of Lemma 3.7.3 in the case of $r=1$, the object $(\mathfrak{X}_0)^{n-2}\times(X^-\otimes_{\mathbb{Z}_p}\mathcal{O}_F)$ maps to $\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-2}\times X^-$.

Remark 3.7.5. If we drop the ordinary hypothesis on both sides of Lemma 3.7.3, the lemma still holds (by the same proof).

Lemma 3.7.6. *Suppose $F/\mathbb{Q}_p$ is split, and form $\widetilde{\mathcal{N}}_{n,r}$ using the framing object $\mathbf{X}^-$. For arbitrary signature $(n-r,r)$, the forgetful map*

$$\begin{aligned} \mathcal{N}(n-r,r) &\longrightarrow \widetilde{\mathcal{N}}_{n,r} \\ (X,\iota,\lambda,\rho) &\longmapsto (X^-,\rho^-) \end{aligned} \tag{3.7.5}$$

is an isomorphism.

Proof. This is immediate from Lemma 3.7.3. Alternative (less elementary) proof: first observe that the forgetful map is an isomorphism on κ-points for any algebraically closed field κ over $\overline{k}$ (see Lemma 3.5.1 and above discussion). As in the proof of Lemma 3.7.2, the claim now follows from Grothendieck–Messing theory. □

Remark 3.7.7. In the situation of Lemma 3.7.6, the open and closed subfunctor $\mathcal{N}_{n,r} \subseteq \widetilde{\mathcal{N}}_{n,r} \cong \mathcal{N}(n-r,r)$ has $\mathcal{N}_{n,r}(\overline{k})$ being a singleton set, corresponding (via Lemma 3.5.1) to the lattice $\mathbf{L} \oplus \mathbf{L}^\perp \subseteq \mathbf{W} \oplus \mathbf{W}^\perp$ (i.e. the locus where the framing ρ is a fiberwise isomorphism).

For $F/\mathbb{Q}_p$ in all cases (inert, ramified, split) and for any $x \in \mathbf{W}$, the local special cycle $\mathcal{Z}(x) \to \mathcal{N}(1,1)$ pulls back along the Serre tensor isomorphism (Lemma 3.7.2) to a certain local special cycle on $\mathcal{N}_{2,1}$ associated with an element $x' \in \mathrm{Hom}^0(\mathbf{X}_0, \mathbf{X}_0)$ (arising from adjunction in the Serre tensor construction). The ramified case is explained in [RSZ17, §6.2]. The inert and split cases may be formulated in a similar way (we omit a more detailed statement, which we will not need). This may be viewed as a local version of [KR14, Proposition 14.5] (see also [Che24c, Section 9.2]).

For $F/\mathbb{Q}_p$ nonsplit (at least if $p \neq 2$), Kudla–Rapoport [KR11, Proposition 8.1] and Rapoport–Smithling–Zhang [RSZ17, Proposition 7.1] use this to describe $\mathcal{Z}(x)$ in terms of certain *quasi-canonical lifting cycles* on $\mathcal{N}_{2,1}$, corresponding to closed immersions $\mathrm{Spf}\, \mathcal{O}_{\breve{E}_s} \to \mathcal{N}_{2,1}$ associated with $(\mathfrak{X}_s, \rho) \in \mathrm{Spf}\, \mathcal{O}_{\breve{E}_s}$ where $(\mathfrak{X}_s, \rho)$ arises from a *quasi-canonical lifting* of $(\mathbf{X}_0, j)$ for suitable $j\colon \mathcal{O}_F \to \mathrm{End}(\mathbf{X}_0)$ (in the notation and sense of Section 5.2 below). This was extended by Li–Zhang [LZ22a] (inert) and Li–Liu [LL22] (ramified) to flat parts of 1-cycles in signature $(n-1,1)$, for arbitrary n in the inert case and even n in the ramified case. We will need this result, which we recall in Section 5.3 below (to the precision we need).

We will need an analogue of the previous paragraph when $F/\mathbb{Q}_p$ is split (allowing $p = 2$). This is accomplished in Section 4 below (statement given in Section 5.3). Our method in the split case is somewhat different from the proofs cited above.

4. More on moduli of p-divisible groups: split

Retain notation from Section 3. Throughout Section 4, we assume $F/\mathbb{Q}_p$ is split.

4.1. **Lifting theory for ordinary p-divisible groups.** We discuss lifting theory for ordinary p-divisible groups over an algebraically closed field κ of characteristic p. The case of height 2 dimension 1 ordinary p-divisible groups is discussed in [Mes72, Appendix]. We spell out the case

of general height and dimension (which reduces to the results in [Mes72, Appendix]). See also the exposition in [Meu07] (or the sketch in [Gro86, §6]), though we will need some additional material on homomorphisms between liftings.

Take integers $r_1, r_2 \geq 0$. The unique ordinary p-divisible group X over κ of height $r_1 + r_2$ and dimension r_1 is $X = \boldsymbol{\mu}_{p^\infty}^{r_1} \times \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2}$.

Next, let R be an adic Noetherian local ring (with maximal ideal being an ideal of definition) with residue field κ. The p-divisible groups $\boldsymbol{\mu}_{p^\infty}$ and $\underline{\mathbb{Q}_p/\mathbb{Z}_p}$ lift uniquely to $\operatorname{Spf} R$ (e.g. by Grothendieck–Messing deformation theory), which we still notate as $\boldsymbol{\mu}_{p^\infty}$ and $\underline{\mathbb{Q}_p/\mathbb{Z}_p}$. If $\mathfrak{X}$ is a lift of X over $\operatorname{Spf} R$, its connected-étale exact sequence must be $0 \to \boldsymbol{\mu}_{p^\infty}^{r_1} \to \mathfrak{X} \to \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2} \to 0$. Classifying lifts $\mathfrak{X}$ is thus the same as classifying such extensions, which are in canonical bijection with $\operatorname{Ext}^1_{\operatorname{Spf} R}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2}, \boldsymbol{\mu}_{p^\infty}^{r_1})$ (using also Drinfeld rigidity, as well as the fact $\operatorname{Hom}(\boldsymbol{\mu}_{p^\infty}, \underline{\mathbb{Q}_p/\mathbb{Z}_p}) = \operatorname{Hom}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}, \boldsymbol{\mu}_{p^\infty}) = 0$). Here, $\operatorname{Ext}^1_{\operatorname{Spf} R}$ is calculated in the abelian category of fppf sheaves of abelian groups over $\operatorname{Spf} R$ (this is also [Mes72, Appendix, Corollary (2.3)]). We typically suppress the R-dependence in $\operatorname{Hom}(-,-)$.

Applying $\operatorname{Hom}(-, \boldsymbol{\mu}_{p^\infty})$ to the short exact sequence of sheaves (fppf sheaves over $\operatorname{Spf} R$)

$$0 \longrightarrow \underline{\mathbb{Z}} \longrightarrow \underline{\mathbb{Z}[1/p]} \longrightarrow \underline{\mathbb{Q}_p/\mathbb{Z}_p} \longrightarrow 0 \tag{4.1.1}$$

gives a boundary morphism $\delta\colon \operatorname{Hom}(\underline{\mathbb{Z}}, \boldsymbol{\mu}_{p^\infty}) \to \operatorname{Ext}^1_{\operatorname{Spf} R}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}, \boldsymbol{\mu}_{p^\infty})$ in the associated long exact sequence. This map δ is an isomorphism [Mes72, Appendix, Proposition (2.5)].[10] By compatibility of Ext with finite direct sums, it follows that the boundary morphism $\delta\colon \operatorname{Hom}(\underline{\mathbb{Z}}^{r_2}, \boldsymbol{\mu}_{p^\infty}^{r_1}) \to \operatorname{Ext}^1_{\operatorname{Spf} R}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2}, \boldsymbol{\mu}_{p^\infty}^{r_1})$ is also an isomorphism.

Given an element $\alpha \in \operatorname{Hom}(\underline{\mathbb{Z}}^{r_2}, \boldsymbol{\mu}_{p^\infty}^{r_1})$, we can identify the extension corresponding to $\delta(\alpha)$ with the bottom row of the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \underline{\mathbb{Z}}^{r_2} & \longrightarrow & \underline{\mathbb{Z}[1/p]}^{r_2} & \longrightarrow & \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow & & \| & & \\ 0 & \longrightarrow & \boldsymbol{\mu}_{p^\infty}^{r_1} & \longrightarrow & \mathfrak{X} & \longrightarrow & \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2} & \longrightarrow & 0 \end{array} \tag{4.1.2}$$

where the rows are exact and the left square is a pushout. This follows from general homological algebra valid in any abelian category (e.g. [SProject, Section 010I] and [SProject, Section 06XP]).

Given $r_1, r_1', r_2, r_2' \in \mathbb{Z}_{\geq 0}$, we have

$$\begin{aligned} \operatorname{Hom}(\boldsymbol{\mu}_{p^\infty}^{r_1} \times \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2}, \boldsymbol{\mu}_{p^\infty}^{r_1'} \times \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2'}) &= \operatorname{Hom}(\boldsymbol{\mu}_{p^\infty}^{r_1}, \boldsymbol{\mu}_{p^\infty}^{r_1'}) \times \operatorname{Hom}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2}, \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2'}) \\ &\cong M_{r_1', r_1}(\mathbb{Z}_p) \times M_{r_2', r_2}(\mathbb{Z}_p), \end{aligned} \tag{4.1.3}$$

since any p-divisible group over κ of height 1 has endomorphism ring $\mathbb{Z}_p$. Here $M_{s,t}(\mathbb{Z}_p)$ denotes $s \times t$ matrices with entries in $\mathbb{Z}_p$. Given

$$\alpha \in \operatorname{Hom}(\underline{\mathbb{Z}}^{r_2}, \boldsymbol{\mu}_{p^\infty}^{r_1}) = \operatorname{Hom}_{\mathbb{Z}_p}(\underline{\mathbb{Z}_p}^{r_2}, \boldsymbol{\mu}_{p^\infty}^{r_1}) \qquad \alpha' \in \operatorname{Hom}(\underline{\mathbb{Z}}^{r_2'}, \boldsymbol{\mu}_{p^\infty}^{r_1'}) = \operatorname{Hom}_{\mathbb{Z}_p}(\underline{\mathbb{Z}_p}^{r_2'}, \boldsymbol{\mu}_{p^\infty}^{r_1'}) \tag{4.1.4}$$

[10] In loc. cit. this is stated for Artinian local rings R, but one can pass to the limit and obtain the statement here (compare [Mes72, Appendix, Remark (2.2)]).

with corresponding lifts $\mathfrak{X}$ and $\mathfrak{X}'$ (of $\boldsymbol{\mu}_{p^\infty}^{r_1} \times \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2}$ and $\boldsymbol{\mu}_{p^\infty}^{r_1'} \times \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{r_2'}$ respectively) over $\operatorname{Spf} R$, a morphism $(f_1, f_2) \in M_{r_1',r_1}(\mathbb{Z}_p) \times M_{r_2',r_2}(\mathbb{Z}_p)$ lifts to a map $f\colon \mathfrak{X} \to \mathfrak{X}'$ if and only if

$$f_1 \circ \alpha = \alpha' \circ f_2, \tag{4.1.5}$$

again by general facts about Ext in abelian categories (compare with the proof of [Mes72, Appendix, Proposition (3.3)], which discusses the case $r_1 = r_2 = r_1' = r_2' = 1$). We will repeatedly use this criterion for lifting to maps $f\colon \mathfrak{X} \to \mathfrak{X}'$. In (4.1.4), the subscripts $\mathbb{Z}_p$ indicate $\mathbb{Z}_p$-linearity (not the base $\operatorname{Spf} \mathbb{Z}_p$).

4.2. **Quasi-canonical lifting cycles: split.** Throughout Section 4.2, we write R for an adic Noetherian local ring (with maximal ideal being an ideal of definition) equipped with a morphism $\operatorname{Spf} R \to \operatorname{Spf} \mathcal{O}_{\breve{F}}$ inducing an isomorphism on residue fields.

Allowing arbitrary signature $(n-r, r)$ for the moment, form the Rapoport–Zink spaces $\mathcal{N}(n-r,r)$, $\mathcal{N}_{n,r}$, and $\widetilde{\mathcal{N}}_{n,r}$ as in Section 3.7. With $(\mathbf{X}, \iota_{\mathbf{X}}, \lambda_{\mathbf{X}})$ denoting the framing object for $\mathcal{N}(n-r,r)$, we take $\mathbf{X}^-$ to be the framing object used to define $\mathcal{N}_{n,r}$ and $\widetilde{\mathcal{N}}_{n,r}$. There are non-canonical isomorphisms $\mathbf{X}^- \cong \boldsymbol{\mu}_{p^\infty}^r \times \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-r}$ and $\mathbf{X}_0^- \cong \underline{\mathbb{Q}_p/\mathbb{Z}_p}$.

Definition 4.2.1. Given a subset $L^- \subseteq \mathbf{W}^- = \operatorname{Hom}^0(\mathbf{X}_0^-, \mathbf{X}^-)$, consider the associated *local special cycle*

$$\mathcal{Y}(L^-) \subseteq \mathcal{N}_{n,r} \tag{4.2.1}$$

which is the subfunctor consisting of pairs (X, ρ) over schemes S over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$ such that, for all $x^- \in L^-$, the quasi-homomorphism

$$\rho^{-1} \circ x_{\overline{S}}^- \circ \rho_{\mathfrak{X}_0^-,\overline{S}}^-\colon \mathfrak{X}_0^- \to X_{\overline{S}} \tag{4.2.2}$$

lifts to a homomorphism $\mathfrak{X}_0^- \to X$.

As in Definition 3.1.6 (also Section 5.1), the notation $\mathfrak{X}_0$ refers to the canonical lifting of $\mathbf{X}_0$ (and $\mathfrak{X}_0 = \mathfrak{X}_0^+ \times \mathfrak{X}_0^-$ is the decomposition via the nontrivial idempotents $e^\pm \in \mathcal{O}_F$, with $\mathfrak{X}_0^+ \cong \boldsymbol{\mu}_{p^\infty}$ and $\mathfrak{X}_0^- \cong \underline{\mathbb{Q}_p/\mathbb{Z}_p}$). Again, $\mathcal{Y}(L^-) \subseteq \mathcal{N}_{n,r}$ is a closed subfunctor (hence a locally Noetherian formal scheme) by [RZ96, Proposition 2.9] for quasi-homomorphisms.

Lemma 4.2.2. *Suppose $L \subseteq \mathbf{W} = \operatorname{Hom}_F^0(\mathbf{X}_0, \mathbf{X})$ is a subset with $L^+ \subseteq \mathbf{L}^+ = \operatorname{Hom}(\mathbf{X}_0^+, \mathbf{X}^+)$. The natural commutative diagram*

$$\begin{array}{ccccc} \mathcal{Y}(L^-) & & \longrightarrow & & \mathcal{Z}(L) \\ \downarrow & \lrcorner & & & \downarrow \\ \mathcal{N}_{n,r} & \hookrightarrow & \widetilde{\mathcal{N}}_{n,r} & \xrightarrow{\sim} & \mathcal{N}(n-r,r) \end{array} \tag{4.2.3}$$

is Cartesian.

Proof. The lower horizontal arrows are as described in Section 3.7 and Lemma 3.7.6 (the composite is an open and closed immersion). The lemma amounts to the claim that, for any $(X, \iota, \lambda, \rho) \in \mathcal{N}(n-r,r)(S)$ (for any scheme S over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$), if $x = x^+ \times x^- \in \operatorname{Hom}_F^0(\mathbf{X}_0, \mathbf{X})$ with $x^+ \in \operatorname{Hom}_{\mathcal{O}_F}(\mathbf{X}_0^+, \mathbf{X}^+)$, then x lifts to a homomorphism $\mathfrak{X}_0 \to X$ if and only if x^- lifts to a homomorphism $\mathfrak{X}_0^- \to X^-$. Stated alternatively, this is the claim that x^+ always lifts to a homomorphism

$\mathfrak{X}_0^+ \to X^+$. Since $\mathcal{N}_{n,r} \cong \operatorname{Spf} \mathcal{O}_{\breve{F}}[\![t_1, \dots, t_{(n-r)r}]\!]$, this is clear because $\mathfrak{X}_0^+ \to X^+$ automatically factors through a homomorphism to the connected part $(X^+)^0 \cong \boldsymbol{\mu}_{p^\infty}^{n-r}$ of X^+, over any base $\operatorname{Spf} R$ where R is Noetherian Henselian local ring (alternative proof: apply (4.1.5)). □

The following lemma is not needed for our main results.

Lemma 4.2.3. *Assume $r = 1$, i.e. signature $(n-1, 1)$. Let $L \subseteq \mathbf{W}$ be an arbitrary subset, with projection to $\mathbf{W}^-$ denoted L^-. Then the following properties hold.*

(1) The structure morphism $\mathcal{Z}(L) \to \operatorname{Spf} \breve{\mathbb{Z}}_p$ is flat.
(2) The inclusion $\mathcal{Z}(L) \hookrightarrow \mathcal{N}(n-1, 1)$ is a local complete intersection and $\mathcal{Z}(L)$ is Cohen–Macaulay.
(3) If $\mathcal{Z}(L) \neq \emptyset$, then L^- generates a $\mathbb{Z}_p$-lattice in $\mathbf{W}^-$ of some finite rank d, and $\mathcal{Z}(L)$ is equidimensional of dimension $n - d$.
(4) If L^- generates a $\mathbb{Z}_p$-lattice of rank 1 in $\mathbf{W}^-$, then $\mathcal{Z}(L) \hookrightarrow \mathcal{N}(n-1, 1)$ is an effective Cartier divisor.

Proof. Given any point $z \in \mathcal{Z}(L)(\overline{k})$, we can use the action $U(\mathbf{W}) \circlearrowright \mathcal{N}(n-r, r)$ from Lemma 3.5.1(4) and find $\gamma \in U(\mathbf{W})$ such that $\gamma \cdot z \in \mathcal{N}_{n,1}(\overline{k}) \subseteq \mathcal{N}(n-1, 1)(\overline{k})$. Thus, we may assume $z \in \mathcal{N}_{n,1}(\overline{k})$, without loss of generality. In this case, we must have $L^+ \subseteq \mathbf{L}^+$ by (3.5.5) and Remark 3.7.7. By Lemma 4.2.2, it is then equivalent to verify the corresponding properties for $\mathcal{Y}(L^-) \hookrightarrow \mathcal{N}_{n,1}$.

We first verify (4). Consider a nonzero element $w \in \mathbf{W}^-$. Our task is to show that $\mathcal{Y}(w) \subseteq \mathcal{N}_{n,1}$ is an effective Cartier divisor. Assuming for the moment that $\mathcal{Y}(w) \neq \mathcal{N}_{n,1}$, this claim then follows from a standard argument as in [KR11, Proposition 3.5] or [How19, Proposition 4.3]: Grothendieck–Messing theory shows that $\mathcal{Y}(w) \hookrightarrow \mathcal{N}_{n,1}$ is the zero locus of a canonical section of a line bundle on its square-zero thickening. This line bundle is precisely $\mathcal{E}^\vee|_{\mathcal{N}_{n,1}}$ for the tautological bundle $\mathcal{E}$ on $\mathcal{N}(n-1, 1)$, and the mentioned canonical section is specified by w.

To see that $\mathcal{Y}(w) \neq \mathcal{N}_{n,1}$, we may assume $w \in \mathbf{L}^-$, as $\mathcal{Y}(w)$ is otherwise empty. We can then extend to a basis $\{w, w_2, \dots, w_{n-1}\}$ for some $\mathbb{Z}_p$-lattice $M^- \subseteq \mathbf{L}^-$. Let $M = M^- \times \mathbf{L}^+$. Applying [Che24b, Lemma 6.7.3] and [Che24b, Lemma 6.7.4] to $\mathcal{Z}(M)$ implies that $\dim \mathcal{Y}(M^-) \leq 1$, via Lemma 4.2.2. In other words, we consider the "horizontal and vertical" union decomposition $\mathcal{Z}(M) = \mathcal{Z}(M)_{\mathscr{H}} \cup \mathcal{Z}(M)_{\mathscr{V}}$ from [Che24b, Lemma 6.7.5], use the fact that $\mathcal{Z}(M)_{\mathscr{H}} \to \operatorname{Spf} \breve{\mathbb{Z}}_p$ is adic finite flat, use the fact that $\mathcal{Z}(M)_{\mathscr{V}} \subseteq \mathcal{N}(n-1, 1)$ is a scheme, and recall that $\mathcal{N}(n-1, 1)$ has discrete underlying reduced subscheme (Lemma 3.5.1). In e.g. [Che24b, Lemma 6.7.5], the notation $\mathcal{Z}(\underline{x})$ instead of $\mathcal{Z}(M)$ was used, for $\underline{x}$ any basis for M.

We also know that $\mathcal{Y}(M^-)$ is nonempty, e.g. by (3.5.5) and Remark 3.7.7. Since $\mathcal{N}_{n,1}$ is equidimensional of dimension n, and since $\mathcal{Y}(M^-) = \mathcal{Y}(w) \cap \mathcal{Y}(w_2) \cap \cdots \cap \mathcal{Y}(w_{n-1})$, we may apply standard dimension theory to conclude $\mathcal{Y}(w) \neq \mathcal{N}_{n,1}$, e.g. as in [SProject, Lemma 00KW]. This verifies (4).

We next verify (2) and (3). If $\operatorname{span}(L^-) \not\subseteq \mathbf{L}^-$, then $\mathcal{Y}(L^-)$ is empty. Assume $\operatorname{span}(L^-) \subseteq \mathbf{L}^-$ is a lattice of rank d, and pick a basis $w_1, \dots, w_d$ for $\operatorname{span}(L^-)$. We have $\mathcal{Y}(L^-) = \mathcal{Y}(w_1) \cap \cdots \cap \mathcal{Y}(w_d)$, and we know that each $\mathcal{Y}(w_i)$ is an effective Cartier divisor by claim (4). Extend to a basis $w_1, \dots, w_{n-1}$ of some lattice $M^- \subseteq \mathbf{L}^-$. As explained in the above argument for claim (4), we know that $\dim \mathcal{Y}(M^-) \leq 1$. Since $\mathcal{Y}(M^-) = \mathcal{Y}(L^-) \cap \mathcal{Y}(w_{d+1}) \cap \cdots \cap \mathcal{Y}(w_{n-1})$, standard dimension

theory again implies that $\dim \mathcal{Y}(L^-) = n - d$. Note that $\mathcal{Y}(L^-)$ is equidimensional, at least in the sense of Section 3.6, because $\mathcal{Y}(L^-)$ has a unique closed point, so we have verified (3). We also find that $\mathcal{Y}(L^-) \subseteq \mathcal{N}_{n,1}$ is a local complete intersection, e.g. in the sense that its ideal sheaf is locally (in fact globally) generated by d elements, corresponding to the Cartier divisors $\mathcal{Y}(w_1), \dots, \mathcal{Y}(w_d)$. This also implies that $\mathcal{Y}(L^-)$ is Cohen–Macaulay, because $\mathcal{N}_{n,1}$ is regular. This verifies (2).

We last verify (1). As above, we may assume that $\mathcal{Y}(L^-)$ is nonempty of dimension $n - d$. Since $\mathcal{Y}(L^-)$ is Cohen–Macaulay, it is enough to check that $\dim \mathcal{Y}(L^-)_{\overline{k}} = n - d - 1$, as miracle flatness will then imply that $\mathcal{Y}(L^-) \to \operatorname{Spf} \breve{\mathbb{Z}}_p$ is flat. We certainly have $\dim \mathcal{Y}(L^-)_{\overline{k}} \geq n - d - 1$ by standard dimension theory, since the special fiber in $\mathcal{N}_{n,1}$ is a principal closed formal subscheme. With notation as in the above verification of (2) and (3), we find $\mathcal{Y}(M^-)_{\overline{k}} = \mathcal{Y}(L^-)_{\overline{k}} \cap \mathcal{Y}(w_{d+1}) \cap \cdots \cap \mathcal{Y}(w_{n-1})$. Since $\dim \mathcal{Y}(M^-)_{\overline{k}} = 0$ because $\mathcal{Y}(M^-)_{\overline{k}} \to \operatorname{Spec} \overline{k}$ is adic and $\mathcal{N}_{n,1}$ has underlying reduced scheme consisting of a point (see references in the above verification of (4)), standard dimension theory again implies $\dim \mathcal{Y}(L^-)_{\overline{k}} = n - d - 1$. This verifies (1). □

Next, choose isomorphisms $(\mathbf{X}^-)^0 \cong \boldsymbol{\mu}_{p^\infty}^r$ and $(\mathbf{X}^-)^{ét} \cong \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-r}$ for the connected and étale parts of $\mathbf{X}^-$ respectively. Any element $(X, \rho) \in \mathcal{N}_{n,r}(\operatorname{Spf} R)$ (i.e. a morphism $\operatorname{Spf} R \to \mathcal{N}_{n,r}$) then corresponds to a class $\alpha \in \operatorname{Ext}^1_{\operatorname{Spf} R}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-r}, \boldsymbol{\mu}_{p^\infty}^r) = \operatorname{Hom}_{\mathbb{Z}_p}(\underline{\mathbb{Z}_p}^{n-r}, \boldsymbol{\mu}_{p^\infty}^r)$ via the lifting theory in Section 4.1.

Lemma 4.2.4. *Fix any isomorphism $\mathbf{X}_0^- \cong \underline{\mathbb{Q}_p/\mathbb{Z}_p}$. Consider $\varphi\colon \operatorname{Spf} R \to \mathcal{N}_{n,r}$, corresponding to $(X, \rho) \in \mathcal{N}_{n,r}(\operatorname{Spf} R)$ and hence a class $\alpha' \in \operatorname{Ext}^1_{\operatorname{Spf} R}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-r}, \boldsymbol{\mu}_{p^\infty}^r)$. Given any subset $L^- \subseteq \mathbf{L}^-$, the morphism φ factors through $\mathcal{Y}(L^-) \subseteq \mathcal{N}_{n,r}$ if and only if the map*

$$x^*\colon \operatorname{Ext}^1_{\operatorname{Spf} R}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-r}, \boldsymbol{\mu}_{p^\infty}^r) \to \operatorname{Ext}^1_{\operatorname{Spf} R}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}, \boldsymbol{\mu}_{p^\infty}^r) \tag{4.2.4}$$

satisfies $x^*(\alpha') = 0$ for all $x \in L^-$.

Proof. In the lemma statement, we have viewed $x \in L^-$ as a morphism $\underline{\mathbb{Q}_p/\mathbb{Z}_p} \to \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-r}$ via the various identifications. The lemma follows from the lifting criterion in (4.1.5) (in the notation of loc. cit., take $\alpha = 0$). □

Next, we restrict to the case of signature $(n-1, 1)$.

Lemma 4.2.5. *Assume that R is moreover a domain and that $\operatorname{Spf} R \to \operatorname{Spf} \mathcal{O}_{\breve{F}}$ is flat. There is a natural map*

$$\left\{ \begin{array}{l} \text{cyclic subgroups of order } p^s \\ \text{in } \operatorname{Ext}^1_{\operatorname{Spf} R}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty}) \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \text{Full rank integral lattices } M \subseteq \mathbf{L} \\ \text{such that } t(M) \leq 1 \text{ and } \operatorname{val}(M) = s \\ \text{and } M^+ = \mathbf{L}^+ \end{array} \right\} \tag{4.2.5}$$

(functorial in R on the left). If R contains a primitive p^s-th root of unity, then the map is a bijection. Otherwise, the left-hand side is empty.

Proof. Recall the identification $\operatorname{Hom}_{\mathbb{Z}_p}(\underline{\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty}) \cong \operatorname{Ext}^1_{\operatorname{Spf} R}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty})$ from Section 4.1. Suppose $\alpha' \in \operatorname{Hom}_{\mathbb{Z}_p}(\underline{\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty})$ generates a cyclic subgroup of order p^s (possible if and only if R contains a primitive p^s-th root of unity). Let $M_{n-1,n-1}(\mathbb{Z}_p)$ act on $\operatorname{Hom}_{\mathbb{Z}_p}(\underline{\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty})$ by

pre-composition. The annihilator of α' is generated (as a one-sided ideal) by an element $f_2 \in M_{n-1,n-1}(\mathbb{Z}_p)$ which has Smith normal form $\mathrm{diag}(1,\dots,1,p^s)$.

We have a canonical identification $\mathbf{L}^- \cong \mathrm{Hom}(\mathbf{X}_0^-, (\mathbf{X}^-)^{\acute{e}t})$. Via the identification $(\mathbf{X}^-)^{\acute{e}t} \cong \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-1}$, we obtain an action of $M_{n-1,n-1}(\mathbb{Z}_p)$ on $\mathbf{L}^-$ (post-composition). We then set $M^- = f_2(\mathbf{L}^-)$, and let $M = \mathbf{L}^+ \oplus M^-$. Note that M^- does not depend on the choice of generator f_2.

Conversely, given a lattice $M \subseteq \mathbf{L}$ as in the lemma statement, select any $f_2 \in M_{n-1,n-1}(\mathbb{Z}_p)$ satisfying $M^- = f_2(\mathbf{L}^-)$, and note that f_2 necessarily has Smith normal form $\mathrm{diag}(1,\dots,1,p^s)$. If R contains a primitive p^s-th root of unity, then f_2 acting on $\mathrm{Hom}_{\mathbb{Z}_p}(\underline{\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty})$ has kernel which is cyclic of order p^s. This gives the inverse map. □

For $s \in \mathbb{Z}_{\geq 0}$, set $\breve{E}_s \coloneqq \breve{F}[\zeta_{p^s}]$ with ring of integers $\mathcal{O}_{\breve{E}_s} = \mathcal{O}_{\breve{F}}[\zeta_{p^s}]$, where ζ_{p^s} is a primitive p^s-th root of unity. Suppose $M \subseteq \mathbf{L}$ is an integral full rank $\mathcal{O}_F$-lattice satisfying $M^+ = \mathbf{L}^+$, with type $t(M) \leq 1$ and $\mathrm{val}(M) = s$. By Lemma 4.2.5 and its proof, there is an associated cyclic subgroup of $\mathrm{Ext}^1_{\mathrm{Spf}\,\mathcal{O}_{\breve{E}_s}}(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty})$. Any generator of this cyclic subgroup defines a morphism $\mathrm{Spf}\,\mathcal{O}_{\breve{E}_s} \to \mathcal{N}_{n,1}$ (via the lifting theory from Section 4.1). Changing the choice of generator corresponds precisely to the action of $\mathrm{Gal}(\breve{E}_s/\breve{F})$ (by Lubin–Tate theory for $\boldsymbol{\mu}_{p^\infty}$). This morphism $\mathrm{Spf}\,\mathcal{O}_{\breve{E}_s} \to \mathcal{N}_{n,1}$ must be a closed immersion: if the morphism factors through $\mathrm{Spf}\,R \to \mathcal{N}_{n,1}$ for some sub $\mathcal{O}_{\breve{F}}$-algebra $R \subseteq \mathcal{O}_{\breve{E}_s}$, then Lemma 4.2.5 implies that $R = \mathcal{O}_{\breve{E}_s}$.

We write $\mathrm{Spf}\,\mathcal{O}_{\breve{E}_s} \cong \mathcal{Z}(M)^\circ \subseteq \mathcal{N}_{n,1}$ for the resulting closed subfunctor, and call it a *quasi-canonical lifting cycle*. This closed subfunctor $\mathcal{Z}(M)^\circ$ does not depend on the choices of isomorphisms $(\mathbf{X}^-)^0 \cong \boldsymbol{\mu}_{p^\infty}$ and $(\mathbf{X}^-)^{\acute{e}t} \cong \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-1}$ appearing in the statement of Lemma 4.2.5.

Lemma 4.2.6. *With M as above, view $\mathcal{Z}(M)^\circ$ as a morphism $\mathrm{Spf}\,\mathcal{O}_{\breve{E}_s} \to \mathcal{N}_{n,1}$ corresponding to $(X,\rho) \in \mathcal{N}_{n,1}(\mathrm{Spf}\,\mathcal{O}_{\breve{E}_s})$.*

If $n = 1$ then $X \cong \boldsymbol{\mu}_{p^\infty}$. If $n \geq 2$ then $X \cong \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-2} \times \mathfrak{X}_s$ (forgetting ρ) where $\mathfrak{X}_s$ is a p-divisible group of height 2 and dimension 1 with $\mathrm{End}(\mathfrak{X}_s) = \mathbb{Z}_p + p^s\mathcal{O}_F$ (a quasi-canonical lifting *in the sense of Section 5.2).*

Proof. Let $\alpha' \in \mathrm{Hom}_{\mathbb{Z}_p}(\underline{\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty})$ be the element corresponding to (X,ρ). If $n = 1$ then $\alpha' = 0$ and $X \cong \boldsymbol{\mu}_{p^\infty}$.

If $n \geq 2$, then (after replacing ρ by $\phi \circ \rho$ for some $\phi \in \mathrm{GL}_{n-1}(\mathbb{Z}_p)$), the lift (X,ρ) of $\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-1} \times \boldsymbol{\mu}_{p^\infty}$ is associated with α' of the form $(0,\dots,0,\zeta_{p^s})$ for $\zeta_{p^s} \in \mathcal{O}_{\breve{E}_s}$ a primitive p^s-th root of unity. For some $\mathfrak{X}_s$ as in the lemma statement, we obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \underline{\mathbb{Z}}^{n-1} & \longrightarrow & \underline{\mathbb{Z}[1/p]}^{n-1} & \longrightarrow & \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-1} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle (0,\dots,0,\zeta_{p^s})} & & \downarrow & & \| & & \\ 0 & \longrightarrow & \boldsymbol{\mu}_{p^\infty} & \longrightarrow & \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-2} \times \mathfrak{X}_s & \longrightarrow & \underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-1} & \longrightarrow & 0\,, \end{array} \tag{4.2.6}$$

using the lifting criterion of (4.1.5) again. □

Lemma 4.2.7. *Suppose $M \subseteq \mathbf{L}$ is an integral full rank $\mathcal{O}_F$-lattice satisfying $M^+ = \mathbf{L}^+$ with type $t(M) \leq 1$ and $\mathrm{val}(M) = s$. Let $L^- \subseteq \mathbf{W}^-$ be any subset. We have $\mathcal{Z}(M)^\circ \subseteq \mathcal{Y}(L^-)$ if and only if $L^- \subseteq M$.*

Proof. If $n = 1$ then $\mathbf{W}^- = 0$ and $\mathcal{Z}(M)^\circ = \mathcal{Y}(L^-) = \mathcal{N}_{n,1}$, so the lemma is trivial in this case. We thus assume $n \geq 2$ below.

It is enough to check the case where L^- consists of a single element, i.e. a quasi-homomorphism $x\colon (\mathbf{X}_0)^- \to \mathbf{X}^-$ (or equivalently, $x\colon (\mathbf{X}_0)^- \to (\mathbf{X}^-)^{\acute{e}t}$ since $\mathbf{X}_0^- \cong \underline{\mathbb{Q}_p/\mathbb{Z}_p}$ is étale). If $x \notin \mathbf{L}^-$ then $\mathcal{Y}(L^-) = \emptyset$ (while $\mathcal{Z}(M)^\circ \neq \emptyset$), so we may assume $x \in \mathbf{L}^-$.

Pick any identification $\mathbf{X}_0^- \cong \underline{\mathbb{Q}_p/\mathbb{Z}_p}$. Set $s = \mathrm{val}(M)$. Use the setup and notation in the proof of Lemma 4.2.5.

View $\mathcal{Z}(M)^\circ$ as a closed immersion $\mathrm{Spf}\, \mathcal{O}_{\breve{E}_s} \to \mathcal{N}_{n,1}$, corresponding to an element $\alpha' \in \mathrm{Hom}_{\mathbb{Z}_p}(\underline{\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty})$. By Lemma 4.2.4, our task is to show that $\alpha' \circ x = 0$ if and only if $x \in M^-$. Since f_2 generates (in $M_{n-1,n-1}(\mathbb{Z}_p)$) the annihilator of α' (as a one-sided ideal), we see that $\alpha' \circ x = 0$ if and only if $x \in f_2(\mathbf{L}^-) = M^-$ (for example, view x as a column vector and observe that $(x, 0, \ldots, 0) \in M_{n-1,n-1}(\mathbb{Z}_p)$ lies in the one-sided ideal generated by f_2). □

Definition 4.2.8. Let $M \subseteq \mathbf{W}$ be a full rank integral $\mathcal{O}_F$-lattice, with type $t(M) \leq 1$ and $\mathrm{val}(M) = s$. Select any $\gamma \in U(\mathbf{W})$ satisfying $\gamma(\mathbf{L}^+) = M^+$ (also write γ for $(\gamma, 1) \in U(\mathbf{W}) \times U(\mathbf{W}^\perp)$, by abuse of notation).

The *quasi-canonical lifting cycle* associated with M is the closed subfunctor

$$\mathrm{Spf}\, \mathcal{O}_{\breve{E}_s} \cong \mathcal{Z}(M)^\circ \coloneqq \gamma(\mathcal{Z}(\gamma^{-1}(M))^\circ) \subseteq \mathcal{N}(n-1, 1) \tag{4.2.7}$$

where $\gamma \in U(\mathbf{W}) \times U(\mathbf{W}^\perp)$ acts on $\mathcal{N}(n-1,1)$ as in Section 3.3.

In the situation of Definition 4.2.8, the closed subfunctor $\mathcal{Z}(M)^\circ$ does not depend on the choice of γ. We have also viewed $\mathcal{N}_{n,1}$ as an open and closed subfunctor of $\mathcal{N}(n-1,1)$ (as in the lower horizontal arrows in Lemma 4.2.2).

Lemma 4.2.9. *If $L \subseteq \mathbf{W}$ is any subset and $M \subseteq \mathbf{W}$ is any full rank integral lattice with $t(M) \leq 1$, we have $\mathcal{Z}(M)^\circ \subseteq \mathcal{Z}(L)$ if and only if $L \subseteq M$.*

Proof. After acting by $U(\mathbf{W})$, it is enough to check the case where $M^+ = \mathbf{L}^+$. In this case, we have $\mathcal{Z}(M)^\circ \subseteq \mathcal{N}_{n,1}$. If $L \not\subseteq \mathbf{L}$, then $\mathcal{Z}(L) \cap \mathcal{Z}(M)^\circ = \emptyset$ by Lemma 3.5.1(5) and Remark 3.7.7 (and $\mathcal{Z}(M)^\circ$ is nonempty). So assume $L \subseteq \mathbf{L}$. Then $\mathcal{Z}(L) \cap \mathcal{N}_{n,1} = \mathcal{Y}(L^-)$ by Lemma 4.2.2. This reduces to the case proved in Lemma 4.2.7. □

Corollary 4.2.10. *Let $L \subseteq \mathbf{W}$ be any subset. Form the horizontal (flat) part of the local special cycle $\mathcal{Z}(L)$, which we denote as $\mathcal{Z}(L)_{\mathscr{H}}$. We have an inclusion of closed formal subschemes*

$$\bigcup_{\substack{L \subseteq M \subseteq M^* \\ t(M) \leq 1}} \mathcal{Z}(M)^\circ \subseteq \mathcal{Z}(L)_{\mathscr{H}} \tag{4.2.8}$$

in $\mathcal{N}(n-1,1)$.

Proof. The union is a scheme-theoretic union. The claim follows from Lemma 4.2.9 because each $\mathcal{Z}(M)^\circ$ is flat over $\mathrm{Spf}\, \mathcal{O}_{\breve{F}}$. □

In the situation of Corollary 4.2.10, we moreover have $\mathcal{Z}(L)_{\mathscr{H}} = \mathcal{Z}(L)$ by Lemma 4.2.3(1). We will not need this.

Lemma 4.2.11. *Let $M \subseteq \mathbf{W}$ and $M' \subseteq \mathbf{W}$ be integral full-rank $\mathcal{O}_F$-lattices with $t(M) \leq 1$ and $t(M') \leq 1$. If $M \neq M'$, then $\mathcal{Z}(M)^\circ \neq \mathcal{Z}(M')^\circ$.*

Proof. Let $N \subseteq \mathbf{W}$ be the unique self-dual full rank lattice such that $N^+ = M^+$. Similarly define N', using M'^+ instead of M^+. On reduced subschemes, we have $\mathcal{Z}(M)^\circ_{\mathrm{red}} = \mathcal{Z}(M')^\circ_{\mathrm{red}}$ if and only if $N = N'$ by Lemma 3.5.1 (more precisely, Remark 3.7.7, Definition 4.2.1, and the action on special cycles in (3.3.4)). So we may assume $N = N'$. Using the $U(\mathbf{W})$ action on $\mathcal{N}(n-1,1)$, we also reduce to the case where $N = \mathbf{L}$.

Set $s = \mathrm{val}(M)$ and $s' = \mathrm{val}(M')$, and view $\mathcal{Z}(M)^\circ$ and $\mathcal{Z}(M')^\circ$ as closed immersions $\varphi\colon \operatorname{Spf} \mathcal{O}_{\breve{E}_s} \to \mathcal{N}(n-1,1)$ and $\varphi'\colon \operatorname{Spf} \mathcal{O}_{\breve{E}_{s'}} \to \mathcal{N}(n-1,1)$. Lemma 4.2.5 implies that $M = M'$ if and only if both $s = s'$ and the morphisms φ, φ' are the same up to $\mathrm{Gal}(\breve{E}_s/\breve{F})$-action (this is equivalent to requiring that the corresponding elements of Ext^1 in that lemma generate the same subgroup). This is satisfied if and only if $\mathcal{Z}(M)^\circ = \mathcal{Z}(M')^\circ$. □

Lemma 4.2.12. *Let $L \subseteq \mathbf{W}$ be a full rank $\mathcal{O}_F$-lattice. Assume that R is moreover a domain and $\operatorname{Spf} R \to \operatorname{Spf} \mathcal{O}_{\breve{F}}$ is flat. Any morphism $\varphi\colon \operatorname{Spf} R \to \mathcal{Z}(L)$ factors through some quasi-canonical lifting cycle $\mathcal{Z}(M)^\circ$.*

Proof. Again, we may act by $U(\mathbf{W})$ on $\mathcal{N}(n-1,1)$ to assume that $\varphi\colon \operatorname{Spf} R \to \mathcal{Z}(L)$ factors through the open and closed component $\mathcal{N}_{n,1} \subseteq \mathcal{N}(n-1,1)$ described in Section 3.7 and above. This implies $L \subseteq \mathbf{L}$, as $\mathcal{Z}(L) \cap \mathcal{N}_{n,1}$ is otherwise empty via Lemma 3.5.1.

Fix isomorphisms as in the statement of Lemma 4.2.5. Then φ corresponds to some $(X, \rho) \in \mathcal{N}_{n,1}$, and this lift of $\mathbf{X}^-$ corresponds to a class $\alpha' \in \mathrm{Ext}^1(\underline{\mathbb{Q}_p/\mathbb{Z}_p}^{n-1}, \boldsymbol{\mu}_{p^\infty})$ via the lifting theory in Section 4.1.

By Lemma 4.2.5, it is enough to show that α' is p^s-torsion for some $s \in \mathbb{Z}_{\geq 0}$ (then φ must factor through $\mathcal{Z}(M)^\circ$ where M is the lattice associated with the cyclic subgroup generated by α'). Select $s \geq 0$ such that $p^s\mathbf{L} \subseteq L$ (such s exists because L is full rank). Then Lemma 4.2.4 implies $p^s a' = 0$, since φ factors through $\mathcal{Z}(L)$ (and hence through $\mathcal{Y}(L^-)$). □

5. Canonical and quasi-canonical liftings

We retain $F/\mathbb{Q}_p$ and accompanying notation as in Section 3. In Sections 5.1 and 5.2, we allow $p = 2$ even if $F/\mathbb{Q}_p$ is ramified. We collect some needed facts about canonical and quasi-canonical lifts in all cases (inert, ramified, split). See also [Gro86], [Wew07], [Meu07]. Our conventions differ slightly from [Wew07], due to the phenomenon explained in [KR11, Footnote 7] (there in the inert case, which we also modify to apply in the ramified case).

5.1. **Canonical liftings.** As in Section 3.1, let $\mathbf{X}_0$ be the unique supersingular p-divisible group of height 2 dimension 1 over $\overline{k}$ if $F/\mathbb{Q}_p$ is nonsplit and replace "supersingular" with "ordinary" if $F/\mathbb{Q}_p$ is split. Let $j\colon \mathcal{O}_F \hookrightarrow \mathrm{End}(\mathbf{X}_0)$ be a ring homomorphism. We reserve the notation $\iota_{\mathbf{X}_0}$ to mean a signature $(1,0)$ action, and allow j to have either signature (i.e. $(1,0)$ or $(0,1)$) for its action on $\operatorname{Lie} \mathbf{X}_0$.

Let $\breve{E}$ be any finite degree field extension of $\breve{F}$, with ring of integers $\mathcal{O}_{\breve{E}}$.

Definition 3.1.6 implies that the pair $(\mathbf{X}_0, j)$ admits a lift $(\mathfrak{X}_0, \iota_{\mathfrak{X}_0}, \rho_{\mathfrak{X}_0})$ over $\operatorname{Spf} \mathcal{O}_{\breve{E}}$, i.e. $(\mathfrak{X}_0, \iota_{\mathfrak{X}_0})$ is a p-divisible group over $\operatorname{Spf} \mathcal{O}_{\breve{E}}$ with $\mathcal{O}_F$-action $\iota_{\mathfrak{X}_0}$, and $\rho_{\mathfrak{X}_0} \colon \mathfrak{X}_{0,\overline{k}} \to \mathbf{X}_0$ is a $\mathcal{O}_F$-linear isomorphism with respect to ι and j.

In the supersingular case, the pair $(\mathfrak{X}_0, \iota_{\mathfrak{X}_0})$ may be described via Lubin–Tate formal groups. In the ordinary case, we have $\mathfrak{X}_0 \cong \boldsymbol{\mu}_{p^\infty} \times \underline{\mathbb{Q}_p/\mathbb{Z}_p}$.

By the *signature* of $(\mathfrak{X}_0, \iota_{\mathfrak{X}_0}, \rho_{\mathfrak{X}_0})$ or $(\mathfrak{X}_0, \iota_{\mathfrak{X}_0})$, we mean the signature of $\iota_{\mathfrak{X}_0}$ acting on $\operatorname{Lie} \mathfrak{X}_0$. This signature is either $(1,0)$ or $(0,1)$. If $F/\mathbb{Q}_p$ is unramified, then $(\mathfrak{X}_0, \iota_{\mathfrak{X}_0}, \rho_{\mathfrak{X}_0})$ must have the same signature as $(\mathbf{X}_0, j)$. In the ramified case, $\iota_{\mathfrak{X}_0}$ can have either signature.

After fixing a signature, the triple $(\mathfrak{X}_0, \iota_{\mathfrak{X}_0}, \rho_{\mathfrak{X}_0})$ is unique up to unique isomorphism, and we call it the *canonical lifting*[11] of $(\mathbf{X}_0, j)$. The canonical lifting over $\operatorname{Spf} \mathcal{O}_{\breve{E}}$ is defined over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$ (i.e. is the base change of the canonical lift over $\operatorname{Spf} \mathcal{O}_{\breve{F}}$).

The map $\iota_{\mathfrak{X}_0} \colon \mathcal{O}_F \to \operatorname{End}(\mathfrak{X}_0)$ is an isomorphism, via the fact $\mathcal{O}_F = \operatorname{End}_{\mathcal{O}_F}(\mathbf{X}_0)$. If $(\mathfrak{X}_0^\sigma, \iota_{\mathfrak{X}_0}^\sigma)$ is as in (3.1.1), we have $\operatorname{Hom}_{\mathcal{O}_F}(\mathfrak{X}_0, \mathfrak{X}_0^\sigma) = 0$ because $\operatorname{End}(\mathfrak{X}_0) = \mathcal{O}_F$.

Example 5.1.1. Assume $F/\mathbb{Q}_p$ is nonsplit, and let $\mathfrak{X}_0$ be the canonical lifting over $\operatorname{Spf} \mathcal{O}_{\breve{E}}$ (of some fixed signature). Despite the identification $\operatorname{End}(\mathfrak{X}_0) \cong \mathcal{O}_F$ as explained above, Drinfeld rigidity for quasi-homomorphisms implies $\operatorname{End}^0(\mathfrak{X}_0) \cong \operatorname{End}^0(\mathbf{X}_0) \cong D$, where D is the quaternion division algebra over $\mathbb{Q}_p$. On the other hand, if $\mathfrak{X}_0'$ denotes the p-divisible group over $\operatorname{Spec} \mathcal{O}_{\breve{E}}$ associated with $\mathfrak{X}_0$ (see e.g. [Che24b, Lemma C.3.1]), we have $\operatorname{End}^0(\mathfrak{X}_0') \cong \operatorname{End}(\mathfrak{X}_0') \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \cong F$. Thus, by our conventions (explained in [Che24b, Appendix C.1]), quasi-homomorphisms do not necessarily lift along the equivalence of p-divisible groups over $\operatorname{Spec} \mathcal{O}_{\breve{E}}$ and $\operatorname{Spf} \mathcal{O}_{\breve{E}}$ from [Che24b, Lemma C.3.1]. See also [Che24b, Remark C.3.5].

5.2. **Quasi-canonical liftings.** Let $\breve{E}$ and $(\mathbf{X}_0, j)$ be as in Section 5.1. For integers $s \geq 0$, let $\mathcal{O}_{F,s} \coloneqq \mathbb{Z}_p + p^s \mathcal{O}_F$ be the order of index p^s in $\mathcal{O}_F$. When $F/\mathbb{Q}_p$ is nonsplit (resp. split) the subgroup $\mathcal{O}_{F,s}^\times \subseteq \mathcal{O}_F^\times$ (resp. $(1 + p^s \mathbb{Z}_p)^\times \subseteq \mathbb{Z}_p^\times$) has an associated finite totally ramified abelian extension $\breve{E}_s$ of $\breve{F}$ by local class field theory. The index is

$$[\breve{E}_s : \breve{F}] = \begin{cases} p^s(1 - \eta^*(p) p^{-1}) & \text{if } s \geq 1 \\ 1 & \text{if } s = 0 \end{cases} \tag{5.2.1}$$

where $\eta^*(p) \coloneqq -1, 0, 1$ in the inert, ramified, and split cases respectively. In the split case, we have $\mathcal{O}_{\breve{E}_s} = \mathcal{O}_{\breve{F}}[\zeta_{p^s}]$ where ζ_{p^s} is a primitive p^s-th root of unity.

In all cases, a *quasi-canonical lifting of level s* of $(\mathbf{X}_0, j)$ is a triple $(\mathfrak{X}_s, \iota_{\mathfrak{X}_s}, \rho_{\mathfrak{X}_s})$ where

$\mathfrak{X}_s$	is a p-divisible group over $\operatorname{Spf} \mathcal{O}_{\breve{E}}$
$\iota_{\mathfrak{X}_s} \colon \mathcal{O}_{F,s} \xrightarrow{\sim} \operatorname{End}(\mathfrak{X}_s)$	is a ring isomorphism
$\rho_{\mathfrak{X}_s} \colon \mathfrak{X}_{s,\overline{k}} \to \mathbf{X}_0$	is an $\mathcal{O}_{F,s}$-linear isomorphism of p-divisible groups over $\overline{k}$.

[11] When j has signature $(1,0)$, what Gross [Gro86] calls a *canonical lifting* is what we call a *canonical lifting of signature* $(1,0)$ here. This change in terminology allows additional flexibility when discussing *quasi-canonical* liftings, to account for e.g. [KR11, Footnote 7]. On the other hand, we do typically mean signature $(1,0)$ for *canonical* liftings, as in Definition 3.1.6.

Note that a quasi-canonical lifting of level $s = 0$ is the same as a canonical lifting. As above, we speak of the *signature* of a quasi-canonical lifting, which means the signature of the action $\iota_{\mathfrak{X}_s}|_{\mathrm{Lie}\,\mathfrak{X}_s}$.

The signature of $(\mathbf{X}_0, j)$ and the signature of a level s quasi-canonical lifting must be

$$\begin{cases} \text{same} & \text{if } F/\mathbb{Q}_p \text{ is inert and } s \text{ is even, or } F/\mathbb{Q}_p \text{ is split} \\ \text{opposite} & \text{if } F/\mathbb{Q}_p \text{ is inert and } s \text{ is odd} \\ \text{either signature} & \text{if } F/\mathbb{Q}_p \text{ is ramified.} \end{cases} \tag{5.2.2}$$

Quasi-canonical liftings of level $s \geq 0$ exist in all such situations, and are defined over $\mathrm{Spf}\,\mathcal{O}_{\breve{E}_s}$. The property of being a level s quasi-canonical lifting is preserved under base change along $\mathrm{Spf}\,\mathcal{O}_{\breve{E}'} \to \mathrm{Spf}\,\mathcal{O}_{\breve{E}}$ for any finite degree field extension $\breve{E}'$ over $\breve{E}$. If $F/\mathbb{Q}_p$ is split, a choice of level s quasi-canonical lifting corresponds to a choice of morphism $\underline{\mathbb{Z}_p} \to \boldsymbol{\mu}_{p^\infty}$ over $\mathrm{Spf}\,\mathcal{O}_{\breve{E}}$ of exact order p^s (i.e. a choice of primitive p^s-th root of unity in $\breve{E}$) via the lifting theory in Section 4.1.

The group $\mathrm{Gal}(\breve{E}_s/\breve{F})$ acts simply transitively on the set of level s quasi-canonical liftings for any fixed signature (if such liftings exist). By Lubin–Tate theory, this action is compatible with the identification $\mathrm{Gal}(\breve{E}_s/\breve{F}) \cong \mathcal{O}_F^\times/\mathcal{O}_{F,s}^\times$ via local class field theory (normalized to send uniformizers to geometric Frobenius) where $a \in \mathcal{O}_F^\times$ acts on the set of quasi-canonical liftings as $(\mathfrak{X}_s, \iota_{\mathfrak{X}_s}, \rho_{\mathfrak{X}_s}) \mapsto (\mathfrak{X}_s, \iota_{\mathfrak{X}_s}, a\rho_{\mathfrak{X}_s})$. In the split case, we have used the isomorphism

$$\begin{aligned} \mathcal{O}_F^\times/\mathcal{O}_{F,s}^\times &\longrightarrow \mathbb{Z}_p^\times/(1+p^s\mathbb{Z}_p)^\times \\ x &\longmapsto e^+(x)e^-(x^{-1}) \end{aligned} \tag{5.2.3}$$

if $(\mathbf{X}_0, j)$ has signature $(1,0)$ and its reciprocal if $(\mathbf{X}_0, j)$ has signature $(0,1)$. In particular, the quasi-canonical liftings of a fixed level s are all isomorphic if the framing $\rho_{\mathfrak{X}_s}$ is forgotten.

Let $(\mathfrak{X}_0, \iota_{\mathfrak{X}_0}, \rho_{\mathfrak{X}_0})$ and $(\mathfrak{X}_s, \iota_{\mathfrak{X}_s}, \rho_{\mathfrak{X}_s})$ be canonical and quasi-canonical lifts over $\mathrm{Spf}\,\mathcal{O}_{\breve{E}}$, for some $(\mathbf{X}_0, j)$ and $(\mathbf{X}_0, j')$ respectively (possibly $j \neq j'$). Then

$$\mathrm{Hom}(\mathfrak{X}_0, \mathfrak{X}_s) \cong \psi_s \cdot \mathcal{O}_F \tag{5.2.4}$$

(no $\mathcal{O}_F$-linearity imposed) is a free $\mathcal{O}_F$-module of rank 1 (where $\mathcal{O}_F$ acts by pre-composition), generated by some isogeny ψ_s of degree p^s. The isogeny ψ_s is defined over $\mathrm{Spf}\,\mathcal{O}_{\breve{E}_s}$. If $\mathfrak{X}_0$ and $\mathfrak{X}_s$ have the same signature, then ψ_s is automatically $\mathcal{O}_{F,s}$-linear.

When $F/\mathbb{Q}_p$ is split, we may take ψ_s to be the map inducing the map $\mathbf{X}_0 \to \mathbf{X}_0$ which is

$$\psi_s|_{\mathbf{X}_0^0} \colon \mathbf{X}_0^0 \xrightarrow{\mathrm{id}} \mathbf{X}_0^0 \qquad \psi_s|_{\mathbf{X}_0^{\acute{e}t}} \colon \mathbf{X}_0^{\acute{e}t} \xrightarrow{\times p^s} \mathbf{X}_0^{\acute{e}t} \tag{5.2.5}$$

on the connected and étale parts, respectively. This follows from the lifting criterion in (4.1.5).

For any generator ψ_s of $\mathrm{Hom}(\mathfrak{X}_0, \mathfrak{X}_s)$, we have

$$\mathrm{length}_{\mathcal{O}_{\breve{E}}}(e^*\Omega^1_{\ker\psi_s/\,\mathrm{Spec}\,\mathcal{O}_{\breve{E}}}) = \frac{1}{2}[\breve{E} : \breve{\mathbb{Q}}_p]\frac{(1-p^{-s})(1-\eta^*(p))}{(1-p^{-1})(p-\eta^*(p))} \tag{5.2.6}$$

where $\eta^*(p) \coloneqq -1, 0, 1$ in the inert, ramified, split cases respectively and where $e\colon \mathrm{Spec}\,\mathcal{O}_{\breve{E}} \to \ker\psi_s$ denotes the identity section. We are passing between $\mathrm{Spf}\,\mathcal{O}_{\breve{E}}$ and $\mathrm{Spec}\,\mathcal{O}_{\breve{E}}$ as in [Che24b, Appendix C.3].

The nonsplit case of (5.2.6) is essentially a computation of Nakkajima and Taguchi [NT91] (see also [KRY04, Proposition 10.3] and its proof). The split case follows from (5.2.5), which implies that $\ker \psi_s$ is étale over $\operatorname{Spec} \overline{k}$ (cf. the closely related [KRY04, Proposition 10.1]).

The following constant $\delta_{\mathrm{tau}}(s) \in \mathbb{Q}$ ("local change of tautological height") will be crucial for the formulation of our local main theorems. With notation as above, we define

$$\delta_{\mathrm{tau}}(s) \coloneqq -\frac{1}{2} v_p(\deg \psi_s) + \frac{1}{[\breve{E} : \breve{\mathbb{Q}}_p]} \operatorname{length}_{\mathcal{O}_{\breve{E}}}(e^* \Omega^1_{\ker \psi_s / \operatorname{Spec} \mathcal{O}_{\breve{E}}}) \tag{5.2.7}$$

$$= -\frac{1}{2} \left(s - \frac{(1-p^{-s})(1-\eta^*(p))}{(1-p^{-1})(p-\eta^*(p))} \right) \tag{5.2.8}$$

for integers $s \in \mathbb{Z}_{\geq 0}$. We used (5.2.6) for the second equality. We also set $\delta_{\mathrm{Fal}}(s) \coloneqq -2\delta_{\mathrm{tau}}(s)$.

The quantity $\delta_{\mathrm{tau}}(s)$ depends only on s and $\eta^*(p)$, and does not depend on the choice of ψ_s. In [Che24b, Part 2], we will explain the relation of $\delta_{\mathrm{tau}}(s)$ and $\delta_{\mathrm{Fal}}(s)$ with local decompositions of "tautological" and Faltings heights of special cycles.

5.3. **Quasi-canonical lifting cycles.** We state how certain local special cycles decompose into *quasi-canonical lifting cycles* (Section 4.2). We continue to use the notation in Section 3.2, now restricting to signature $(n-1,1)$. We also assume $p \neq 2$ unless $F/\mathbb{Q}_p$ is split (in the inert case, this is so that we may cite [LZ22a, Theorem 4.2.1]).

Suppose $M^\flat \subseteq \mathbf{W}$ is an integral $\mathcal{O}_F$-lattice of rank $n-1$ with $t(M^\flat) \leq 1$. Set $s = \lfloor \mathrm{val}(M^\flat) \rfloor$ (notation as in Section 2.2). There is an associated *quasi-canonical lifting cycle* $\mathcal{Z}(M^\flat)^\circ \subseteq \mathcal{N}(n-1,1)$, which is a certain closed subfunctor such that

$$\mathcal{Z}(M^\flat)^\circ \cong \begin{cases} \operatorname{Spf} \mathcal{O}_{\breve{E}_s} & \text{if } F/\mathbb{Q}_p \text{ is unramified} \\ \operatorname{Spf} \mathcal{O}_{\breve{E}_s} \sqcup \operatorname{Spf} \mathcal{O}_{\breve{E}_s} & \text{if } F/\mathbb{Q}_p \text{ is ramified.} \end{cases} \tag{5.3.1}$$

Suppose $\varphi\colon \operatorname{Spf} \mathcal{O}_{\breve{E}_s} \to \mathcal{N}(n-1,1)$ is a morphism representing any component of $\mathcal{Z}(M^\flat)^\circ$, with corresponding tuple $(X, \iota, \lambda, \rho) \in \mathcal{N}(n-1,1)(\operatorname{Spf} \mathcal{O}_{\breve{E}_s})$. If $n = 1$, then $M^\flat = 0$ and $X \cong \mathfrak{X}_0^\sigma$. If $n \geq 2$, then there exists a polarization-preserving $\mathcal{O}_F$-linear isomorphism (forgetting ρ)

$$X \cong (\mathfrak{X}_0)^{n-2} \times (\mathfrak{X}_s \otimes_{\mathbb{Z}_p} \mathcal{O}_F) \tag{5.3.2}$$

for some level s quasi-canonical lift $\mathfrak{X}_s$ (and $\mathfrak{X}_0$ being the canonical lift), where $\mathfrak{X}_s \otimes_{\mathbb{Z}_p} \mathcal{O}_F$ is equipped with the polarization as in (3.7.2), where $\mathfrak{X}_0^{n-2}$ has the diagonal polarization $\lambda_{\mathfrak{X}_0}^{n-2}$ for some principal polarization $\lambda_{\mathfrak{X}_0}$ on $\mathfrak{X}_0$ if $F/\mathbb{Q}_p$ is unramified, and where $\mathfrak{X}_0^{n-2}$ has a product polarization as in (3.1.6) (with respect to some principal polarization $\lambda_{\mathfrak{X}_0}$ on $\mathfrak{X}_0$) if $F/\mathbb{Q}_p$ is ramified.

For the inert case of the above assertions, see [LZ22a, §4.2] (we are using the same notation), and also [KR11, Proposition 8.1] (there for $n = 2$).

For the ramified case, see [RSZ17, Proposition 7.1] (there for $n = 2$) and also the proof of [LL22, Proposition 2.44] (also [LL22, Definition 2.45]; we are using their notation but with $\mathcal{N}$ replaced by $\mathcal{Z}$). If moreover $n = 2$, the two components of $\mathcal{Z}(M^\flat)^\circ$ correspond to the two connected components of $\mathcal{N}(1,1)$ as in Lemma 3.5.1, particularly part (5), i.e. $\mathcal{Z}(M^\flat)^\circ \to \mathcal{N}(1,1)$ is surjective on underlying topological spaces. For general n, there exists a closed embedding $\mathcal{N}(1,1) \to \mathcal{N}(n-$

$1,1)$ such that $\mathcal{Z}(M^\flat)^\circ \to \mathcal{N}(n-1,1)$ factors as a quasi-canonical lifting cycle through $\mathcal{N}(1,1)$, via [LL22, Remark 2.52].

For the split case, $\mathcal{Z}(M^\flat)^\circ$ was defined in Definition 4.2.8. The assertion $X \cong (\mathfrak{X}_0)^{n-2} \times (\mathfrak{X}_s \otimes_{\mathbb{Z}_p} \mathcal{O}_F)$ follows from Lemma 4.2.6 (note that X in loc. cit. is X^- in the present notation) and Remark 3.7.4.

Proposition 5.3.1. *Let $L^\flat \subseteq \mathbf{W}$ be a non-degenerate $\mathcal{O}_F$-lattice of rank $n-1$. Form the horizontal (flat) part of the local special cycle $\mathcal{Z}(L^\flat)$, which we denote as $\mathcal{Z}(L^\flat)_{\mathscr{H}}$. We have an equality of closed formal subschemes*

$$\mathcal{Z}(L^\flat)_{\mathscr{H}} = \bigcup_{\substack{L^\flat \subseteq M^\flat \subseteq M^{\flat *} \\ t(M^\flat) \leq 1}} \mathcal{Z}(M^\flat)^\circ \tag{5.3.3}$$

in $\mathcal{N}(n-1,1)$, where the union runs over full rank lattices $M^\flat \subseteq L_F^\flat$.

Proof. The union is the scheme-theoretic union.

The inert case is [LZ22a, Theorem 4.2.1]. The ramified case is [LL22, Lemma 2.54] (if $F/\mathbb{Q}_p$ is ramified, the condition $t(M^\flat) \leq 1$ implies $t(M^\flat) = 1$ since we have assumed n is even in the ramified case).

For the split case, the inclusion $\supseteq$ is Corollary 4.2.10. By Lemma 4.2.12, the inclusion $\subseteq$ will hold if we can verify that $\mathcal{Z}(L^\flat)_{\mathscr{H}} \cong \operatorname{Spf} R$ for some finite flat $\mathcal{O}_{\breve{F}}$-algebra R with $R \otimes_{\mathcal{O}_{\breve{F}}} \breve{F}$ reduced (with R not necessarily a domain). This is [Che24b, Lemma 6.7.4], proved via global special cycles. $\square$

Part 2. Local main theorems

6. Local Whittaker functions

We will treat Eisenstein series and local Whittaker functions in greater detail in [Che24c, Part 1]. Here in Section 6, we fix some notation and state key properties (e.g. functional equations) which we need as ingredients for proving our main local theorems on arithmetic Siegel–Weil in the non-Archimedean cases (Section 7).

Let F be a degree 2 finite étale algebra over a non-Archimedean local field F_0. Write $\mathcal{O}_F$ and $\mathcal{O}_{F_0}$ for the corresponding rings of integers. Let q be the residue cardinality of $\mathcal{O}_{F_0}$, and let p be the residue characteristic. We write $\eta\colon F_0^\times \to \{\pm 1\}$ for the quadratic character associated to the extension F/F_0. Let $\Delta \subseteq \mathcal{O}_{F_0}$ be the discriminant ideal associated to F/F_0.

6.1. **Additional notation on $U(m,m)$.** Recall the group $U(m,m)$ defined in (1.4.2). There we defined $U(m,m)$ with respect to the degree 2 finite flat map $\mathbb{Z} \to \mathcal{O}_F$, but we can make the same definition with respect to any finite locally free morphism $A \to B$ of commutative rings such that B is equipped with an involution $b \mapsto \overline{b}$ over A. We set $H \coloneqq U(m,m)$. With the exception of Section 6.4, we take the case where $A \to B$ is $\mathcal{O}_{F_0} \to \mathcal{O}_F$ for the remainder of the article.

We consider

$$P \coloneqq \left\{ h = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \in H \right\} \qquad M \coloneqq \left\{ m(a) = \begin{pmatrix} a & 0 \\ 0 & {}^t\overline{a}^{-1} \end{pmatrix} : a \in \mathrm{Res}_{B/A}\,\mathrm{GL}_m \right\} \tag{6.1.1}$$

$$w \coloneqq \begin{pmatrix} 0 & 1_m \\ -1_m & 0 \end{pmatrix} \qquad N \coloneqq \left\{ n(b) = \begin{pmatrix} 1_m & b \\ 0 & 1_m \end{pmatrix} : b \in \mathrm{Herm}_m \right\} \tag{6.1.2}$$

where P, M, and N are subgroups of H. We have $P(R) = M(R)N(R)$ for all A-algebras R. We consider the standard open compact subgroup

$$K^\circ \coloneqq H(\mathcal{O}_{F_0}) \subseteq H(F_0) \tag{6.1.3}$$

and have $H(F_0) = P(F_0)K^\circ$.

6.2. **Principal series.** Characters are assumed continuous and unitary unless specified otherwise. Given $s \in \mathbb{C}$ and a character $\chi\colon F^\times \to \mathbb{C}^\times$, we form the local *degenerate principal series*

$$I(s,\chi) \coloneqq \mathrm{Ind}_{P(F_0)}^{H(F_0)}(\chi \cdot |-|_F^{s+m/2}). \tag{6.2.1}$$

This is an unnormalized induction, consisting of smooth and K°-finite functions $\Phi\colon H(F_0) \to \mathbb{C}$ satisfying

$$\Phi(m(a)n(b)h, s) = \chi(a)|\det a|_F^{s+m/2}\Phi(h,s) \tag{6.2.2}$$

for all $m(a) \in M(F_0)$ and $n(b) \in N(F_0)$ and $h \in H(F_0)$. Here we wrote $\chi(a) \coloneqq \chi(\det a)$ for short. A section $\Phi(h,s)$ of $I(s,\chi)$ is *standard* if $\Phi(k,s)$ is independent of s for any fixed $k \in K^\circ$. We say that Φ is *spherical* if $\Phi(hk,s) = \Phi(h,s)$ for any $k \in K^\circ$. We write Φ° for the *normalized spherical (standard) section*, i.e. the spherical standard section satisfying $\Phi^\circ(1,s) = 1$ for all $s \in \mathbb{C}$.

6.3. **Normalized local Whittaker functions.** Now assume F_0 has characteristic 0. Let $\chi\colon F^\times \to \mathbb{C}^\times$ and $\psi\colon F_0 \to \mathbb{C}^\times$ be characters with ψ nontrivial and unramified. Given a standard section $\Phi \in I(s,\chi)$, and given $T \in \mathrm{Herm}_m(F_0)$ with $\det T \neq 0$, there is a *local Whittaker function* defined by the absolutely convergent integral

$$W_T(h,s,\Phi) \coloneqq \int_{N(F_0)} \Phi(w^{-1}n(b)h,s)\psi(-\mathrm{tr}(Tb))\, dn(b) \tag{6.3.1}$$

for $h \in H(F_0)$ and $s \in \mathbb{C}$ with $\mathrm{Re}(s) > m/2$, where $dn(b)$ is the self-dual Haar measure with respect to the pairing $b, b' \mapsto \psi(\mathrm{tr}(bb'))$ on $\mathrm{Herm}_m(F_0) \cong N(F_0)$. For any fixed h, the function $W_T(h,s,\Phi)$ admits holomorphic continuation to $s \in \mathbb{C}$ [Kar79, Corollary 3.6.1][KS97]. Extending linearly defines $W_T(h,s,\Phi)$ whenever Φ is a finite meromorphic linear combination of standard sections.

Next, fix $n \in \mathbb{Z}$ and assume $\chi|_{F_0^\times} = \eta^n$. Set $s_0 \coloneqq (n-m)/2$. We also assume χ is unramified. Note that n must then be even if F/F_0 is ramified.

We define a *normalized local Whittaker function*

$$W_T^*(h,s)_n^\circ \coloneqq \Lambda_T(s)_n^\circ W_T(h,s,\Phi^\circ) \tag{6.3.2}$$

using a *local normalizing factor*, which we define as

$$\Lambda_T(s)_n^\circ \coloneqq |\Delta|_{F_0}^{-m(m-1)/4} \left(\prod_{j=0}^{m-1} L(2s+m-j, \eta^{j+n}) \right) |(\det T)\Delta^{\lfloor m/2 \rfloor}|_{F_0}^{-s-s_0}. \tag{6.3.3}$$

For this particular (shifted) product of L-factors, compare [HKS96, §6]. The local L-factors $L(s,\eta^j)$ are the standard ones as in [Tat79, §3]; for the reader's convenience, we recall

$$L(s,\xi) = \begin{cases} (1-\xi(\varpi_0)q^{-s})^{-1} & \text{if } \xi \text{ is unramified} \\ 1 & \text{if } \xi \text{ is ramified} \end{cases}$$

for any quasi-character ξ of F_0, where ϖ_0 denotes a uniformizer of $\mathcal{O}_{F_0}$.

We have a functional equation

$$W_T^*(h,-s)_n^\circ = \eta((-1)^{m(m-1)/2}\det T)^{n-m-1} W_T^*(h,s)_n^\circ, \tag{6.3.4}$$

see [Che24c, Section 5.1]. We use the shorthand

$$W_T^*(s)_n^\circ \coloneqq W_T^*(1_{2m},s)_n^\circ. \tag{6.3.5}$$

If F/F_0 is ramified, we assume for the rest of Section 6 that $\mathcal{O}_{F_0}$ has odd residue characteristic. There is a *normalized local density* (Laurent) polynomial $\mathrm{Den}^*(X,T)_n \in \mathbb{Z}[1/q][X, X^{-1/2}]$ such that

$$W_T^*(s+s_0)_n^\circ = \mathrm{Den}^*(q^{-2s},T)_n \tag{6.3.6}$$

for all $s \in \mathbb{C}$. We also consider the variant $\mathrm{Den}(X,T)_n \in \mathbb{Z}[1/q][X]$ given by

$$|(\det T)\Delta^{\lfloor m/2 \rfloor}|_{F_0}^{s+s_0} W_T^*(s+s_0)_n^\circ = \mathrm{Den}(q^{-2s},T)_n \tag{6.3.7}$$

for all $s \in \mathbb{C}$. We have

$$\begin{aligned} \mathrm{Den}(X,T)_{n+1} &= \mathrm{Den}(q^{-1}X,T)_n && \text{if } F/F_0 \text{ is split} \\ \mathrm{Den}(X,T)_{n+1} &= \mathrm{Den}(-q^{-1}X,T)_n && \text{if } F/F_0 \text{ is inert} \\ \mathrm{Den}(X,T)_{n+2} &= \mathrm{Den}(q^{-2}X,T)_n && \text{if } F/F_0 \text{ is ramified.} \end{aligned} \tag{6.3.8}$$

For our main non-Archimedean local theorems, it is important for us to compute $\mathrm{Den}(X,L)_n$ using Cho–Yamauchi type formulas. Implicit in this are formulas interpolating local Whittaker functions in terms of local densities defined by a limiting procedure. This will be explained/recalled in [Che24c, Lemma 4.4.2].

Remark 6.3.1. On the right-hand side of (6.3.6), we mean evaluating $\mathrm{Den}^*(X,T)_n$ at $X^{1/2} = q^{-s}$. We similarly abuse notation elsewhere. For example, $\mathrm{Den}^*(qX,T)_n \in \mathbb{Z}[1/q^{1/2}][X,X^{-1/2}]$ is obtained from $\mathrm{Den}^*(X,T)_n$ by replacing $X^{1/2}$ with $q^{1/2}X^{1/2}$. The notation $\frac{d}{dX}\colon \mathbb{Q}[X,X^{-1/2}] \to \mathbb{Q}[X,X^{-1/2}]$ means the $\mathbb{Q}$-linear map $X^{j/2} \mapsto (j/2)X^{j/2-1}$.

If L is a non-degenerate Hermitian $\mathcal{O}_F$-lattice which admits a basis with Gram matrix T, we set

$$\mathrm{Den}^*(X,L)_n \coloneqq \mathrm{Den}^*(X,T)_n \qquad \mathrm{Den}(X,L)_n \coloneqq \mathrm{Den}(X,T)_n. \tag{6.3.9}$$

We have

$$\mathrm{Den}^*(X,L)_n = (q^{2s_0}X^{-1/2})^{\lfloor \mathrm{val}(L)\rfloor}\mathrm{Den}(X,L)_n. \tag{6.3.10}$$

The local densities satisfy a certain cancellation property: if L° is a self-dual Hermitian lattice of rank d, then for any non-degenerate Hermitian lattice L and every integer $r \in \mathbb{Z}$ (assume r is even if F/F_0 is ramified), we have

$$\mathrm{Den}(X, L \oplus L^\circ)_{r+d} = \mathrm{Den}(X,L)_r \qquad \mathrm{Den}^*(X, L\oplus L^\circ)_{r+d} = \mathrm{Den}^*(X,L)_r \tag{6.3.11}$$

where $L \oplus L^\circ$ is the orthogonal direct sum. This is explained at the end of [Che24c, Section 4.5].

The functional equation in (6.3.4) implies

$$\mathrm{Den}(q^{2s_0}X^{-1},L)_n = \varepsilon(L)^{n-m-1}X^{-\lfloor \mathrm{val}(L)\rfloor}\mathrm{Den}(q^{2s_0}X,L)_n \tag{6.3.12}$$

$$\mathrm{Den}^*(q^{2s_0}X^{-1},L)_n = \varepsilon(L)^{n-m-1}\mathrm{Den}^*(q^{2s_0}X,L)_n. \tag{6.3.13}$$

Both $\varepsilon(L)$ and $\mathrm{val}(L)$ were defined in Section 2.2. In the case where $\chi|_{F_0^\times}$ is trivial, these non-Archimedean functional equations are essentially [Ike08, Corollary 3.2].

If $m = n$ and F/F_0 is unramified, our $\mathrm{Den}(X,L)_n$ agrees with the polynomial $\mathrm{Den}(X,L)$ from [LZ22a, §3] and [FYZ24]. If $m = n$ and F/F_0 is ramified (so n is even, by the assumption that χ is unramified and $\chi|_{F_0^\times} = \eta^n$), our $\mathrm{Den}(X^2,L)_n$ is the $\mathrm{Den}(X,L)$ of [LL22, §2].

For any m and n, we have

$$W_T^*(s)_n^\circ = 1 \qquad \mathrm{Den}^*(X,T)_n = \mathrm{Den}(X,T)_n = 1 \tag{6.3.14}$$

in the following two cases: (1) if T defines a self-dual Hermitian lattice when m is even and F/F_0 is unramified, and (2) if T defines an almost self-dual Hermitian lattice when m is odd and F/F_0 is ramified.

6.4. **Limits.** In Section 6.4, we take $m = n$ with $m \geq 1$. We consider nonsingular $T \in \mathrm{Herm}_n(F_0)$ of the form $T = \mathrm{diag}(t, T^\flat)$ where $T^\flat \in \mathrm{Herm}_{n-1}(F_0)$ with $T^\flat$ nonsingular, and we study the local Whittaker function $W_T^*(s)_n^\circ$ as $t \to 0$. The following limiting identities will be crucial for the proofs of our main local theorems. We collect them here for easier comparison between the inert/ramified/split and Archimedean cases.

Proofs of the non-Archimedean cases will appear in Section 7 below. The Archimedean case is (a special case of) one formulation of our main Archimedean local theorem, whose proof appears in [Che24a].

The Archimedean case involves an analogous normalized Archimedean local Whittaker function $W_T^*(s)_n^\circ$. Since the present paper focuses on the non-Archimedean case, we have skipped the definition of $W_T^*(s)_n^\circ$ (it is defined in a similar way as Section 6.3, but for a certain scalar weight standard section of the principal series, and for a different normalizing factor involving Gamma functions, etc.). A definition may be found in [Che24a, Section 3.1] (also [Che24c, Section 4.2]).

Although the following limit formulas look very similar in all cases, our proofs of these identities are fairly different in the non-Archimedean vs Archimedean cases.

If F_0 is non-Archimedean and F/F_0 is inert, Proposition 7.5.2 implies

$$\frac{d}{ds}\Big|_{s=-1/2} W_{T^\flat}^*(s)_n^\circ = \lim_{t\to 0}\left(\frac{d}{ds}\Big|_{s=0} W_T^*(s)_n^\circ + (\log|t|_{F_0} - \log q) W_{T^\flat}^*(-1/2)_n^\circ\right) \tag{6.4.1}$$

if the limit is taken over nonzero $t \in F_0$ with $\varepsilon(\mathrm{diag}(t, T^\flat)) = -1$.

If F_0 is non-Archimedean and F/F_0 is split, Proposition 7.5.2 implies

$$\frac{d}{ds}\Big|_{s=-1/2} W_{T^\flat}^*(s)_n^\circ = \lim_{t\to 0}\left(\log q \cdot W_T^*(0)_n^\circ + (\log|t|_{F_0} - \log q)\cdot W_{T^\flat}^*(-1/2)_n^\circ\right) \tag{6.4.2}$$

if the limit is taken over nonzero $t \in F_0$.

If F_0 is non-Archimedean and F/F_0 is ramified, Proposition 7.5.2 implies

$$2\frac{d}{ds}\Big|_{s=-1/2} W_{T^\flat}^*(s)_n^\circ = \lim_{t\to 0}\left(\frac{d}{ds}\Big|_{s=0} W_T^*(s)_n^\circ + (\log|t|_{F_0} - \log q)\cdot W_{T^\flat}^*(-1/2)_n^\circ\right) \tag{6.4.3}$$

if the limit is taken over nonzero $t \in F_0$ with $\varepsilon(\mathrm{diag}(t, T^\flat)) = -1$.

If F/F_0 is $\mathbb{C}/\mathbb{R}$, [Che24a, Proposition 4.1.2] gives

$$\frac{d}{ds}\Big|_{s=-1/2} W_{T^\flat}^*(s)_n^\circ = \lim_{t\to 0^\pm}\left(\frac{d}{ds}\Big|_{s=0} W_T^*(s)_n^\circ + (\log|t|_{F_0} + \log(4\pi) - \Gamma'(1)) W_{T^\flat}^*(-1/2)_n^\circ\right) \tag{6.4.4}$$

where the sign on $0^\pm$ is $-$ if $T^\flat$ is positive definite, and is $+$ if $T^\flat$ is not positive definite. In this Archimedean setting, [Che24a, Proposition 4.1.2] also gives a similar limiting formula for nonsingular $T^\flat$ not positive definite of arbitrary size.

7. Non-Archimedean local main theorems

Let F_0 be a non-Archimedean local field of characteristic 0, residue cardinality q, and residue characteristic p. Let F be a finite étale F_0-algebra of degree 2. We use notation $\breve{F}$ and $\breve{F}_0$ as in Sections 3–5 (there with $F_0 = \mathbb{Q}_p$). We thus have $[\breve{F} : \breve{F}_0] = 1$ if F/F_0 is unramified, and $[\breve{F} : \breve{F}_0] = 2$ if F/F_0 is ramified.

Notation on Hermitian lattices from Section 2.2 will be used freely. For a non-degenerate Hermitian $\mathcal{O}_F$-lattice L, we use the shorthand $\mathrm{val}'(L) \coloneqq \lfloor \mathrm{val}(L) \rfloor \in \mathbb{Z}_{\geq 0}$, as well as $\mathrm{val}'(x) \coloneqq \lfloor \mathrm{val}(x) \rfloor$ for any $x \in L$. Explicitly, we have $\mathrm{val}'(L) = \mathrm{val}(L) - 1/2$ if F/F_0 is ramified and L has odd rank, and $\mathrm{val}'(L) = \mathrm{val}(L)$ otherwise. Fix an integer $n \geq 1$, and assume n is even if F/F_0 is ramified.

If $F_0 = \mathbb{Q}_p$, we form the associated Rapoport–Zink space $\mathcal{N} \coloneqq \mathcal{N}(n-1, 1)$ (Section 3.1). Recall the space of local special quasi-homomorphisms $\mathbf{W} \subseteq \mathbf{V}$ (Section 3.2). Recall that $\mathbf{W}$ and $\mathbf{V}$ are non-degenerate Hermitian F-modules, with $\mathrm{rank}_F \mathbf{W} = \mathrm{rank}_F \mathbf{V} = n$ if $F/\mathbb{Q}_p$ is nonsplit, and $\mathrm{rank}_F \mathbf{W} = n - 1$ and $\mathrm{rank}_F \mathbf{V} = n$ if $F/\mathbb{Q}_p$ is split. Recall that $\varepsilon(\mathbf{V}) = -1$ if $F/\mathbb{Q}_p$ is nonsplit, and that $\varepsilon(\mathbf{V}) = 1$ if $F/\mathbb{Q}_p$ is split.

7.1. **Statement of local main theorem.** We first define the geometric side of our main local identity, taking $F_0 = \mathbb{Q}_p$. We also assume $p \neq 2$ if $F/\mathbb{Q}_p$ is nonsplit. Let $L^\flat \subseteq \mathbf{W}$ be any non-degenerate Hermitian $\mathcal{O}_F$-lattice of rank $n-1$. Form the associated local special cycle $\mathcal{Z}(L^\flat) \subseteq \mathcal{N}$. Recall that the flat part $\mathcal{Z}(L^\flat)_{\mathscr{H}} \subseteq \mathcal{Z}(L^\flat)$ decomposes into quasi-canonical lifting cycles $\mathcal{Z}(M^\flat)^\circ$ for certain lattices $M^\flat$ (Proposition 5.3.1). Recall also the derived vertical local special cycle ${}^{\mathbb{L}}\mathcal{Z}(L^\flat)_{\mathscr{V}} \in \mathrm{gr}^{n-1}_{\mathcal{N}} K_0'(\mathcal{Z}(L^\flat)_{\overline{k}})_{\mathbb{Q}}$ (Section 3.6).

Definition 7.1.1. Given a non-degenerate Hermitian $\mathcal{O}_F$-lattice $L^\flat \subseteq \mathbf{W}$ of rank $n-1$, the associated *local intersection number* is

$$\mathrm{Int}(L^\flat)_n \coloneqq \mathrm{Int}_{\mathscr{H}}(L^\flat)_n + \mathrm{Int}_{\mathscr{V}}(L^\flat)_n \tag{7.1.1}$$

where

$$\mathrm{Int}_{\mathscr{H}}(L^\flat)_n \coloneqq \sum_{\substack{L^\flat \subseteq M^\flat \subseteq M^{\flat *} \\ t(M^\flat) \leq 1}} \mathrm{Int}_{\mathscr{H}}(M^\flat)^\circ_n \tag{7.1.2}$$

with the sum running over full rank lattices $M^\flat \subseteq L^\flat_F$, where

$$\mathrm{Int}_{\mathscr{H}}(M^\flat)^\circ_n \coloneqq 2 \deg \mathcal{Z}(M^\flat)^\circ \cdot \delta_{\mathrm{tau}}(\mathrm{val}'(M^\flat)) \tag{7.1.3}$$

for any non-degenerate integral lattice $M^\flat \subseteq \mathbf{W}$ with $t(M^\flat) \leq 1$, and where

$$\mathrm{Int}_{\mathscr{V}}(L^\flat)_n \coloneqq 2[\breve{F} : \breve{\mathbb{Q}}_p]^{-1} \deg_{\overline{k}}({}^{\mathbb{L}}\mathcal{Z}(L^\flat)_{\mathscr{V}} \cdot \mathcal{E}^\vee). \tag{7.1.4}$$

We relate these local intersection numbers with global intersection numbers at the end of [Che24b, Sections 6.8 and 6.9].

The quantity

$$\deg \mathcal{Z}(M^\flat)^\circ = \begin{cases} [\breve{F} : \breve{\mathbb{Q}}_p] p^{\mathrm{val}'(M^\flat)} (1 - \eta^*(p) p^{-1}) & \text{if } \mathrm{val}'(M^\flat) \geq 1 \\ [\breve{F} : \breve{\mathbb{Q}}_p] & \text{if } \mathrm{val}'(M^\flat) = 0 \end{cases} \tag{7.1.5}$$

is the degree of $\mathcal{Z}(M^\flat)^\circ \to \mathrm{Spf}\, \mathcal{O}_{\breve{F}}$, where $\eta^*(p) \coloneqq -1, 0, 1$ in the inert, ramified, and split cases respectively, as in (5.2.1); the extra factor of $[\breve{F} : \breve{F}_0]$ accounts for the two components of $\mathcal{Z}(M^\flat)^\circ$ when F/F_0 is ramified, as in (5.3.1). Recall that $\delta_{\mathrm{tau}}(s)$ is the "local change of tautological height" defined in (5.2.7), and recall that $\mathcal{E}^\vee$ is the dual tautological bundle on $\mathcal{N}$ (Definition 3.1.7). As in (3.6.14), we understand $\mathcal{E}^\vee = [\mathcal{O}_{\mathcal{N}}] - [\mathcal{E}] \in K_0'(\mathcal{N})$ in (7.1.4), so that ${}^{\mathbb{L}}\mathcal{Z}(L^\flat)_{\mathscr{V}} \cdot \mathcal{E}^\vee \in$

$F^n_{\mathcal{N}}K'_0(\mathcal{Z}(L^\flat)_{\overline{k}})_{\mathbb{Q}}$. For $L^\flat$ as above, recall that $\mathcal{Z}(L^\flat)_{\overline{k}}$ is a scheme proper over $\operatorname{Spec}\overline{k}$ [Che24b, Lemma 6.7.3], so there is a degree map $\deg_{\overline{k}}\colon F^n_{\mathcal{N}}K'_0(\mathcal{Z}(L^\flat)_{\overline{k}}) \to \mathbb{Z}$.

We refer to $\mathrm{Int}_{\mathscr{H}}(L^\flat)_n$ and $\mathrm{Int}_{\mathscr{V}}(L^\flat)_n$ respectively as the "horizontal" and "vertical" parts of the local intersection number.

We next define the automorphic side of our main local identity. For this, we allow F_0 to be an arbitrary finite extension of $\mathbb{Q}_p$ (allowing $p = 2$ if F/F_0 is unramified). If $L^\flat$ is a non-degenerate Hermitian $\mathcal{O}_F$-lattice of rank $n-1$, we set

$$\partial\mathrm{Den}^*(L^\flat)_n \coloneqq -2[\breve{F} : \breve{F}_0]\frac{d}{dX}\bigg|_{X=1}\mathrm{Den}^*(q^2X, L^\flat)_n \tag{7.1.6}$$

where $\mathrm{Den}^*(X, L^\flat)_n \in \mathbb{Z}[1/q][X, X^{-1/2}]$ is a normalized local density (6.3.6). We are abusing notation as in Remark 6.3.1, i.e. $\mathrm{Den}^*(q^2X, L^\flat)_n$ means to evaluate $\mathrm{Den}^*(X, L^\flat)_n$ at $X^{1/2}$ being $qX^{1/2}$. We also set

$$\mathrm{Den}^*(L^\flat)_n \coloneqq [\breve{F} : \breve{F}_0] \cdot \mathrm{Den}^*(q^2, L^\flat)_n. \tag{7.1.7}$$

For non-degenerate integral Hermitian $\mathcal{O}_F$-lattices $M^\flat$ of rank $n-1$ with $t(M^\flat) \le 1$, we define $\partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n$ inductively[12] by the relation

$$\partial\mathrm{Den}^*(M^\flat)_n = \sum_{M^\flat \subseteq N^\flat \subseteq N^{\flat *}} \partial\mathrm{Den}^*_{\mathscr{H}}(N^\flat)^\circ_n \tag{7.1.8}$$

where the sum runs over lattices $N^\flat \subseteq M^\flat_F$. That is, we define $\partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n$ by induction on $\mathrm{val}(M^\flat)$, and have $\partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n \coloneqq \partial\mathrm{Den}^*(M^\flat)_n$ if $M^\flat$ is maximal integral.

Given any non-degenerate integral Hermitian $\mathcal{O}_F$-lattice $L^\flat$ of rank $n-1$, we then define $\partial\mathrm{Den}^*_{\mathscr{V}}(L^\flat)_n$ so that the relation

$$\partial\mathrm{Den}^*(L^\flat)_n = \Bigg(\sum_{\substack{L^\flat \subseteq M^\flat \subseteq M^{\flat *} \\ t(M^\flat)\le 1}} \partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n\Bigg) + \partial\mathrm{Den}^*_{\mathscr{V}}(L^\flat)_n \tag{7.1.9}$$

is satisfied, where the sum runs over lattices $M^\flat \subseteq L^\flat_F$.

Theorem 7.1.2. *Suppose $F_0 = \mathbb{Q}_p$ and that $p \neq 2$ unless $F/\mathbb{Q}_p$ is split. For any non-degenerate Hermitian $\mathcal{O}_F$-lattice $L^\flat \subseteq \mathbf{W}$ of rank $n-1$, we have*

$$\mathrm{Int}(L^\flat)_n = \partial\mathrm{Den}^*(L^\flat)_n. \tag{7.1.10}$$

Moreover, we have

$$\mathrm{Int}_{\mathscr{H}}(M^\flat)^\circ_n = \partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n \qquad \mathrm{Int}_{\mathscr{V}}(L^\flat)_n = \partial\mathrm{Den}^*_{\mathscr{V}}(L^\flat)_n \tag{7.1.11}$$

where $M^\flat \subseteq \mathbf{W}$ is any non-degenerate integral Hermitian $\mathcal{O}_F$-lattice of rank $n-1$ with $t(M^\flat) \le 1$.

On account of the decompositions in (7.1.1), (7.1.2), and (7.1.9), it is clearly enough to prove the refined identities in (7.1.11). The theorem is also clear if $L^\flat$ is not integral, since both sides of (7.1.10) are zero in this case (the special cycle $\mathcal{Z}(L^\flat)$ will be empty, and $\mathrm{Den}(X, L^\flat)_n$ will be identically zero as discussed in [Che24c, Section 4.4]).

[12]The symbol $\circ$ indicates "primitive" here for quasi-canonical lifting cycles, while $\circ$ indicates "spherical" in Section 6.3. There is no notation clash as written, but we hope this remark helps to avoid confusion.

We also record a special value formula (as observed in the inert case by Li and Zhang [LZ22a, Corollary 4.6.1]) for later use. Its proof will appear in Section 7.2.

Lemma 7.1.3. *Suppose $F_0 = \mathbb{Q}_p$ and that $p \neq 2$ unless $F/\mathbb{Q}_p$ is split. For any non-degenerate Hermitian $\mathcal{O}_F$-lattice $L^\flat \subseteq \mathbf{W}$ of rank $n-1$, we have*

$$\deg \mathcal{Z}(L^\flat)_{\mathscr{H}} = \mathrm{Den}^*(L^\flat)_n. \tag{7.1.12}$$

In the preceding lemma statement, $\deg \mathcal{Z}(L^\flat)_{\mathscr{H}}$ means the degree of the adic finite flat morphism $\mathcal{Z}(L^\flat)_{\mathscr{H}} \to \mathrm{Spf}\, \mathcal{O}_{\breve{F}}$ of formal schemes.

7.2. **Horizontal identity.** We will need Cho–Yamauchi formulas for local densities (unitary version, as proved in [LZ22a, Theorem 3.5.1] (inert) [FYZ24, Theorem 2.2(3)] (split) [LL22, Lemma 2.15] (ramified)). For this, we allow F_0 to be an arbitrary finite extension of $\mathbb{Q}_p$ (allowing $p = 2$ if F/F_0 is unramified). Then, if L is any non-degenerate Hermitian $\mathcal{O}_F$-lattice of rank n (still assuming n even if F/F_0 is ramified), we have

$$\mathrm{Den}(X, L)_n = \sum_{L \subseteq M \subseteq M^*} X^{\ell(M/L)} \mathrm{Den}(X, M)^\circ_n \tag{7.2.1}$$

$$\mathrm{Den}(X, M)^\circ_n \coloneqq \prod_{i=0}^{t(M)-1} (1 - \eta^{*i}(\varpi_0) q^i X) \tag{7.2.2}$$

where $\eta^*(\varpi_0) \coloneqq \eta^{*i}(\varpi_0) \coloneqq -1, 0, 1$ if i is odd and $\eta^{*i}(\varpi_0) \coloneqq 1$ if i is even, in the inert, ramified, split cases respectively, and $\mathrm{Den}(X, L)_n \in \mathbb{Z}[X]$ is the local density polynomial normalized as in Section 6.3. The displayed sum runs over lattices $M \subseteq L_F$.

Suppose $L^\flat$ is a Hermitian $\mathcal{O}_F$-lattice of rank $n-1$ (still assuming n even if F/F_0 is ramified). If F/F_0 is unramified, we have

$$\mathrm{Den}(X, L^\flat)_n = \mathrm{Den}(\eta^*(\varpi_0) q^{-1} X, L^\flat)_{n-1} \qquad \mathrm{Den}(X, L^\flat)^\circ_n \coloneqq \mathrm{Den}(\eta^*(\varpi_0) q^{-1} X, L^\flat)^\circ_{n-1} \tag{7.2.3}$$

where the left equality is by (6.3.8), and the right equality applies only if $L^\flat$ is also integral.

If F/F_0 is ramified, we have

$$\mathrm{Den}(X, L^\flat)_n = \sum_{L^\flat \subseteq M^\flat \subseteq M^{\flat *}} (q^{-1} X)^{\ell(M^\flat / L^\flat)} \mathrm{Den}(X, M^\flat)^\circ_n \tag{7.2.4}$$

$$\mathrm{Den}(X, M^\flat)^\circ_n \coloneqq \prod_{i=0}^{\frac{t(M^\flat)-3}{2}} (1 - q^{2i} X) \tag{7.2.5}$$

where the sum runs over lattices $M^\flat \subseteq L_F^\flat$ (may be verified using [LL22, Lemma 2.15]).

If $M^\flat$ is a non-degenerate integral Hermitian $\mathcal{O}_F$-lattice of rank $n-1$, set

$$\mathrm{Den}^*(M^\flat)^\circ_n \coloneqq [\breve{F} : \breve{F}_0] q^{\mathrm{val}'(M^\flat)} \mathrm{Den}(1, M^\flat)^\circ_n. \tag{7.2.6}$$

Combining (7.2.2) and (7.2.3) (unramified cases) and inspecting (7.2.5) (ramified case) shows that $\mathrm{Den}(1, M^\flat)^\circ_n = 0$ unless $t(M^\flat) \leq 1$. For F/F_0 ramified, we are using the fact that $t(M^\flat)$ must be odd, because it has the same parity as $n-1$ (Section 2.2).

We have

$$[\breve{F} : \breve{F}_0]\mathrm{Den}^*(q^2, L^\flat)_n = [\breve{F} : \breve{F}_0]\mathrm{Den}^*(1, L^\flat)_n = \sum_{L^\flat \subseteq M^\flat \subseteq M^{\flat *}} \mathrm{Den}^*(M^\flat)^\circ_n \tag{7.2.7}$$

where the sum runs over lattices $M^\flat \subseteq L^\flat_F$. The first equality follows from the functional equation (6.3.13), and the second equality follows from the Cho–Yamauchi formulas along with (6.3.10). Note $\mathrm{Den}^*(M^\flat)_n = \mathrm{Den}^*(M^\flat)^\circ_n$ if $M^\flat$ is maximal integral.

Proof of Lemma 7.1.3. Specialize (7.2.7) to the case $F_0 = \mathbb{Q}_p$. As explained above, we have $\mathrm{Den}^*(M^\flat)^\circ_n = 0$ if $t(M^\flat) \geq 2$, for all $M^\flat$ in (7.2.7). It thus suffices to verify the refined identity $\deg \mathcal{Z}(M^\flat)^\circ = \mathrm{Den}^*(M^\flat)^\circ_n$ for every $M^\flat$ of rank $n-1$ with $t(M^\flat) \leq 1$. This identity follows from comparing (7.2.6) with (7.1.5), upon using (7.2.2) and (7.2.3) (unramified cases) and (7.2.5) (ramified case) to calculate $\mathrm{Den}(1, M^\flat)^\circ_n = (1 - \eta^*(p)p^{-1})$ if $\mathrm{val}'(M^\flat) \geq 1$ and $\mathrm{Den}(1, M^\flat)^\circ_n = 1$ if $\mathrm{val}'(M^\flat) = 0$. In the unramified cases, we use the fact that $\mathrm{val}'(M^\flat) = 0$ if and only if $t(M^\flat) = 0$. □

Proposition 7.2.1. *Assume $F_0 = \mathbb{Q}_p$ and that $p \neq 2$ unless $F/\mathbb{Q}_p$ is split. For any integral rank $n-1$ non-degenerate Hermitian $\mathcal{O}_F$-lattice $M^\flat \subseteq \mathbf{W}$ with $t(M^\flat) \leq 1$, we have*

$$\mathrm{Int}_{\mathscr{H}}(M^\flat)^\circ_n = \partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n \tag{7.2.8}$$

Proof. By definition, the quantity $\mathrm{Int}_{\mathscr{H}}(M^\flat)^\circ_n$ depends only on $\mathrm{val}(M^\flat)$. Since $t(M^\flat) \leq 1$, we may write $M^\flat = L^{\flat\prime} \oplus L^{\flat\prime\prime}$ (orthogonal direct sum) where $L^{\flat\prime}$ is self-dual of rank $n-2$ and $\mathrm{val}(L^{\flat\prime\prime}) = \mathrm{val}(M^\flat)$ (in the unramified case, this follows upon diagonalizing $M^\flat$; in the ramified case, this follows from picking a "standard basis" as in [LL22, Lemma 2.12]). Using the cancellation property of local densities explained in (6.3.11), we thus reduce to the case $n = 2$ (which we now assume).

By the inductive decompositions in (7.1.2) and (7.1.8), it is enough to show $\mathrm{Int}_{\mathscr{H}}(M^\flat)_n = \partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)_n$ (induct on $\mathrm{val}(M^\flat)$). We have $\partial\mathrm{Den}^*(M^\flat)_n = \partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)_n$ by construction, since $t(M^\flat) \leq 1$ (i.e. compare (7.1.8) and (7.1.9)).

Set $b = \mathrm{val}'(M^\flat)$. Using the Cho–Yamauchi formulas, we find

$$\mathrm{Den}^*(q^2X, M^\flat)_n = X^{-b/2}\sum_{j=0}^{b}(qX)^j \qquad \partial\mathrm{Den}^*(M^\flat)_n = [\breve{F} : \breve{F}_0]\sum_{j=0}^{b}(b-2j)q^j \tag{7.2.9}$$

in all cases. The preceding formulas are valid even if $F_0 \neq \mathbb{Q}_p$ (and also valid if $p = 2$ whenever F/F_0 is unramified), hence why we wrote q instead of p.

We have

$$\begin{aligned}\mathrm{Int}_{\mathscr{H}}(M^\flat)_n &= 2[\breve{F} : \breve{\mathbb{Q}}_p] \sum_{M^\flat \subseteq N^\flat \subseteq N^{\flat *}} p^s(1 - \eta^*(p)p^{-1})\delta_{\mathrm{tau}}(s) \\ &= -[\breve{F} : \breve{\mathbb{Q}}_p] \sum_{M^\flat \subseteq N^\flat \subseteq N^{\flat *}} p^s(1 - \eta^*(p)p^{-1}) \left(s - \frac{(1-p^{-s})(1-\eta^*(p))}{(1-p^{-1})(p-\eta^*(p))}\right)\end{aligned} \tag{7.2.10}$$

where the sum runs over lattices $N^\flat \subseteq M^\flat_F$, and where $s \coloneqq \mathrm{val}'(N^\flat)$.

We prove the identity $\mathrm{Int}_{\mathscr{H}}(M^\flat)_n = \partial\mathrm{Den}^*(M^\flat)_n$ by induction on b. The case $b = 0$ is clear, as both quantities are 0.

Next suppose $b \geq 1$. Let $M^{\flat\prime}$ be a rank one non-degenerate lattice with $\mathrm{val}'(M^{\flat\prime}) = b-1$. Let $M^{\flat\prime\prime}$ be a rank one non-degenerate lattice with $\mathrm{val}'(M^{\flat\prime\prime}) = b-2$.

We have

$$\partial\mathrm{Den}^*(M^\flat)_n - \partial\mathrm{Den}^*(M^{\flat\prime})_n = [\breve{F} : \breve{F}_0](-bq^b + \sum_{j=0}^{b-1} q^j) \tag{7.2.11}$$

$$\partial\mathrm{Den}^*(M^\flat)_n - \partial\mathrm{Den}^*(M^{\flat\prime\prime})_n = [\breve{F} : \breve{F}_0](-bq^b - bq^{b-1} + 2\sum_{j=0}^{b-1} q^j). \tag{7.2.12}$$

If $F/\mathbb{Q}_p$ is inert, we find

$$\begin{aligned}\mathrm{Int}_{\mathscr{H}}(M^\flat)_n - \mathrm{Int}_{\mathscr{H}}(M^{\flat\prime\prime})_n &= -[\breve{F} : \breve{\mathbb{Q}}_p]p^b(1+p^{-1})\left(b - 2\frac{(1-p^{-b})}{(1-p^{-1})(p+1)}\right) \\ &= [\breve{F} : \breve{\mathbb{Q}}_p](-bp^b - bp^{b-1} + 2\sum_{j=0}^{b-1} p^j). \end{aligned} \tag{7.2.13}$$

If $F/\mathbb{Q}_p$ is ramified, we find

$$\mathrm{Int}_{\mathscr{H}}(M^\flat)_n - \mathrm{Int}_{\mathscr{H}}(M^{\flat\prime})_n = -[\breve{F} : \breve{\mathbb{Q}}_p]p^b\left(b - \frac{(1-p^{-b})}{(1-p^{-1})p}\right) \tag{7.2.14}$$

$$= [\breve{F} : \breve{\mathbb{Q}}_p](-bp^b + \sum_{j=0}^{b-1} p^j). \tag{7.2.15}$$

If $F/\mathbb{Q}_p$ is split, we find

$$\mathrm{Int}_{\mathscr{H}}(M^\flat)_n - \mathrm{Int}_{\mathscr{H}}(M^{\flat\prime})_n = -[\breve{F} : \breve{\mathbb{Q}}_p]\sum_{j=0}^{b} p^j(1-p^{-1})j \tag{7.2.16}$$

$$= [\breve{F} : \breve{\mathbb{Q}}_p](-bp^b + \sum_{j=0}^{b-1} p^j). \tag{7.2.17}$$

This proves the proposition in all cases, by induction on b. □

Corollary 7.2.2. *Theorem 7.1.2 holds when $n = 2$.*

Proof. If $n = 2$, Proposition 7.2.1 shows $\mathrm{Int}_{\mathscr{H}}(L^\flat)_n = \partial\mathrm{Den}^*_{\mathscr{H}}(L^\flat)_n = \partial\mathrm{Den}^*(L^\flat)_n$. We have $\mathrm{gr}^1_{\mathcal{N}} K'_0(\mathcal{Z}(L^\flat)_{\overline{k}})_{\mathbb{Q}} = 0$ because $\mathcal{Z}(L^\flat)_{\overline{k}}$ is a scheme and because the reduced subscheme $\mathcal{N}_{\mathrm{red}} \subseteq \mathcal{N}$ is 0-dimensional (a disjoint union of copies of $\mathrm{Spec}\,\overline{k}$), see Lemma 3.5.1. Hence $\mathrm{Int}_{\mathscr{V}}(L^\flat)_n = 0$ since ${}^{\mathbb{L}}\mathcal{Z}(L^\flat)_{\mathscr{V}} \in \mathrm{gr}^1_{\mathcal{N}} K'_0(\mathcal{Z}(L^\flat)_{\overline{k}})_{\mathbb{Q}}$. □

7.3. **Induction formula.** Throughout Sections 7.3 and 7.5, we allow F_0 to be an arbitrary finite extension of $\mathbb{Q}_p$ (allowing $p = 2$ if F/F_0 is unramified). We take the following setup for the rest of Section 7 (i.e. the notations n, V, L, L', L'', $L^\flat$, x, x', and x'' are all reserved unless otherwise indicated).

Setup 7.3.1. Let V be a non-degenerate Hermitian F-module of rank n, with pairing $(-,-)$. Assume n is even if F/F_0 is ramified. Let $L^\flat \subseteq V$ be a non-degenerate Hermitian $\mathcal{O}_F$-lattice

of rank $n-1$. Let $x, x', x'' \in V$ be nonzero and orthogonal to $L^\flat$ with $\langle x\rangle \subseteq \langle x'\rangle \subseteq \langle x''\rangle$ and $\mathrm{length}_{\mathcal{O}_F}(\langle x'\rangle/\langle x\rangle) = \mathrm{length}_{\mathcal{O}_F}(\langle x''\rangle/\langle x'\rangle) = 1$. Set

$$L \coloneqq L^\flat \oplus \langle x\rangle \qquad L' \coloneqq L^\flat \oplus \langle x'\rangle \qquad L'' \coloneqq L^\flat \oplus \langle x''\rangle \tag{7.3.1}$$

The notation L'' and x'' will only appear in our proof of the induction formula (Proposition 7.3.2) for F/F_0 split.

Proposition 7.3.2 (Induction formula)**.** *If* $\mathrm{val}(x) > 2[\breve{F} : \breve{F}_0]^{-1} a_{\max}(L^\flat)$*, we have*

$$\mathrm{Den}(X, L)_n = \begin{cases} X^2\mathrm{Den}(X, L')_n + (1-X)\mathrm{Den}(q^2X, L^\flat)_n & \text{if } F/F_0 \text{ is inert} \\ X\mathrm{Den}(X, L')_n + (1-X)\mathrm{Den}(q^2X, L^\flat)_n & \text{if } F/F_0 \text{ is ramified} \\ X\mathrm{Den}(X, L')_n + \mathrm{Den}(q^2X, L^\flat)_n & \text{if } F/F_0 \text{ is split.} \end{cases} \tag{7.3.2}$$

In the inert case, this is [Ter13, Theorem 5.1] (strictly speaking, with a blanket $p \neq 2$ assumption), which is a unitary analogue of [Kat99, Theorem 2.6(1)] and [CY20, Corollary 4.10] (orthogonal groups); see also [LZ22a, Proposition 3.7.1] (there allowing $p = 2$) for a statement closer to ours.

Using the Cho–Yamauchi formulas, we give a uniform proof of the inert and ramified cases (Lemma 7.3.6). Our lower bounds on $\mathrm{val}(x)$ should be nonsharp in general, e.g. the above references in the inert case show the induction formula with the weaker assumption $\mathrm{val}(x) > a_{\max}(L^\flat)$, but this makes no difference for the proof of Theorem 7.1.2, where we will take $\mathrm{val}(x) \to \infty$ (Proposition 7.5.2).

The case when F/F_0 is split is more difficult for us, and the same method only shows a weaker version of the induction formula (stated in Lemma 7.3.6), which is insufficient for our purposes. Extracting the induction formula from this weak version is the subject of Section 7.4. For the proof of Theorem 7.1.2, only the statement of the induction formula and the definitions in (7.3.3) and (7.3.4) will be needed.

We first record a few preparatory lemmas. As in Section 2.2, we fix a uniformizer $\varpi \in \mathcal{O}_F$ and a generator $u \in \mathcal{O}_F$ of the different ideal such that $\varpi^\sigma = -\varpi$ and $u^\sigma = -u$. We say a quantity *stabilizes*, e.g. for $\mathrm{val}(x) > C$ (for some constant C) if that quantity does not depend on x if $\mathrm{val}(x) > C$. When F/F_0 is nonsplit, given an $\mathcal{O}_F$-module M and $m \in M$, we say e.g. that m is *exact ϖ^e-torsion* for $e \geq 0$ if $\varpi^e m = 0$ but $\varpi^{e-1} m \neq 0$ (and if $e = 0$, the only exact ϖ^e-torsion element is 0). We use similar terminology for $\mathcal{O}_{F_0}$-modules and exact ϖ_0^e-torsion elements, etc..

Lemma 7.3.3. *Let M be a non-degenerate integral Hermitian $\mathcal{O}_F$-lattice of rank m. Suppose elements $w_1, \dots, w_r \in M$ have $\mathcal{O}_F$-span M. Write T for the associated Gram matrix. Then $t(M) + \mathrm{rank}((uT) \otimes \mathcal{O}_F/\varpi) = m$.*

Proof. If F/F_0 is split, the rank continues to make sense because T is Hermitian (e.g. diagonalize the Hermitian form). In the unramified cases, the lemma follows by diagonalizing the Hermitian form. In the ramified case, the lemma follows by putting M into "standard form" (i.e. an orthogonal direct sum of rank one lattices and rank two hyperbolic lattices) as in [LL22, Definition 2.11]. We are allowing m even or odd. □

Lemma 7.3.4. *Let M be a non-degenerate integral Hermitian $\mathcal{O}_F$-lattice of rank m. Suppose $M_F = W' \oplus W''$ is an orthogonal decomposition with W'' of rank 1.*

(1) We have $t(M) - 1 \leq t(M \cap W') \leq t(M) + 1$.

(2) Let $M' \subseteq W'$ *and* $M'' \subseteq W''$ *be the images of* M *under the projections* $M_F \to W'$ *and* $M_F \to W''$. *Assume that* M' *and* M'' *are integral and that* $\mathrm{val}(M'') > 0$. *Then we have* $t(M) = t(M') + 1$.

Proof.

(1) The ramified case follows from [LL22, Lemma 2.23(2)]. The inert case when $t(M) = 0$ is [LZ22a, Lemma 4.5.1]. The same proof works in general for arbitrary F/F_0 in arbitrary characteristic: select any basis $w_1, \ldots, w_{m-1}$ of $M \cap W'$ with Gram matrix denoted $T^\flat$, extend to a basis $w_1, \ldots, w_m$ of M with Gram matrix denoted T, then use the formulas $t(M) + \mathrm{rank}((uT) \otimes \mathcal{O}_F/\varpi) = m$ and $t(M \cap W') + \mathrm{rank}((uT^\flat) \otimes \mathcal{O}_F/\varpi) = m - 1$ from Lemma 7.3.3.

(2) Let $\underline{w} = [w_1, \ldots, w_m]$ be any basis of M, and let $T = (\underline{w}, \underline{w})$ be the corresponding Gram matrix. Let $\underline{w}' = [w_1', \ldots, w_m']$ be the projection of $\underline{w}$ to W', with Gram matrix $T' = (\underline{w}', \underline{w}')$. Since $\mathrm{val}(M'') > 0$, we see $(uT) \otimes \mathcal{O}_F/\varpi = (uT') \otimes \mathcal{O}_F/\varpi$. Applying Lemma 7.3.3 twice (once for M and $r = m$ and once for M' and $r = m$) proves the claim. □

Set

$$\mathrm{Den}_{L^\flat,x}(X)_n^\circ \coloneqq \sum_{\substack{L \subseteq M \subseteq M^* \\ M \cap L_F^\flat = L^\flat}} X^{\ell(M/L)} \mathrm{Den}(X, M)_n^\circ \tag{7.3.3}$$

$$\mathrm{Den}_{L^\flat,x}(X)_n \coloneqq \mathrm{Den}(X, L)_n = \sum_{L^\flat \subseteq M^\flat \subseteq M^{\flat *}} \mathrm{Den}_{M^\flat,x}(X)_n^\circ \tag{7.3.4}$$

where the first sum runs over lattices $M \subseteq V$ and the second sum runs over lattices $M^\flat \subseteq L_F^\flat$. Note that the only dependence on x is on $\mathrm{val}(x)$ (since $\mathrm{Den}_{L^\flat,x}(X)_n^\circ$ only depends on the isomorphism class of the Hermitian lattice L).

Lemma 7.3.5. *The polynomial*

$$f_x(X) \coloneqq \begin{cases} \mathrm{Den}_{L^\flat,x}(X)_n^\circ - X^2 \mathrm{Den}_{L^\flat,x'}(X)_n^\circ & \text{if } F/F_0 \text{ is inert} \\ \mathrm{Den}_{L^\flat,x}(X)_n^\circ - X \mathrm{Den}_{L^\flat,x'}(X)_n^\circ & \text{if } F/F_0 \text{ is ramified} \\ \mathrm{Den}_{L^\flat,x}(X)_n^\circ - 2X \mathrm{Den}_{L^\flat,x'}(X)_n^\circ + X^2 \mathrm{Den}_{L^\flat,x''}(X)_n^\circ & \text{if } F/F_0 \text{ is split} \end{cases} \tag{7.3.5}$$

(an element of $\mathbb{Z}[X]$*) stabilizes for* $\mathrm{val}(x) > 2[\breve{F} : \breve{F}_0]^{-1} a_{\max}(L^\flat)$.

Proof. If F/F_0 is split, write $\varpi = \varpi_1 \varpi_2$ for $\varpi_i \in \mathcal{O}_F$ with $\mathrm{val}(\varpi_1 x) = \mathrm{val}(\varpi_2 x) = \mathrm{val}(x) + 1$. We may assume $x'' = \varpi^{-1} x$ in this case. Note $\mathrm{Den}_{L^\flat,x'}(X)_n^\circ = \mathrm{Den}_{L^\flat,\varpi_1^{-1}x}(X)_n^\circ = \mathrm{Den}_{L^\flat,\varpi_2^{-1}x}(X)_n^\circ$.

Inspecting (7.3.3) shows

$$f_x(X) = \sum_{\substack{L\subseteq M\subseteq M^* \\ M\cap L_F^\flat = L^\flat \\ \varpi^{-1}x\notin M}} X^{\ell(M/L)}\mathrm{Den}(X,M)_n^\circ \qquad \text{if } F/F_0 \text{ is nonsplit} \tag{7.3.6}$$

$$f_x(X) = \sum_{\substack{L\subseteq M\subseteq M^* \\ M\cap L_F^\flat = L^\flat \\ \varpi_1^{-1}x\notin M \\ \varpi_2^{-1}x\notin M}} X^{\ell(M/L)}\mathrm{Den}(X,M)_n^\circ \qquad \text{if } F/F_0 \text{ is split} \tag{7.3.7}$$

where the sums run over lattices $M \subseteq V$. For each such M, we know $L^\flat \subseteq M$ is a saturated sublattice, hence $M = L^\flat \oplus \langle \xi \rangle$ (not necessarily orthogonal direct sum) for some $\xi \in V$.

If $L^\flat$ is not integral, then the lemma is trivial as the polynomials of the lemma statement are 0. We assume $L^\flat$ is integral for the rest of the proof.

If F/F_0 is nonsplit, each lattice M appearing in (7.3.6) is of the form $M = L^\flat \oplus \langle y + \varpi^{-e}x\rangle$ for a uniquely determined element $y \in L^{\flat *}/L^\flat$, where $e \in \mathbb{Z}_{\geq 0}$ is such that $y \in L^{\flat *}/L^\flat$ is of exact ϖ^e-torsion. Conversely, an element $y \in L^{\flat *}/L^\flat$ gives rise to an M appearing in (7.3.6) if and only if $\mathrm{val}(y + \varpi^{-e}x) \geq 0$. If $\mathrm{val}(x) > 2[\breve{F} : \breve{F}_0]^{-1}a_{\max}(L^\flat)$, then $\mathrm{val}(\varpi^{-e}x) > 0$, so $\mathrm{val}(y + \varpi^{-e}x) \geq 0$ holds if and only if $\mathrm{val}(y) \geq 0$.

If F/F_0 is split, the preceding paragraph holds upon replacing ϖ^{-e} with $\varpi_1^{-e_1}\varpi_2^{-e_2}$ for $e_1, e_2 \in \mathbb{Z}_{\geq 0}$ such that $y \in L^{\flat *}/L^\flat$ is of exact $\varpi_1^{e_1}\varpi_2^{e_2}$-torsion (i.e. $\varpi_1^{e_1}\varpi_2^{e_2}y \in L^\flat$ but $\varpi_1^{e_1-1}\varpi_2^{e_2}y \notin L^\flat$ and $\varpi_1^{e_1}\varpi_2^{e_2-1}y \notin L^\flat$).

In the previous notation, we thus have

$$f_x(X) = \sum_{\substack{y\in L^{\flat *}/L^\flat \\ \mathrm{val}(y)\geq 0}} X^{\ell((L^\flat+\langle y\rangle)/L^\flat)}\mathrm{Den}(X,M)_n^\circ \tag{7.3.8}$$

where the sum runs over y, with $M = L^\flat \oplus \langle y + \varpi^{-e}x\rangle$ in the nonsplit case and $M = L^\flat \oplus \langle y + \varpi_1^{-e_1}\varpi_2^{-e_2}x\rangle$ in the split case.

In the notation of (7.3.8), we have $t(M) = t(L^\flat + \langle y\rangle) + 1$ by Lemma 7.3.4(2) (using $\mathrm{val}(\varpi^{-e}x) > 0$ in the nonsplit case and $\mathrm{val}(\varpi_1^{-e_1}\varpi_2^{-e_2}x) > 0$ in the split case). Hence we have

$$\mathrm{Den}(X,M)_n^\circ = \prod_{i=0}^{t(L^\flat+\langle y\rangle)} (1 - \eta^{*i}(\varpi_0)q^iX) \tag{7.3.9}$$

(see definition in (7.2.2)), and now the right-hand side of (7.3.8) clearly depends only on $L^\flat$ (and not on x). □

Lemma 7.3.6. *With notation as above, assume* $\mathrm{val}(x) > 2[\breve{F} : \breve{F}_0]^{-1}a_{\max}(L^\flat)$. *We have*

$$(1-X)\mathrm{Den}(q^2X, L^\flat)_n = \begin{cases} \mathrm{Den}(X,L) - X^2\mathrm{Den}(X,L') & \text{if } F/F_0 \text{ is inert} \\ \mathrm{Den}(X,L) - X\mathrm{Den}(X,L') & \text{if } F/F_0 \text{ is ramified} \\ \mathrm{Den}(X,L) - 2X\mathrm{Den}(X,L') + X^2\mathrm{Den}(X,L'') & \text{if } F/F_0 \text{ is split.} \end{cases} \tag{7.3.10}$$

Proof. Combining (7.3.8) and (7.3.9), we find that the right-hand side of (7.3.10) is given by

$$\sum_{L^\flat \subseteq M^\flat \subseteq M^{\flat *}} X^{\ell(M^\flat/L^\flat)} \sum_{\substack{y \in M^{\flat *}/M^\flat \\ \mathrm{val}(y) \geq 0}} X^{\ell((M^\flat + \langle y \rangle)/M^\flat)} \prod_{i=0}^{t(M^\flat + \langle y \rangle)} (1 - \eta^{*i}(\varpi_0) q^i X) \tag{7.3.11}$$

in all cases, where the outer sum runs over lattices $M^\flat \subseteq L_F^\flat$.

Collecting the terms with $M^\flat + \langle y \rangle = N^\flat$ for fixed integral lattices $N^\flat \subseteq L_F^\flat$, we find that (7.3.11) is equal to

$$\sum_{L^\flat \subseteq N^\flat \subseteq N^{\flat *}} X^{\ell(N^\flat/L^\flat)} \prod_{i=0}^{t(N^\flat)} (1 - \eta^{*i}(\varpi_0) q^i X) \sum_{\substack{L^\flat \subseteq M^\flat \subseteq N^\flat \\ N^\flat/M^\flat \text{cyclic}}} (\text{number of generators of } N^\flat/M^\flat)$$

where the outer sum runs over lattices $N^\flat \subseteq L_F^\flat$ and the inner sum runs over lattices $M^\flat$. We have

$$\begin{aligned} \sum_{\substack{L^\flat \subseteq M^\flat \subseteq N^\flat \\ N^\flat/M^\flat \text{cyclic}}} (\text{number of generators of } N^\flat/M^\flat) &= \sum_{\substack{N^{\flat *} \subseteq M^{\flat *} \subseteq L^{\flat *} \\ M^{\flat *}/N^{\flat *} \text{cyclic}}} (\text{number of generators of } M^{\flat *}/N^{\flat *}) \\ &= |L^{\flat *}/N^{\flat *}| = |N^\flat/L^\flat| = q^{\ell(N^\flat/L^\flat)} \end{aligned}$$

so (7.3.11) is equal to

$$\sum_{L^\flat \subseteq N^\flat \subseteq N^{\flat *}} (qX)^{\ell(N^\flat/L^\flat)} \prod_{i=0}^{t(N^\flat)} (1 - \eta^{*i}(\varpi_0) q^i X). \tag{7.3.12}$$

Inspecting the Cho–Yamauchi formulas (and surrounding discussion) at the beginning of Section 7.2 shows that the displayed expression is equal to $(1 - X)\mathrm{Den}(q^2 X, L^\flat)_n$ in all cases (if F/F_0 is ramified, note that $t(N^\flat)$ is always odd because $N^\flat$ has rank $n - 1$, which we have assumed is odd in the ramified case). □

7.4. **More on induction formula: split.** Now assume that F/F_0 is split. Suppose we can show that $\mathrm{Den}(X, L) - X\mathrm{Den}(X, L')$ stabilizes for $\mathrm{val}(x) > 2a_{\max}(L^\flat)$. This implies $\mathrm{Den}(X, L) - X\mathrm{Den}(X, L') = \mathrm{Den}(X, L') - X\mathrm{Den}(X, L'')$, and substituting into Lemma 7.3.6 would then give $(1 - X)(\mathrm{Den}(X, L) - X\mathrm{Den}(X, L')) = (1 - X)\mathrm{Den}(q^2 X, L^\flat)_n$. Dividing by $(1 - X)$ exactly yields the split case of the induction formula, Proposition 7.3.2.

The rest of Section 7.4 is devoted to verifying the stabilization of $\mathrm{Den}(X, L) - X\mathrm{Den}(X, L')$. We define some more notation (only used in Section 7.4). Fix a uniformizer ϖ_0 of $\mathcal{O}_{F_0}$, and consider the elements

$$\varpi_1 = (\varpi_0, 1) \qquad \varpi_2 = (1, -\varpi_0) \qquad \mathrm{id}_1 = (1, 0) \qquad \mathrm{id}_2 = (0, 1) \tag{7.4.1}$$

in $\mathcal{O}_F = \mathcal{O}_{F_0} \times \mathcal{O}_{F_0}$. Given an $\mathcal{O}_F$-module M, we set $M_1 \coloneqq \mathrm{id}_1 \cdot M$ and $M_2 \coloneqq \mathrm{id}_2 \cdot M$ (so $M = M_1 \oplus M_2$). We similarly write $y_1 \coloneqq \mathrm{id}_1 \cdot y$ and $y_2 \coloneqq \mathrm{id}_2 \cdot y$ for $y \in M$. If M is a non-degenerate Hermitian $\mathcal{O}_F$-lattice, we set $M_1^* \coloneqq \mathrm{id}_2 \cdot M^*$ and $M_2^* \coloneqq \mathrm{id}_1 \cdot M^*$. If M is moreover integral, the Hermitian pairing induces an identification $M_2^*/M_1 \cong \mathrm{Hom}_{\mathcal{O}_{F_0}}(M_1^*/M_2, F_0/\mathcal{O}_{F_0})$.

For integers $t \geq 0$, we set

$$\mathfrak{m}(t, X) \coloneqq \prod_{i=0}^{t-1}(1 - q^i X) \tag{7.4.2}$$

so that $\mathrm{Den}(X, M)_n^\circ = \mathfrak{m}(t(M), X)$ for any integral non-degenerate Hermitian $\mathcal{O}_F$-lattice M of rank n. If $\mathcal{T}$ is a finite length $\mathcal{O}_{F_0}$-module, we set

$$t_0(\mathcal{T}) \coloneqq \dim_{\mathbb{F}_q}(\mathcal{T} \otimes_{\mathcal{O}_{F_0}} \mathbb{F}_q) \qquad \ell_0(\mathcal{T}) \coloneqq \mathrm{length}_{\mathcal{O}_{F_0}}(\mathcal{T}). \tag{7.4.3}$$

Lemma 7.4.1. *Consider the polynomial*

$$h_{\mathrm{diff},x}(X) \coloneqq \sum_{\substack{L \subseteq M \subseteq M^* \\ M_1 \cap L_F^\flat = L_1^\flat \\ M_2 \cap L_F^\flat \neq L_2^\flat \\ M/L \text{ is cyclic} \\ \varpi_1^{-1} x \notin M}} X^{\ell(M/L)} \mathfrak{m}(t(M), X) \tag{7.4.4}$$

where the sum runs over lattices $M \subseteq V$ (satisfying the displayed conditions). This sum stabilizes for $\mathrm{val}(x) > 2a_{\max}(L^\flat)$.

Proof. Each lattice M in the sum is of the form $M = L + \langle \xi \rangle$ for a unique element $\xi = y + \varpi_1^{-e_1}\varpi_2^{-e_2}x \in L^*/L$ with $y \in L^{\flat *}$, such that $\mathrm{val}(\xi) \geq 0$, and with $e_1, e_2 \in \mathbb{Z}_{\geq 0}$.

Assume $\mathrm{val}(x) > 2a_{\max}(L^\flat)$. We claim that $\mathrm{val}(y) \geq 0$ (in the notation above). The additional conditions on M imply that $y_1 \in L_2^{\flat *}/L_1^\flat$ is of exact $\varpi_1^{e_1}$-torsion and that $\varpi_2^{e_2} y_2 \notin L^\flat$. We thus have $e_1 \leq a_{\max}(L^\flat)$ and $e_2 < a_{\max}(L^\flat)$, so $\mathrm{val}(\varpi_1^{-e_1}\varpi_2^{-e_2}x) > 0$ when $\mathrm{val}(x) > 2a_{\max}(L^\flat)$. This implies that $\mathrm{val}(y) \geq 0$ as well.

Consider the F-linear (non-unitary) automorphism $\phi\colon V \to V$ which is the identity on $L_F^\flat$ and sends $x \mapsto \varpi_2 x$. Then $M \mapsto \phi(M)$ is a bijection from the set of lattices appearing in the sum for $h_{\mathrm{diff},x}(X)$ to the set of lattices appearing in the sum for $h_{\mathrm{diff},\varpi_2 x}(X)$ (we remind the reader that L depends on x as well).

In the above setup, an application of Lemma 7.3.4(2) shows $t(M) = t(\phi(M)) = t(L^\flat + \langle y \rangle) + 1$. We also find $\ell(M/L) = \ell(\phi(M)/(L^\flat \oplus \langle \varpi_2 x \rangle)) = \ell((L^\flat + \langle y \rangle)/L^\flat)$. This shows $h_{\mathrm{diff},x}(X) = h_{\mathrm{diff},\varpi_2 x}(X)$ (compare the M term and the $\phi(M)$ term). This proves the lemma, as the x-dependence of $h_{\mathrm{diff},x}(X)$ is only on $\mathrm{val}(x)$. □

Lemma 7.4.2. *Let $\mathcal{T}$ be a finite length $\mathcal{O}_{F_0}$-module, and suppose $\mathcal{T}$ is ϖ_0^e-torsion. For any integer $b > e$, form the $\mathcal{O}_{F_0}$-module $A = \mathcal{T} \oplus (\varpi_0^{-b}\mathcal{O}_{F_0}/\mathcal{O}_{F_0})$. Consider $u = t + w \in A$ with $t \in \mathcal{T}$ and $w \in \varpi_0^{-b}\mathcal{O}_{F_0}/\mathcal{O}_{F_0}$ both of exact ϖ_0^r-torsion. There is a (non-canonical) isomorphism*

$$A/(u) \cong (\mathcal{T}/(t)) \oplus (\varpi_0^{-b}\mathcal{O}_{F_0}/\mathcal{O}_{F_0}). \tag{7.4.5}$$

Proof. This follows from the structure theorem for finitely generated modules over the discrete valuation ring $\mathcal{O}_{F_0}$. For example, we can select elements $t_1, \ldots, t_c \in \mathcal{T}$ such that $\mathcal{T} = \langle t_1 \rangle \oplus \cdots \oplus \langle t_c \rangle$ and such that $t = \varpi_0^s t_1$ for some $s \geq 0$. The case $r = 0$ is trivial, so take $r \geq 1$. Then $r + s \leq e$. If $w' \in \varpi_0^{-b}\mathcal{O}_{F_0}/\mathcal{O}_{F_0}$ is such that $\varpi_0^s w' = w$ there is an isomorphism

$$\mathcal{T} \oplus (\varpi_0^{-b}\mathcal{O}_{F_0}/\mathcal{O}_{F_0}) \to A \tag{7.4.6}$$

sending $t_1 \mapsto t_1 + w'$, $t_i \mapsto t_i$ for $i \geq 2$, and $z \mapsto z$ (for any generator z of $\varpi_0^{-b}\mathcal{O}_{F_0}/\mathcal{O}_{F_0}$). This isomorphism takes t to $t + w$. □

Given a finite torsion cyclic $\mathcal{O}_{F_0}$-module $N \cong \mathcal{O}_{F_0}/\varpi_0^a\mathcal{O}_{F_0}$, we set $\mathrm{ord}(N) \coloneqq a$.

Lemma 7.4.3. *Let $\mathcal{T}$ be a finite length $\mathcal{O}_{F_0}$-module, and assume $\mathcal{T}$ is ϖ_0^e-torsion for some $e \geq 0$. For any integer $b \geq 0$, form the $\mathcal{O}_{F_0}$-module $A_b \coloneqq \mathcal{T} \oplus (\varpi_0^{-b}\mathcal{O}_{F_0}/\mathcal{O}_{F_0})$. The polynomial*

$$\alpha_b \coloneqq \sum_{\substack{\text{cyclic submodules} \\ N \subseteq A_b}} X^{\mathrm{ord}(N)}\mathfrak{m}(t_0(A_b/N), X) \tag{7.4.7}$$

stabilizes for $b > e$.

Proof. Applying $- \otimes_{\mathcal{O}_{F_0}} \mathbb{F}_q$ to the exact sequence

$$0 \to N \to A_b \to A_b/N \to 0 \tag{7.4.8}$$

shows

$$t_0(A_b/N) = \begin{cases} t_0(A_b) & \text{if } N \subseteq \varpi_0 A_b \\ t_0(A_b) - 1 & \text{if } N \not\subseteq \varpi_0 A_b \end{cases} \tag{7.4.9}$$

for any cyclic submodule $N \subseteq A_b$. We also have $t_0(A_b) = t_0(\mathcal{T}) + 1$ if $b \geq 1$.

There is a natural inclusion $A_b \to A_{b+1}$. For any cyclic submodule $N \subseteq A_b$, we have

$$t_0(A_b/N) = \begin{cases} t_0(A_{b+1}/N) - 1 & \text{if } N = \langle t + \varpi_0^{-b}\rangle \text{ with } t \in \varpi_0\mathcal{T} \\ t_0(A_{b+1}/N) & \text{otherwise} \end{cases} \tag{7.4.10}$$

where $\varpi_0^{-b} \in \varpi_0^{-b}\mathcal{O}_{F_0}/\mathcal{O}_{F_0}$. Assume $b > e$. Then, in the first case above, the element $t \in \mathcal{T}$ is uniquely determined by N (using $b > e$). The cyclic submodules $N \subseteq A_{b+1}$ with $N \not\subseteq A_b$ are of the form $N = \langle t + \varpi_0^{-b-1}\rangle$ for a unique $t \in \mathcal{T}$.

We thus have

$$\alpha_{b+1} - \alpha_b \tag{7.4.11}$$

$$= \sum_{\substack{t \in \mathcal{T} \\ N = \langle t + \varpi_0^{-b-1}\rangle}} X^{\mathrm{ord}(N)}\mathfrak{m}(t_0(A_{b+1}/N), X) + \sum_{\substack{t \in \varpi_0\mathcal{T} \\ N = \langle t + \varpi_0^{-b}\rangle}} X^{\mathrm{ord}(N)}\mathfrak{m}(t_0(A_{b+1}/N), X) \tag{7.4.12}$$

$$- \sum_{\substack{t \in \varpi_0\mathcal{T} \\ N = \langle t + \varpi_0^{-b}\rangle}} X^{\mathrm{ord}(N)}\mathfrak{m}(t_0(A_b/N), X) \tag{7.4.13}$$

where the sums run over $t \in \mathcal{T}$ or $t \in \varpi_0\mathcal{T}$, as indicated. We compute

$$\sum_{\substack{t \in \mathcal{T} \\ N = \langle t + \varpi_0^{-b-1}\rangle}} X^{\mathrm{ord}(N)}\mathfrak{m}(t_0(A_{b+1}/N), X) = |\mathcal{T}|X^{b+1}\mathfrak{m}(t_0(\mathcal{T}), X) \tag{7.4.14}$$

where $|\mathcal{T}|$ is the cardinality of $\mathcal{T}$. For any integer $a \geq 0$, we have the identity $\mathfrak{m}(a+1,X)-\mathfrak{m}(a,X) = -q^a X\mathfrak{m}(a,X)$, so we compute

$$\sum_{\substack{t\in\varpi_0\mathcal{T}\\ N=\langle t+\varpi_0^{-b}\rangle}} X^{\mathrm{ord}(N)}\mathfrak{m}(t_0(A_{b+1}/N),X) - \sum_{\substack{t\in\varpi_0\mathcal{T}\\ N=\langle t+\varpi_0^{-b}\rangle}} X^{\mathrm{ord}(N)}\mathfrak{m}(t_0(A_b/N),X) \tag{7.4.15}$$

$$= -|\varpi_0\mathcal{T}|q^{t_0(\mathcal{T})}X^{b+1}\mathfrak{m}(t_0(\mathcal{T}),X). \tag{7.4.16}$$

But the exact sequence

$$0 \to \varpi_0\mathcal{T} \to \mathcal{T} \to \mathcal{T}/\varpi_0\mathcal{T} \to 0 \tag{7.4.17}$$

shows that $|\mathcal{T}| = |\varpi_0\mathcal{T}|q^{t_0(\mathcal{T})}$ since $t_0(\mathcal{T}) = \dim_{\mathbb{F}_q}\mathcal{T}/\varpi_0\mathcal{T}$ by definition. Substituting into (7.4.11) shows $\alpha_{b+1}-\alpha_b = 0$. □

Lemma 7.4.4. *The polynomial* $\mathrm{Den}_{L^\flat,x}(X)_n^\circ - X\mathrm{Den}_{L^\flat,x'}(X)_n^\circ$ *stabilizes for* $\mathrm{val}(x) > 2a_{\max}(L^\flat)$.

Proof. As the x' dependence of $\mathrm{Den}_{L^\flat,x'}(X)_n^\circ$ is only on $\mathrm{val}(x')$, we may assume $x' = \varpi_1^{-1}x$ without loss of generality. Assume $\mathrm{val}(x) > 2a_{\max}(L^\flat)$. The lemma is trivial if $L^\flat$ is not integral (the polynomial is 0), so assume $L^\flat$ is integral.

Inspecting (7.3.3) shows that $\mathrm{Den}_{L^\flat,x}(X)_n^\circ - X\mathrm{Den}_{L^\flat,x'}(X)_n^\circ$ is equal to

$$\sum_{\substack{L\subseteq M\subseteq M^*\\ M\cap L_F^\flat = L^\flat\\ \varpi_1^{-1}x\notin M}} X^{\ell(M/L)}\mathfrak{m}(t(M),X) \tag{7.4.18}$$

where the sum runs over lattices $M \subseteq V$. For each M in the above sum, we know that $L^\flat \subseteq M$ is a saturated sublattice, so there is a direct sum decomposition $M = L^\flat \oplus \langle\xi\rangle$ (not necessarily orthogonal) for some $\xi \in V$. This implies that each M/L is cyclic as an $\mathcal{O}_F$-module. Similar reasoning was used at the beginning of the proof of Lemma 7.3.5. By Lemma 7.4.1, it is enough to show that

$$\sum_{\substack{L\subseteq M\subseteq M^*\\ M_1\cap L_F^\flat = L_1^\flat\\ M/L \text{ is cyclic}\\ \varpi_1^{-1}x\notin M}} X^{\ell(M/L)}\mathfrak{m}(t(M),X) \tag{7.4.19}$$

stabilizes for $\mathrm{val}(x) > 2a_{\max}(L^\flat)$, where the sum runs over lattices $M \subseteq V$ (because the difference between (7.4.19) and (7.4.18) is (7.4.4)). In this proof, the property of being cyclic is always understood with respect to $\mathcal{O}_F$.

We find that (7.4.19) equals

$$\sum_{\substack{L_1\subseteq M_1\subseteq L_2^*\\ M_1\cap L_F^\flat = L_1^\flat\\ \varpi_1^{-1}x_1\notin M_1}} \sum_{\substack{L_2\subseteq M_2\subseteq M_1^*\\ M_2/L_2 \text{ is cyclic}}} X^{\ell(M/L)}\mathfrak{m}(t(M),X) \tag{7.4.20}$$

where the outer sum runs over lattices $M_1 \subseteq V_1$, the right-most sum runs over lattices $M_2 \subseteq V_2$, and $M = M_1 \oplus M_2$. Note that the lattices M_1 always satisfy M_1/L_1 being cyclic, because $M_1 \cap L_F^\flat = L_1^\flat$ implies $M_1 = L_1^\flat \oplus \langle y_1 + \varpi_1^{-e_1}x_1\rangle$ where $y_1 \in L_2^{\flat *}/L_1^\flat$ is of exact $\varpi_1^{e_1}$-torsion. Here we are using the fact that $\varpi_1^{-1}x \notin L$, by the convention in Setup 7.3.1.

To prove the lemma, it is enough to check that (7.4.20) does not change if x is replaced with $\varpi_2 x$. The set of lattices $M_1 \subseteq V_1$ appearing in the outer sum is indexed by elements $y_1 \in L_2^{\flat *}/L_1^\flat$ (since e_1 is determined by y_1, in the above notation), and hence does not change if x is replaced by $\varpi_2 x$ (here using $\mathrm{val}(x) > a_{\max}(L^\flat)$ to ensure $M_1 \subset L_2^*$ for any choice of y_1). Note also that $\ell_0(M_1/L_1) = e_1$ and hence does not change when x is replaced by $\varpi_2 x$.

For the rest of the proof, fix an M_1 as in the outer sum of (7.4.20). We will show that the inner sum of (7.4.20) does not change if x is replaced by $\varpi_2 x$.

Set $A = M_1^*/L_2$. The inner sum is

$$X^{\ell_0(M_1/L_1)} \sum_{\substack{\text{cyclic submodules} \\ N \subseteq A}} X^{\mathrm{ord}(N)} \mathfrak{m}(t_0(A/N), X). \tag{7.4.21}$$

We already discussed that the factor $X^{\ell_0(M_1/L_1)}$ does not change when x is replaced by $\varpi_2 x$. On the other hand, we have $A \cong \mathrm{Hom}_{\mathcal{O}_{F_0}}(L_2^*/M_1, F_0/\mathcal{O}_{F_0})$ so $A \cong L_2^*/M_1$ (non-canonically). If $b \coloneqq \mathrm{val}(x)$ and $\mathcal{T} \coloneqq L_2^{\flat *}/L_1^\flat$, then Lemma 7.4.2 shows $A \cong (\mathcal{T}/\langle y_1 \rangle) \oplus (\varpi_0^{-b}\mathcal{O}_{F_0}/\mathcal{O}_{F_0})$, where y_1 is associated to M_1 as above (since the submodule $(M_1/L_1) \subseteq L_2^*/L_1$ is cyclic and generated by $y_1 + \varpi_1^{-e_1}x_1$ where y_1 is of exact $\varpi_1^{e_1}$-torsion).

Now Lemma 7.4.3 implies that the sum in (7.4.21) does not change if x is replaced by $\varpi_2 x$. □

Proof of Proposition 7.3.2 in split case. Assume F/F_0 is split. As remarked at the beginning of Section 7.4, it is enough to show that $\mathrm{Den}(X, L) - X\mathrm{Den}(X, L')$ stabilizes for $\mathrm{val}(x) > 2a_{\max}(L^\flat)$. We have

$$\mathrm{Den}(X, L) - X\mathrm{Den}(X, L') = \sum_{L^\flat \subseteq M^\flat \subseteq M^{\flat *}} \mathrm{Den}_{M^\flat, x}(X)_n^\circ - X\mathrm{Den}_{M^\flat, x'}(X)_n^\circ \tag{7.4.22}$$

by definition as in (7.3.4), so Lemma 7.4.4 proves the claimed stabilization. □

7.5. **Limits.** We continue in the setup of Section 7.3 but now assume $\varepsilon(V) = -1$ if F/F_0 is nonsplit. Recall also the definitions in (7.3.3) and (7.3.4).

Let $M^\flat \subseteq L_F^\flat$ be any non-degenerate Hermitian $\mathcal{O}_F$-lattice of rank $n-1$ with $t(M^\flat) \leq 1$. If F/F_0 is nonsplit, set

$$\partial\mathrm{Den}_{L^\flat}(x)_n \coloneqq -[\breve{F} : \breve{F}_0]\frac{d}{dX}\bigg|_{X=1} \mathrm{Den}(X, L)_n \quad \partial\mathrm{Den}_{M^\flat, \mathscr{H}}(x)_n^\circ \coloneqq -[\breve{F} : \breve{F}_0]\frac{d}{dX}\bigg|_{X=1} \mathrm{Den}_{M^\flat, x}(X)_n^\circ$$

$$\partial\mathrm{Den}_{L^\flat, \mathscr{H}}(x)_n \coloneqq \sum_{\substack{L \subseteq N \subseteq N^* \\ N^\flat = N \cap L_F^\flat \\ t(N^\flat) \leq 1}} \partial\mathrm{Den}_{N^\flat, \mathscr{H}}(x)_n^\circ \qquad \partial\mathrm{Den}_{L^\flat, \mathscr{V}}(x)_n \coloneqq \partial\mathrm{Den}_{L^\flat}(x)_n - \partial\mathrm{Den}_{L^\flat, \mathscr{H}}(x)_n.$$

If F/F_0 is split, set

$$\mathrm{Den}_{L^\flat}(x)_n \coloneqq \mathrm{Den}(X, L)_n\Big|_{X=1} \qquad \mathrm{Den}_{M^\flat, \mathscr{H}}(x)_n^\circ \coloneqq \mathrm{Den}_{M^\flat, x}(X)_n^\circ\Big|_{X=1}$$

$$\mathrm{Den}_{L^\flat, \mathscr{H}}(x)_n \coloneqq \sum_{\substack{L \subseteq N \subseteq N^* \\ N^\flat = N \cap L_F^\flat \\ t(N^\flat) \leq 1}} \mathrm{Den}_{N^\flat, \mathscr{H}}(x)_n^\circ \qquad \mathrm{Den}_{L^\flat, \mathscr{V}}(x)_n \coloneqq \mathrm{Den}_{L^\flat}(x)_n - \mathrm{Den}_{L^\flat, \mathscr{H}}(x)_n.$$

The above sums run over lattices $N \subseteq V$ (so $N^\flat$ varies). These definitions also apply for any $x \notin L_F^\flat$ (not necessarily perpendicular to $L_F^\flat$), as long as we take $L = L^\flat + \langle x \rangle$.

Lemma 7.5.1. *If F/F_0 is split, then $\mathrm{Den}_{L^\flat,\mathscr{V}}(x)_n = 0$ for all x.*

Proof. Inspecting (7.2.2) shows that $\mathrm{Den}(1,M)_n^\circ = 0$ unless $M = M^*$. Lemma 7.3.4 implies $\mathrm{Den}_{N^\flat,x}(X)_n^\circ|_{X=1} = 0$ unless $t(N^\flat) \leq 1$, i.e. $\mathrm{Den}_{L^\flat}(x)_n = \mathrm{Den}_{L^\flat,\mathscr{H}}(x)_n$. □

Given $x \in V$ with $(x,x) \neq 0$, we set $\mathrm{val}''(x) \coloneqq \mathrm{val}'(x)$ if F/F_0 is not inert and $\mathrm{val}''(x) \coloneqq (\mathrm{val}(x)-1)/2$ if F/F_0 is inert, as shorthand. We say a limit *stabilizes* if the argument of the limit becomes constant. For the following proposition in the case of F/F_0 inert, the limit in fact stabilizes for $\mathrm{val}(x) > a_{\max}(L^\flat)$ by the same proof, using the sharper bound in [LZ22a, Proposition 3.7.1]. The same applies to Corollary 7.5.3.

Proposition 7.5.2. *If F/F_0 is nonsplit, we have*

$$\partial\mathrm{Den}^*(L^\flat)_n = 2[\breve{F} : \breve{F}_0]^{-1} \lim_{x \to 0} \left(\partial\mathrm{Den}_{L^\flat}(x')_n - \mathrm{val}''(x)\mathrm{Den}^*(L^\flat)_n \right). \tag{7.5.1}$$

If F/F_0 is split, we have

$$\partial\mathrm{Den}^*(L^\flat)_n = \lim_{x \to 0} \left(\mathrm{Den}_{L^\flat}(x')_n - \mathrm{val}(x)\mathrm{Den}^*(L^\flat)_n \right). \tag{7.5.2}$$

The expressions are 0 if $L^\flat$ is not integral, and all limits stabilize for $\mathrm{val}(x) \gg 0$. If $L^\flat$ is integral, then the limits stabilize when $\mathrm{val}(x) > 2[\breve{F} : \breve{F}_0]^{-1} a_{\max}(L^\flat)$.

Proof. We emphasize that we are following Setup 7.3.1; in particular, we have $x' \to 0$ as $x \to 0$. Assume $L^\flat$ is integral (as the lemma is otherwise clear) and assume $\mathrm{val}(x) > 2[\breve{F} : \breve{F}_0]^{-1} a_{\max}(L^\flat)$. The key input is the induction formula from Proposition 7.3.2.

For the reader's convenience, we recall the formulas

$$\mathrm{Den}^*(X, L^\flat)_n = (qX^{-1/2})^{\mathrm{val}'(L^\flat)} \mathrm{Den}(X, L^\flat)_n \tag{7.5.3}$$

$$\partial\mathrm{Den}^*(L^\flat)_n \coloneqq -2[\breve{F} : \breve{F}_0] \frac{d}{dX}\bigg|_{X=1} \mathrm{Den}^*(q^2X, L^\flat)_n \qquad \mathrm{Den}^*(L^\flat)_n \coloneqq [\breve{F} : \breve{F}_0] \cdot \mathrm{Den}^*(q^2, L^\flat)_n,$$

where the first equation is specialized from (6.3.10), and the other two equations are copied from (7.1.6) and (7.1.7).

Case F/F_0 is nonsplit: Multiplying the induction formula from Proposition 7.3.2 by $X^{-\mathrm{val}'(L^\flat)/2}$, we obtain

$$(1-X)\mathrm{Den}^*(q^2X, L^\flat)_n = X^{-\mathrm{val}'(L^\flat)/2}\mathrm{Den}(X, L)_n - X^{-\mathrm{val}'(L^\flat)/2+2[\breve{F}:\breve{F}_0]^{-1}}\mathrm{Den}(X, L')_n \tag{7.5.4}$$

since the left-hand side is also $X^{-\mathrm{val}'(L^\flat)/2}(1-X)\mathrm{Den}(q^2X, L^\flat)_n$.

Taking one derivative of (7.5.4) at $X = 1$ and multiplying by $-[\breve{F} : \breve{F}_0]$ yields

$$\mathrm{Den}^*(L^\flat)_n = \partial\mathrm{Den}_{L^\flat}(x)_n - \partial\mathrm{Den}_{L^\flat}(x')_n. \tag{7.5.5}$$

Here we used $\mathrm{Den}(1, L)_n = \mathrm{Den}(1, L')_n = 0$ because $\varepsilon(V) = -1$ causes a sign in the functional equation (6.3.12). Taking two derivatives of (7.5.4) at $X = 1$ and multiplying by $[\breve{F} : \breve{F}_0]$ yields

the identity

$$\mathrm{val}'(L^\flat)\partial\mathrm{Den}_{L^\flat}(x)_n + [\breve{F}:\breve{F}_0]\frac{d^2}{dX^2}\bigg|_{X=1}\mathrm{Den}(X,L)_n \tag{7.5.6}$$
$$= (\mathrm{val}'(L^\flat) - 4[\breve{F}:\breve{F}_0]^{-1})\partial\mathrm{Den}_{L^\flat}(x')_n + [\breve{F}:\breve{F}_0]\frac{d^2}{dX^2}\bigg|_{X=1}\mathrm{Den}(X,L')_n + \partial\mathrm{Den}^*(L^\flat)_n.$$

Again using $\varepsilon(V) = -1$, we apply the functional equation for $\mathrm{Den}(X,L)$ from (6.3.12) to find

$$\frac{d^2}{dX^2}\bigg|_{X=1}\mathrm{Den}(X,L)_n = (\mathrm{val}(L)-1)\frac{d}{dX}\bigg|_{X=1}\mathrm{Den}(X,L)_n \tag{7.5.7}$$
$$= -(\mathrm{val}(L)-1)[\breve{F}:\breve{F}_0]^{-1}\partial\mathrm{Den}_{L^\flat}(x)_n \tag{7.5.8}$$

(the second equality is by definition) and similarly for L'. We also have

$$\mathrm{val}'(L^\flat) = \mathrm{val}(L) - 2[\breve{F}:\breve{F}_0]^{-1}\mathrm{val}''(x) - 1 \qquad \mathrm{val}(L) = \mathrm{val}(L') + 2[\breve{F}:\breve{F}_0]^{-1}. \tag{7.5.9}$$

Substituting all displayed equations into (7.5.6) proves the claim.

Case F/F_0 is split: Multiplying both sides of the induction formula from Proposition 7.3.2 by $X^{-\mathrm{val}(L^\flat)/2}$, we obtain

$$\mathrm{Den}^*(q^2X, L^\flat)_n = X^{-\mathrm{val}(L^\flat)/2}\mathrm{Den}(X,L)_n - X^{-\mathrm{val}(L^\flat)/2+1}\mathrm{Den}(X,L')_n \tag{7.5.10}$$

since the left-hand side is also $X^{-\mathrm{val}(L^\flat)/2}\mathrm{Den}(q^2X, L^\flat)_n$.

Evaluating (7.5.10) at $X = 1$ yields

$$\mathrm{Den}^*(L^\flat)_n = \mathrm{Den}(1,L)_n - \mathrm{Den}(1,L')_n. \tag{7.5.11}$$

Taking one derivative of (7.5.10) at $X = 1$ and multiplying by -2, we find

$$\mathrm{val}(L^\flat)\mathrm{Den}(1,L)_n - 2\frac{d}{dX}\bigg|_{X=1}\mathrm{Den}(X,L)_n \tag{7.5.12}$$
$$= (\mathrm{val}(L^\flat)-2)\mathrm{Den}(1,L')_n - 2\frac{d}{dX}\bigg|_{X=1}\mathrm{Den}(X,L')_n + \partial\mathrm{Den}^*(L^\flat)_n.$$

The functional equation (6.3.12) implies

$$2\frac{d}{dX}\bigg|_{X=1}\mathrm{Den}(X,L)_n = \mathrm{val}(L)\mathrm{Den}(1,L)_n \tag{7.5.13}$$

and similarly for L'. We also have

$$\mathrm{val}(L^\flat) = \mathrm{val}(L) - \mathrm{val}(x) \qquad \mathrm{val}(L) = \mathrm{val}(L') + 1. \tag{7.5.14}$$

Substituting all displayed equations into (7.5.12) proves the claim. □

Corollary 7.5.3. *Let $M^\flat \subseteq L_F^\flat$ be any full rank integral lattice with $t(M^\flat) \leq 1$. If F/F_0 is nonsplit, the following formulas hold.*

(1) $\partial\mathrm{Den}^*_{\mathscr{V}}(L^\flat)_n = 2[\breve{F}:\breve{F}_0]^{-1}\lim_{x\to 0}\partial\mathrm{Den}_{L^\flat,\mathscr{V}}(x)_n$

(2) $\partial\mathrm{Den}^*_{\mathscr{H}}(L^\flat)_n = 2[\breve{F}:\breve{F}_0]^{-1}\lim_{x\to 0}\Big(\partial\mathrm{Den}_{L^\flat,\mathscr{H}}(x')_n - \mathrm{val}''(x)\mathrm{Den}^*(L^\flat)_n\Big)$

(3) $\partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n = 2[\breve{F}:\breve{F}_0]^{-1}\lim_{x\to 0}\Big(\partial\mathrm{Den}_{M^\flat,\mathscr{H}}(x')^\circ_n - \mathrm{val}''(x)\mathrm{Den}^*(M^\flat)^\circ_n\Big)$.

If F/F_0 is split, the following formulas hold.

(1) $\partial\mathrm{Den}^*_{\mathscr{V}}(L^\flat)_n = \lim_{x\to 0}\mathrm{Den}_{L^\flat,\mathscr{V}}(x)_n$

(2) $\partial\mathrm{Den}^*_{\mathscr{H}}(L^\flat)_n = \lim_{x\to 0}\Big(\mathrm{Den}_{L^\flat,\mathscr{H}}(x')_n - \mathrm{val}(x)\mathrm{Den}^*(L^\flat)_n\Big)$

(3) $\partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n = \lim_{x\to 0}\Big(\mathrm{Den}_{M^\flat,\mathscr{H}}(x')^\circ_n - \mathrm{val}(x)\mathrm{Den}^*(M^\flat)^\circ_n\Big)$.

All limits stabilize for $\mathrm{val}(x)\gg 0$. *The expressions (1) and (2) are* 0 *if* $L^\flat$ *is not integral. If* $L^\flat$ *is integral, then the limits in (1) and (2) stabilize for* $\mathrm{val}(x) > 2[\breve{F}:\breve{F}_0]^{-1}a_{\max}(L^\flat)$. *The limit in (3) stabilizes for* $\mathrm{val}(x) > 2[\breve{F}:\breve{F}_0]^{-1}a_{\max}(M^\flat)$.

Proof. Denote the result of Proposition 7.5.2 as (0). We have (3) $\implies$ (2) in all cases, nonsplit or split, by summing over $M^\flat$ containing $L^\flat$. We have (0) $\implies$ (3) by taking $L^\flat = M^\flat$ and inducting on $\mathrm{val}(M^\flat)$, starting with the base cases of $M^\flat$ being maximal integral with $t(M^\flat)\le 1$; in that case we have $\partial\mathrm{Den}^*(M^\flat)_n = \partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)_n = \partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n$, and similarly for $\mathrm{Den}^*(M^\flat)_n$ as well as $\partial\mathrm{Den}_{M^\flat}(x')_n$ (nonsplit) and $\mathrm{Den}_{M^\flat}(x')_n$ (split). Since (0) = (1) + (2) as a sum of equations, we conclude that (0) $\implies$ (1) as well. □

The following lemma is the geometric counterpart of Corollary 7.5.3(1), upon specializing to $\alpha = {}^{\mathbb{L}}\mathcal{Z}(L^\flat)_{\mathscr{V}}$ for a non-degenerate Hermitian $\mathcal{O}_F$-lattice $L^\flat\subseteq\mathbf{W}$. The notations $\alpha\cdot\mathcal{E}^\vee$ and $\alpha\cdot{}^{\mathbb{L}}\mathcal{Z}(w)$ were explained in (3.6.14).

Lemma 7.5.4. *Take* $F_0=\mathbb{Q}_p$, *and assume* $p\neq 2$ *if* $F/\mathbb{Q}_p$ *is ramified. Let* $\mathcal{Z}\to\mathrm{Spec}\,\overline{k}$ *be a proper scheme equipped with a closed immersion* $\mathcal{Z}\hookrightarrow\mathcal{N}$. *Given any* $\alpha\in\mathrm{gr}_1K_0'(\mathcal{Z})_{\mathbb{Q}}$, *we have*

$$\deg_{\overline{k}}(\alpha\cdot\mathcal{E}^\vee) = \lim_{w\to 0}\deg_{\overline{k}}(\alpha\cdot{}^{\mathbb{L}}\mathcal{Z}(w)) \tag{7.5.15}$$

where the limit runs over $w\in\mathbf{W}$. *The limit stabilizes for* w *satisfying* $\mathcal{Z}\subseteq\mathcal{Z}(w)$.

Proof. We may assume $F/\mathbb{Q}_p$ is nonsplit, as otherwise $\mathrm{gr}_1K_0'(\mathcal{Z})=0$ for dimension reasons (Section 3.5) and the lemma is trivial.

For any fixed $w\in\mathbf{W}$, there exists $e\gg 0$ such that $\mathcal{Z}\subseteq\mathcal{Z}(p^ew)$, because over a quasi-compact base scheme, p^e times any quasi-homomorphism is a homomorphism for $e\gg 0$. Hence $\mathcal{Z}\subseteq\mathcal{Z}(w)$ for all $w\in\mathbf{W}$ lying in a sufficiently small neighborhood of 0.

Assume $w\in\mathbf{W}$ is such that $\mathcal{Z}\subseteq\mathcal{Z}(w)$. Write $\mathcal{I}(w)\subseteq\mathcal{O}_{\mathcal{N}}$ for the ideal sheaf of the Cartier divisor $\mathcal{Z}(w)$ (Section 3.6). As recalled in (3.6.9), Grothendieck–Messing theory as in [How19, Definition 4.2] (inert) and [LL22, Lemma 2.39] (ramified) gives a canonical isomorphism $\mathcal{E}\otimes\mathcal{O}_{\mathcal{Z}(w)}\cong\mathcal{I}(w)\otimes\mathcal{O}_{\mathcal{Z}(w)}$. From this, we immediately obtain $\beta\otimes\mathcal{E}=\beta\otimes\mathcal{I}(w)$ in $K_0'(\mathcal{Z})_{\mathbb{Q}}$ for any $\beta\in K_0'(\mathcal{Z})_{\mathbb{Q}}$. Taking β to be any lift of α, we find $\alpha-\alpha\otimes\mathcal{E}=\alpha-\alpha\otimes\mathcal{I}(w)$ as elements of $\mathrm{gr}_0\,K_0'(\mathcal{Z})_{\mathbb{Q}}$.

A related "linear invariance" argument appears in the proof of [LL22, Lemma 2.55(3)]. □

Proof of Theorem 7.1.2. The horizontal part of the theorem was already verified in Proposition 7.2.1, so it remains to show $\mathrm{Int}_{\mathscr{V}}(L^\flat)_n = \partial\mathrm{Den}^*_{\mathscr{V}}(L^\flat)_n$.

If $F/\mathbb{Q}_p$ is split, then ${}^{\mathbb{L}}\mathcal{Z}(L^\flat)_{\mathscr{V}}=0$ and so $\mathrm{Int}_{\mathscr{V}}(L^\flat)_n=0$. Applying Corollary 7.5.3(1) with $V=\mathbf{V}$, we find $\partial\mathrm{Den}^*_{\mathscr{V}}(L^\flat)_n=0$ since $\mathrm{Den}_{L^\flat,\mathscr{V}}(x)_n=0$ for all x (Lemma 7.5.1).

Next assume $F/\mathbb{Q}_p$ is nonsplit. For any $w\in\mathbf{W}$ not in $L^\flat_F$, we have $\deg_{\overline{k}}({}^{\mathbb{L}}\mathcal{Z}(L^\flat)_{\mathscr{V}}\cdot{}^{\mathbb{L}}\mathcal{Z}(w)) = \partial\mathrm{Den}_{L^\flat,\mathscr{V}}(w)_n$ by [LZ22a, Theorem 8.2.1] (inert) and [LL22, Theorem 2.7] (the "vertical" parts of the main results of loc. cit.). Lemma 7.5.4 implies $\mathrm{Int}_{\mathscr{V}}(L^\flat)_n = 2[\breve{F}:\breve{F}_0]^{-1}\lim_{w\to 0}\partial\mathrm{Den}_{L^\flat,\mathscr{V}}(w)_n$.

Restricting to w perpendicular to $L_F^\flat$, the limiting formula in Corollary 7.5.3(1) now implies $\mathrm{Int}_{\mathscr{V}}(L^\flat)_n = \partial\mathrm{Den}^*_{\mathscr{V}}(L^\flat)_n$. □

Remark 7.5.5. Suppose $F_0 = \mathbb{Q}_p$, suppose $F/\mathbb{Q}_p$ is nonsplit, and assume $p \neq 2$. Let $M^\flat \subseteq \mathbf{V}$ be a non-degenerate integral $\mathcal{O}_F$-lattice of rank $n-1$ with $t(M^\flat) \leq 1$. As above, let $\mathcal{Z}(M^\flat)^\circ \subseteq \mathcal{N}$ be the associated quasi-canonical lifting cycle.

For any nonzero $w \in \mathbf{V}$ not in $M_F^\flat$, we have $\deg_{\overline{k}}(\mathcal{Z}(M^\flat)^\circ \cap \mathcal{Z}(w)) = \partial\mathrm{Den}_{M^\flat,\mathscr{H}}(w)^\circ_n$ by [KR11, Proposition 8.4] (inert, see also [LZ22a, Corollary 5.4.6, Theorem 6.1.3]) and [LL22, Corollary 2.46] (ramified), i.e. the "horizontal" parts of the main results of loc. cit..

The "horizontal part" of our main theorem showed $\mathrm{Int}_{\mathscr{H}}(M^\flat)^\circ_n = \partial\mathrm{Den}^*_{\mathscr{H}}(M^\flat)^\circ_n$ (Proposition 7.2.1). Using also the special value formula in Lemma 7.1.3, our limiting result Corollary 7.5.3(3) is equivalent to the geometric statement

$$\begin{aligned} &2\deg \mathcal{Z}(M^\flat)^\circ \cdot \delta_{\mathrm{tau}}(\mathrm{val}'(M^\flat)) \\ &= 2[\breve{F} : \breve{\mathbb{Q}}_p]^{-1} \lim_{x \to 0} \Big(\deg_{\overline{k}}(\mathcal{Z}(M^\flat)^\circ \cap \mathcal{Z}(x)) - \mathrm{val}''(\varpi x) \deg \mathcal{Z}(M^\flat)^\circ \Big) \end{aligned} \tag{7.5.16}$$

(limiting over nonzero x perpendicular to $M^\flat$) where $\delta_{\mathrm{tau}}(\mathrm{val}'(M^\flat))$ is the "local change of tautological height", as in (5.2.7) (which is $-1/2$ times the "local change of Faltings height" $\delta_{\mathrm{Fal}}(\mathrm{val}'(M^\flat))$).

To prove our main theorem, we verified (7.5.16) indirectly by the computations in Section 7.2. Direct computations are also possible.

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Department of Mathematics, Fine Hall - Washington Road. Princeton, NJ 08544, USA
Email address: rcchen.math@gmail.com